

Human review packet: KR21Monoculture

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: KR21Monoculture
Paper id: KR21Monoculture
Source version: arXiv:2101.05853
Source record: audit/paper_statement_map.json
Rows: 57
Generated: 2026-08-18
Source URL: [official source](#)

Review status: 0/57 source-claim screens; 0/56 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper KR21Monoculture --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

EconCSLib.Probability.StrictlyWellOrderedNoise

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (f : \mathbb{R} \rightarrow \mathbb{R}) \{ \{a \ b \ c \ d : \mathbb{R}\} \}, b < a \rightarrow d < c \rightarrow f(a - c) * f(b - d) > f(a - d) * f(b - c)$

Exact Lean library declaration

```
def StrictlyWellOrderedNoise (f : ℝ → ℝ) : Prop :=  
  ∀ {a b c d : ℝ}, b < a → d < c →  
    f (a - c) * f (b - d) > f (a - d) * f (b - c)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.WeaklyWellOrderedNoise

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (f : \mathbb{R} \rightarrow \mathbb{R}) \{ \{a \ b \ c \ d : \mathbb{R}\} \}, b < a \rightarrow d < c \rightarrow f(a - d) * f(b - c) \leq f(a - c) * f(b - d)$

Exact Lean library declaration

```
def WeaklyWellOrderedNoise (f : ℝ → ℝ) : Prop :=  
  ∀ {a b c d : ℝ}, b < a → d < c →  
    f (a - d) * f (b - c) ≤ f (a - c) * f (b - d)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.gaussianNoiseKernel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa \ x : \mathbb{R}) \rightarrow \text{Real.exp } (-\kappa * x^2)$

Exact Lean library declaration

```
def gaussianNoiseKernel (κ : ℝ) (x : ℝ) : ℝ :=  
  Real.exp (-κ * x ^ 2)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.gaussianVarianceFromStd

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\sigma : \mathbb{R}) \rightarrow \text{NNReal.mk } (\sigma^2) \dots$

Exact Lean library declaration

```
def gaussianVarianceFromStd (σ : ℝ) : ℝ≥0 :=  
  NNReal.mk (σ ^ 2) (sq_nonneg σ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.independentGaussianPairMeasureWithStd

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(σ xi xj : ℝ) →  
  (ProbabilityTheory.gaussianReal xi (EconCSLib.Probability.gaussianVarianceFromStd σ)).prod  
  (ProbabilityTheory.gaussianReal xj (EconCSLib.Probability.gaussianVarianceFromStd σ))
```

Exact Lean library declaration

```
noncomputable def independentGaussianPairMeasureWithStd  
  (σ xi xj : ℝ) : Measure (ℝ × ℝ) :=  
  (ProbabilityTheory.gaussianReal xi (gaussianVarianceFromStd σ)).prod  
  (ProbabilityTheory.gaussianReal xj (gaussianVarianceFromStd σ))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.laplacianNoiseKernel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(lam x : ℝ) → Real.exp (-lam * |x|)
```

Exact Lean library declaration

```
def laplacianNoiseKernel (lam : ℝ) (x : ℝ) : ℝ :=  
  Real.exp (-lam * |x|)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.Candidate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow \text{Fin } (n + 2)$

Exact Lean library declaration

abbrev Candidate $(n : \mathbb{N}) := \text{Fin } (n + 2)$

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.Ranking

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow \text{Equiv.Perm } (\text{EconCSLib.SocialChoice.Ranking.Candidate } n)$

Exact Lean library declaration

abbrev Ranking $(n : \mathbb{N}) := \text{Equiv.Perm } (\text{Candidate } n)$

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.firstChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (π : EconCSLib.SocialChoice.Ranking.Ranking n) → π 0
```

Exact Lean library declaration

```
def firstChoice {n : ℕ} (π : Ranking n) : Candidate n :=  
  π 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.secondChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (π : EconCSLib.SocialChoice.Ranking.Ranking n) → π 1
```

Exact Lean library declaration

```
def secondChoice {n : ℕ} (π : Ranking n) : Candidate n :=  
  π 1
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.bestRemainingAfter

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (π : EconCSLib.SocialChoice.Ranking.Ranking n) →  
  (c : EconCSLib.SocialChoice.Ranking.Candidate n) →  
    if EconCSLib.SocialChoice.Ranking.firstChoice π = c then EconCSLib.SocialChoice.Ranking.secondChoice π  
    else EconCSLib.SocialChoice.Ranking.firstChoice π
```

Exact Lean library declaration

```
def bestRemainingAfter {n : ℕ} (π : Ranking n) (c : Candidate n) : Candidate n :=  
  if firstChoice π = c then secondChoice π else firstChoice π
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.secondMoverUtility

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
  (π σ : EconCSLib.SocialChoice.Ranking.Ranking n) →  
    value (EconCSLib.SocialChoice.Ranking.bestRemainingAfter π (EconCSLib.SocialChoice.Ranking.firstChoice σ))
```

Exact Lean library declaration

```
def secondMoverUtility {n : ℕ} (value : Candidate n → ℝ)  
  (π σ : Ranking n) : ℝ :=  
  value (bestRemainingAfter π (firstChoice σ))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{DecidableEq } \alpha] \rightarrow (\mu : \text{PMF } \alpha) \rightarrow (f : \alpha \rightarrow \mathbb{R}) \rightarrow \sum a, (\mu a).toReal * f a$

Exact Lean library declaration

```
noncomputable def pmfExp {α : Type*} [Fintype α] [DecidableEq α]
  (μ : PMF α) (f : α → ℝ) : ℝ :=
  ∑ a, (μ a).toReal * f a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfPairExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      [inst_2 : Fintype β] →  
        [inst_3 : DecidableEq β] →  
          (μ : PMF α) → (ν : PMF β) → (f : α → β → ℝ) → EconCSLib.pmfExp μ fun a => EconCSLib.pmfExp ν fun b => f a b
```

Exact Lean library declaration

```
noncomputable def pmfPairExp {α β : Type*}  
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]  
  (μ : PMF α) (ν : PMF β) (f : α → β → ℝ) : ℝ :=  
  pmfExp μ (fun a => pmfExp ν (fun b => f a b))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (μ₂ μ₁ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
  (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
    EconCSLib.pmfPairExp μ₂ μ₁ fun π σ => EconCSLib.SocialChoice.Ranking.secondMoverUtility value π σ
```

Exact Lean library declaration

```
def expectedSecondMoverIndependent {n : ℕ}  
  (μ₂ μ₁ : PMF (Ranking n)) (value : Candidate n → ℝ) : ℝ :=  
  pmfPairExp μ₂ μ₁ (fun π σ => secondMoverUtility value π σ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
    (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
      EconCSLib.pmfExp μ fun π => value (EconCSLib.SocialChoice.Ranking.secondChoice π)
```

Exact Lean library declaration

```
def expectedSecondMoverShared {n : ℕ}  
  (μ : PMF (Ranking n)) (value : Candidate n → ℝ) : ℝ :=  
  pmfExp μ (fun π => value (secondChoice π))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.PrefersIndependentReranking

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),  
  EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared μ value <  
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent μ μ value
```

Exact Lean library declaration

```
def PrefersIndependentReranking {n : ℕ}  
  (μ : PMF (Ranking n)) (value : Candidate n → ℝ) : Prop :=  
  expectedSecondMoverShared μ value < expectedSecondMoverIndependent μ μ value
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.PrefersWeakerCompetition

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (μBetter μWorse : PMF (EconCSLib.SocialChoice.Ranking.Ranking n))
  (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
  EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent μWorse μBetter value <
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent μWorse μWorse value
```

Exact Lean library declaration

```
def PrefersWeakerCompetition {n : ℕ}
  (μBetter μWorse : PMF (Ranking n)) (value : Candidate n → ℝ) : Prop :=
  expectedSecondMoverIndependent μWorse μBetter value <
    expectedSecondMoverIndependent μWorse μWorse value
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.RankingPair

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow \text{EconCSLib.SocialChoice.Ranking.Ranking } n \times \text{EconCSLib.SocialChoice.Ranking.Ranking } n$

Exact Lean library declaration

abbrev RankingPair $(n : \mathbb{N}) := \text{Ranking } n \times \text{Ranking } n$

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rankOf

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
(π : EconCSLib.SocialChoice.Ranking.Ranking n) → (c : EconCSLib.SocialChoice.Ranking.Candidate n) → (Equiv.symm π) c
```

Exact Lean library declaration

```
def rankOf {n : ℕ} (π : Ranking n) (c : Candidate n) : Candidate n :=  
  π.symm c
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (ρ : EconCSLib.SocialChoice.Ranking.Ranking n) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
  {a b : EconCSLib.SocialChoice.Ranking.Candidate n},
  EconCSLib.SocialChoice.Ranking.rankOf ρ a < EconCSLib.SocialChoice.Ranking.rankOf ρ b → value b < value a
```

Exact Lean library declaration

```
def StrictlyOrderedBy {n : ℕ} (ρ : Ranking n) (value : Candidate n → ℝ) : Prop :=
  ∀ {a b : Candidate n}, rankOf ρ a < rankOf ρ b → value b < value a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.WeaklyOrderedBy

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (ρ : EconCSLib.SocialChoice.Ranking.Ranking n) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
  {a b : EconCSLib.SocialChoice.Ranking.Candidate n},
  EconCSLib.SocialChoice.Ranking.rankOf ρ a < EconCSLib.SocialChoice.Ranking.rankOf ρ b → value b ≤ value a
```

Exact Lean library declaration

```
def WeaklyOrderedBy {n : ℕ} (ρ : Ranking n) (value : Candidate n → ℝ) : Prop :=
  ∀ {a b : Candidate n}, rankOf ρ a < rankOf ρ b → value b ≤ value a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.disagreementEvent

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (a : EconCSLib.SocialChoice.Ranking.RankingPair n),  
  EconCSLib.SocialChoice.Ranking.firstChoice a.1 ≠ EconCSLib.SocialChoice.Ranking.firstChoice a.2
```

Exact Lean library declaration

```
def disagreementEvent {n : ℕ} : RankingPair n → Prop :=  
  fun p => firstChoice p.1 ≠ firstChoice p.2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.decidableDisagreementEvent

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (a : EconCSLib.SocialChoice.Ranking.RankingPair n) → id inferInstance
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfPairProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{β : Type u_2} →
  [inst : Fintype α] →
    [inst_1 : DecidableEq α] →
      [inst_2 : Fintype β] →
        [inst_3 : DecidableEq β] →
          (μ : PMF α) →
            (ν : PMF β) →
              (p : α × β → Prop) →
                [inst_4 : DecidablePred p] → EconCSLib.pmfPairExp μ ν fun a b => if p (a, b) then 1 else 0
```

Exact Lean library declaration

```
noncomputable def pmfPairProb {α β : Type*}
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]
  (μ : PMF α) (ν : PMF β) (p : α × β → Prop) [DecidablePred p] : ℝ :=
  pmfPairExp μ ν (fun a b => if p (a, b) then 1 else 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.disagreementProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
(μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
EconCSLib.pmfPairProb μ μ EconCSLib.SocialChoice.Ranking.disagreementEvent
```

Exact Lean library declaration

```
def disagreementProb {n : ℕ} (μ : PMF (Ranking n)) : ℝ :=  
  pmfPairProb μ μ disagreementEvent
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.expectedFirstMoverUtility

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
    (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
      EconCSLib.pmfExp μ fun π => value (EconCSLib.SocialChoice.Ranking.firstChoice π)
```

Exact Lean library declaration

```
def expectedFirstMoverUtility {n : ℕ}  
  (μ : PMF (Ranking n)) (value : Candidate n → ℝ) : ℝ :=  
  pmfExp μ (fun π => value (firstChoice π))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      (μ : PMF α) → (p : α → Prop) → [inst_2 : DecidablePred p] → EconCSLib.pmfExp μ fun a => if p a then 1 else 0
```

Exact Lean library declaration

```
noncomputable def pmfProb {α : Type*} [Fintype α] [DecidableEq α]  
  (μ : PMF α) (p : α → Prop) [DecidablePred p] : ℝ :=  
  pmfExp μ (fun a => if p a then 1 else 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.firstChoiceProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
  (c : EconCSLib.SocialChoice.Ranking.Candidate n) →  
    EconCSLib.pmfProb μ fun π => c = EconCSLib.SocialChoice.Ranking.firstChoice π
```

Exact Lean library declaration

```
def firstChoiceProb {n : ℕ}  
  (μ : PMF (Ranking n)) (c : Candidate n) : ℝ :=  
  EconCSLib.pmfProb μ (fun π => c = firstChoice π)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.firstChoiceMissProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (μ : PMF (EconCSLib.SocialChoice.Ranking.Ranking n)) →  
    (c : EconCSLib.SocialChoice.Ranking.Candidate n) → 1 - EconCSLib.SocialChoice.Ranking.firstChoiceProb μ c
```

Exact Lean library declaration

```
def firstChoiceMissProb {n : ℕ}  
  (μ : PMF (Ranking n)) (c : Candidate n) : ℝ :=  
  1 - firstChoiceProb μ c
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.instMeasurableSpaceRanking

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow \mathbb{T}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.invertedPair

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {n : ℕ} (ρ π : EconCSLib.SocialChoice.Ranking.Ranking n)
  (ab : EconCSLib.SocialChoice.Ranking.Candidate n × EconCSLib.SocialChoice.Ranking.Candidate n),
  EconCSLib.SocialChoice.Ranking.rankOf ρ ab.1 < EconCSLib.SocialChoice.Ranking.rankOf ρ ab.2 ∧
  EconCSLib.SocialChoice.Ranking.rankOf π ab.2 < EconCSLib.SocialChoice.Ranking.rankOf π ab.1
```

Exact Lean library declaration

```
def invertedPair {n : ℕ} (ρ π : Ranking n) (ab : Candidate n × Candidate n) : Prop :=
  rankOf ρ ab.1 < rankOf ρ ab.2 ∧ rankOf π ab.2 < rankOf π ab.1
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.inversionFinset

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (ρ π : EconCSLib.SocialChoice.Ranking.Ranking n) → {ab | EconCSLib.SocialChoice.Ranking.invertedPair ρ π ab}
```

Exact Lean library declaration

```
noncomputable def inversionFinset {n : ℕ} (ρ π : Ranking n) :  
  Finset (Candidate n × Candidate n) := by  
  classical  
  exact Finset.univ.filter (fun ab => invertedPair ρ π ab)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.kendallTau

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (ρ π : EconCSLib.SocialChoice.Ranking.Ranking n) → (EconCSLib.SocialChoice.Ranking.inversionFinset ρ π).card
```

Exact Lean library declaration

```
noncomputable def kendallTau {n : ℕ} (ρ π : Ranking n) : ℕ :=  
  (inversionFinset ρ π).card
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rerankingGainOnPair

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
  (π σ : EconCSLib.SocialChoice.Ranking.Ranking n) →  
    if EconCSLib.SocialChoice.Ranking.firstChoice π = EconCSLib.SocialChoice.Ranking.firstChoice σ then 0  
    else value (EconCSLib.SocialChoice.Ranking.firstChoice π) - value (EconCSLib.SocialChoice.Ranking.secondChoice π)
```

Exact Lean library declaration

```
def rerankingGainOnPair {n : ℕ} (value : Candidate n → ℝ)  
  (π σ : Ranking n) : ℝ :=  
  if firstChoice π = firstChoice σ then 0  
  else value (firstChoice π) - value (secondChoice π)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.pairRerankingGain

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) →  
  (a : EconCSLib.SocialChoice.Ranking.RankingPair n) →  
    EconCSLib.SocialChoice.Ranking.rerankingGainOnPair value a.1 a.2
```

Exact Lean library declaration

```
def pairRerankingGain {n : ℕ} (value : Candidate n → ℝ) :  
  RankingPair n → ℝ :=  
  fun p => rerankingGainOnPair value p.1 p.2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rankByScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (score : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) → Tuple.sort fun i => -score i
```

Exact Lean library declaration

```
noncomputable def rankByScore {n : ℕ} (score : Candidate n → ℝ) : Ranking n :=  
  Tuple.sort (fun i : Candidate n => -score i)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rankingPMFOfMeasure

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
(μ : MeasureTheory.Measure Ω) →
[inst_1 : MeasureTheory.IsProbabilityMeasure μ] →
(rank : Ω → EconCSLib.SocialChoice.Ranking.Ranking n) →
(hrank : Measurable rank) → (MeasureTheory.Measure.map rank μ).toPMF
```

Exact Lean library declaration

```
def rankingPMFOfMeasure
  {n : ℕ} {Ω : Type*} [MeasurableSpace Ω]
  (μ : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure μ]
  (rank : Ω → Ranking n) (hrank : Measurable rank) : PMF (Ranking n) :=
  @MeasureTheory.Measure.toPMF (Ranking n) _ _ _ (μ.map rank)
  (MeasureTheory.Measure.isProbabilityMeasure_map hrank.aemeasurable)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking012

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.refl (EconCSLib.SocialChoice.Ranking.Candidate 1)

Exact Lean library declaration

```
def rum3Ranking012 : Ranking 1 :=  
  Equiv.refl (Candidate 1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking021

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.swap 1 2

Exact Lean library declaration

```
def rum3Ranking021 : Ranking 1 :=  
  Equiv.swap (1 : Candidate 1) (2 : Candidate 1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking102

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.swap 0 1

Exact Lean library declaration

```
def rum3Ranking102 : Ranking 1 :=  
  Equiv.swap (0 : Candidate 1) (1 : Candidate 1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking120

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.trans (Equiv.swap 1 2) (Equiv.swap 0 1)

Exact Lean library declaration

```
def rum3Ranking120 : Ranking 1 :=  
  (Equiv.swap (1 : Candidate 1) (2 : Candidate 1)).trans  
  (Equiv.swap (0 : Candidate 1) (1 : Candidate 1))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking201

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.trans (Equiv.swap 0 2) (Equiv.swap 0 1)

Exact Lean library declaration

```
def rum3Ranking201 : Ranking 1 :=  
  (Equiv.swap (0 : Candidate 1) (2 : Candidate 1)).trans  
    (Equiv.swap (0 : Candidate 1) (1 : Candidate 1))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3Ranking210

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

Equiv.swap 0 2

Exact Lean library declaration

```
def rum3Ranking210 : Ranking 1 :=  
  Equiv.swap (0 : Candidate 1) (2 : Candidate 1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.SocialChoice.Ranking.rum3RankByScores

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(s1 s2 s3 : ℝ) →
  if h0 : s2 ≤ s1 ∧ s3 ≤ s1 then
    if s3 ≤ s2 then EconCSLib.SocialChoice.Ranking.rum3Ranking012 else EconCSLib.SocialChoice.Ranking.rum3Ranking021
  else
    if h1 : s1 < s2 ∧ s3 ≤ s2 then
      if s3 ≤ s1 then EconCSLib.SocialChoice.Ranking.rum3Ranking102 else EconCSLib.SocialChoice.Ranking.rum3Ranking120
    else
      if s2 ≤ s1 then EconCSLib.SocialChoice.Ranking.rum3Ranking201 else EconCSLib.SocialChoice.Ranking.rum3Ranking210
```

Exact Lean library declaration

```
noncomputable def rum3RankByScores (s1 s2 s3 : ℝ) : Ranking 1 :=
  if h0 : s2 ≤ s1 ∧ s3 ≤ s1 then
    if s3 ≤ s2 then rum3Ranking012 else rum3Ranking021
  else if h1 : s1 < s2 ∧ s3 ≤ s2 then
    if s3 ≤ s1 then rum3Ranking102 else rum3Ranking120
  else
    if s2 ≤ s1 then rum3Ranking201 else rum3Ranking210
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finCons

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{k : ℕ} →
(head : α) →
(tail : Fin k → α) →
(a : Fin (k + 1)) →
  match a with
  | ⟨0, _h⟩ => head
  | ⟨n.succ, h⟩ => tail ⟨n, ...⟩
```

Exact Lean library declaration

```
def finCons {α : Type*} {k : ℕ} (head : α) (tail : Fin k → α) :
  Fin (k + 1) → α
| ⟨0, _h⟩ => head
| ⟨n + 1, h⟩ => tail ⟨n, Nat.succ_lt_succ_iff.mp h⟩

@[simp]
theorem finCons_zero {α : Type*} {k : ℕ} (head : α) (tail : Fin k → α) :
  finCons head tail ⟨0, Nat.succ_pos k⟩ = head := rfl
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteAvailableWeight

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      (baseWeight : α → ℝ) → (forbidden : Finset α) →  $\sum$  a, if a ∈ forbidden then 0 else baseWeight a
```

Exact Lean library declaration

```
noncomputable def finiteAvailableWeight {α : Type*} [Fintype α] [DecidableEq α]  
  (baseWeight : α → ℝ) (forbidden : Finset α) : ℝ :=  
   $\sum$  a : α, if a ∈ forbidden then 0 else baseWeight a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteFreshList

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(α : Type u_1) →  
(k : ℕ) → (forbidden : Finset α) → { list // Function.Injective list ∧ ∀ (slot : Fin k), list slot ∉ forbidden }
```

Exact Lean library declaration

```
abbrev finiteFreshList (α : Type*) (k : ℕ) (forbidden : Finset α) :=  
  {list : Fin k → α // Function.Injective list ∧ ∀ slot, list slot ∉ forbidden}
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteFreshListCons

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
[inst : DecidableEq α] →
{k : ℕ} →
{forbidden : Finset α} →
(head : { a // a ∉ forbidden }) →
(tail : EconCSLib.finiteFreshList α k (insert (↑head) forbidden)) → (EconCSLib.finCons ↑head ↑tail, ...)
```

Exact Lean library declaration

```
def finiteFreshListCons {α : Type*} [DecidableEq α] {k : ℕ}
{forbidden : Finset α}
(head : {a // a ∉ forbidden})
(tail : finiteFreshList α k (insert head.1 forbidden)) :
finiteFreshList α (k + 1) forbidden :=
{finCons head.1 tail.1,
by
  refine finCons_injective ?_ tail.2.1
  intro i htail_eq_head
  exact tail.2.2 i (by
    rw [htail_eq_head]
    simp)
,
by
  intro slot
  cases slot using Fin.cases with
  | zero =>
    exact head.2
  | succ i =>
    intro hforbidden
    exact tail.2.2 i
    (Finset.mem_insert.mpr (0r.inr hforbidden)))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteFreshListNil

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\alpha : \text{Type } u_1) \rightarrow (\text{forbidden} : \text{Finset } \alpha) \rightarrow \langle \text{fun } i \Rightarrow i.\text{elim0}, \multimap \rangle$

Exact Lean library declaration

```
def finiteFreshListNil (α : Type*) (forbidden : Finset α) :  
  finiteFreshList α 0 forbidden :=  
  ⟨fun i => Fin.elim0 i,  
    by  
      intro i _j _h  
      exact Fin.elim0 i,  
    by  
      intro i  
      exact Fin.elim0 i)⟩
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteWeightedPMF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u 1} →  
[inst : Fintype α] →  
(weight : α → ℝ) →  
  (hweight_nonneg : ∀ (a : α), 0 ≤ weight a) →  
    (htotal_pos : 0 < ∑ a, weight a) → PMF.ofFintype (fun a => ENNReal.ofReal (weight a / ∑ b, weight b)) ...
```

Exact Lean library declaration

```
noncomputable def finiteWeightedPMF {α : Type*} [Fintype α]  
  (weight : α → ℝ)  
  (hweight_nonneg : ∀ a, 0 ≤ weight a)  
  (htotal_pos : 0 < ∑ a, weight a) : PMF α :=  
  PMF.ofFintype  
    (fun a => ENNReal.ofReal (weight a / ∑ b, weight b))  
  (by  
    have hterm_nonneg : ∀ a : α, 0 ≤ weight a / ∑ b, weight b := by  
      intro a  
      exact div_nonneg (hweight_nonneg a) (le_of_lt htotal_pos)  
    have hsum_real :  
      (∑ a : α, weight a / ∑ b, weight b) = 1 := by  
        rw [← Finset.sum_div]  
        field_simp [ne_of_gt htotal_pos]  
    calc  
      ∑ a : α, ENNReal.ofReal (weight a / ∑ b, weight b)  
      = ENNReal.ofReal (∑ a : α, weight a / ∑ b, weight b) := by  
        symm  
        exact ENNReal.ofReal_sum_of_nonneg  
          (s := (Finset.univ : Finset α))  
          (f := fun a => weight a / ∑ b, weight b)  
          (by intro a ; exact hterm_nonneg a)  
      = ENNReal.ofReal 1 := by rw [hsum_real]  
      = 1 := by norm_num)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteWeightedPMFAvailable

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
[inst : FIntype α] →
[inst_1 : DecidableEq α] →
(baseWeight : α → ℝ) →
(forbidden : Finset α) →
(hbase_nonneg : ∀ (a : α), 0 ≤ baseWeight a) →
(havailable_pos : 0 < EconCSLib.finiteAvailableWeight baseWeight forbidden) →
EconCSLib.finiteWeightedPMF (fun a => baseWeight ↑a) ... ..
```

Exact Lean library declaration

```
noncomputable def finiteWeightedPMFAvailable
{α : Type*} [FIntype α] [DecidableEq α]
(baseWeight : α → ℝ) (forbidden : Finset α)
(hbase_nonneg : ∀ a, 0 ≤ baseWeight a)
(havailable_pos : 0 < finiteAvailableWeight baseWeight forbidden) :
PMF {a // a ∉ forbidden} :=
finiteWeightedPMF
(fun a : {a // a ∉ forbidden} => baseWeight a.1)
(by intro a; exact hbase_nonneg a.1)
(by
  rw [← finiteAvailableWeight_eq_subtype_sum baseWeight forbidden]
  exact havailable_pos)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteWithoutReplacementPMF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
[inst : Fintype α] →
[inst_1 : DecidableEq α] →
(baseWeight : α → ℝ) →
(hbase_nonneg : ∀ (a : α), 0 ≤ baseWeight a) →
(havailable :
  ∀ (forbidden : Finset α),
  forbidden.card < Fintype.card α → 0 < EconCSLib.finiteAvailableWeight baseWeight forbidden) →
(k : ℕ) →
(forbidden : Finset α) →
(a : forbidden.card + k ≤ Fintype.card α) →
WellFounded.Nat.fix
  (fun x => PSigma.casesOn x fun k forbidden => PSigma.casesOn forbidden fun forbidden a => k)
  (fun _x a =>
    PSigma.casesOn (motive := fun _x =>
      ((y : (k : ℕ) ×' (forbidden : Finset α) ×' forbidden.card + k ≤ Fintype.card α) →
        InvImage (fun x1 x2 => x1 < x2)
          (fun x =>
            PSigma.casesOn x fun k forbidden => PSigma.casesOn forbidden fun forbidden a => k)
            y _x →
              PMF (EconCSLib.finiteFreshList α y.1 y.2.1)) →
              PMF (EconCSLib.finiteFreshList α _x.1 _x.2.1))
        x
      (fun k forbidden a =>
        PSigma.casesOn (motive := fun forbidden =>
          ((y : (k : ℕ) ×' (forbidden : Finset α) ×' forbidden.card + k ≤ Fintype.card α) →
            InvImage (fun x1 x2 => x1 < x2)
              (fun x =>
                PSigma.casesOn x fun k forbidden => PSigma.casesOn forbidden fun forbidden a => k)
                y (k, forbidden) →
                  PMF (EconCSLib.finiteFreshList α y.1 y.2.1)) →
                  PMF (EconCSLib.finiteFreshList α (k, forbidden).1 (k, forbidden).2.1))
              forbidden
            (fun forbidden a a_1 =>
              (match (motive :=
                (x : ℕ) →
                (x_1 : Finset α) →
                (x_2 : x_1.card + x ≤ Fintype.card α) →
                (y :
                  (k : ℕ) ×'
                    (forbidden : Finset α) ×' forbidden.card + k ≤ Fintype.card α) →
                  InvImage (fun x1 x2 => x1 < x2)
                    (fun x =>
                      PSigma.casesOn x fun k forbidden =>
                        PSigma.casesOn forbidden fun forbidden a => k)
                      y {x, {x_1, x_2}} →
                        PMF (EconCSLib.finiteFreshList α y.1 y.2.1)) →
                        PMF (EconCSLib.finiteFreshList α x x_1)
                    k, forbidden, a with
                  | 0, forbidden, _hbudget => fun x => PMF.pure (EconCSLib.finiteFreshListNil α forbidden)
                  | k.succ, forbidden, _hbudget => fun x =>
                    (EconCSLib.finiteWeightedPMFAvailable baseWeight forbidden hbase_nonneg ...).bind
                    fun head =>
                    PMF.map (EconCSLib.finiteFreshListCons head)
                    (x {k, {insert (↑head) forbidden, ...}} ...))
                  a_1
                a)
              a)
            (k, {forbidden, a}))
          a)
        (k, {forbidden, a}))
    a)
  (k, {forbidden, a}))
```

Exact Lean library declaration

```
noncomputable def finiteWithoutReplacementPMF
{α : Type*} [Fintype α] [DecidableEq α]
(baseWeight : α → ℝ)
(hbase_nonneg : ∀ a, 0 ≤ baseWeight a)
(havailable : ∀ forbidden : Finset α,
  forbidden.card < Fintype.card α →
  0 < finiteAvailableWeight baseWeight forbidden) :
(k : ℕ) → (forbidden : Finset α) →
forbidden.card + k ≤ Fintype.card α →
PMF (finiteFreshList α k forbidden)
| 0, forbidden, _hbudget => PMF.pure (finiteFreshListNil α forbidden)
| k + 1, forbidden, _hbudget =>
  (finiteWeightedPMFAvailable
    baseWeight forbidden hbase_nonneg
    (havailable forbidden (by omega)))
    .bind fun head =>
    (finiteWithoutReplacementPMF
      baseWeight hbase_nonneg havailable
      k (insert head.1 forbidden)
      (by
        rw [Finset.card_insert_of_notMem head.2]
        omega)).map (finiteFreshListCons head)
termination_by k forbidden _hbudget => k
```

--
The first draw of the recursive without-replacement sampler has exactly the available weighted law from the current forbidden set.


```

-/
theorem finiteWithoutReplacementPMF_head_prob
  {α : Type*} [Fintype α] [DecidableEq α]
  (baseWeight : α → ℝ)
  (hbase_nonneg : ∀ a, 0 ≤ baseWeight a)
  (havailable : ∀ forbidden : Finset α,
    forbidden.card < Fintype.card α →
      0 < finiteAvailableWeight baseWeight forbidden)
  {k : ℕ} (forbidden : Finset α)
  (hbudget : forbidden.card + (k + 1) ≤ Fintype.card α)
  (head₀ : {a // a ∉ forbidden}) :
  pmfProb
    (finiteWithoutReplacementPMF
      baseWeight hbase_nonneg havailable
      (k + 1) forbidden hbudget)
    (fun sample =>
      sample.1 (0, Nat.succ_pos k) = head₀.1) =
  (finiteWeightedPMFAvailable
    baseWeight forbidden hbase_nonneg
    (havailable forbidden (by omega))) head₀).toReal := by
classical
rw [finiteWithoutReplacementPMF]
rw [pmfProb_bind]
let headLaw :=
  finiteWeightedPMFAvailable
    baseWeight forbidden hbase_nonneg
    (havailable forbidden (by omega))
have hinner : ∀ head : {a // a ∉ forbidden},
  pmfProb
    ((finiteWithoutReplacementPMF
      baseWeight hbase_nonneg havailable
      k (insert head.1 forbidden)
      (by
        rw [Finset.card_insert_of_notMem head.2]
        omega)).map (finiteFreshListCons head))
    (fun sample =>
      sample.1 (0, Nat.succ_pos k) = head₀.1) =
    if head = head₀ then (1 : ℝ) else 0 := by
intro head
rw [pmfProb_map]
by_cases hhead : head = head₀
· subst head
  simpa using
    (pmfProb_eq_one_of_forall _ _ (by
      intro tail
      rfl))
· have hval_ne : head.1 ≠ head₀.1 := by
  intro hval
  exact hhead (Subtype.ext hval)
  simpa [hhead] using
    (pmfProb_eq_zero_of_no_mass _ _ (by
      intro tail htail
      exact False.elim (hval_ne (by simpa using htail)))
    )
calc
  pmfExp headLaw
    (fun head =>
      pmfProb
        ((finiteWithoutReplacementPMF
          baseWeight hbase_nonneg havailable
          k (insert head.1 forbidden)
          (by
            rw [Finset.card_insert_of_notMem head.2]
            omega)).map (finiteFreshListCons head))
        (fun sample =>
          sample.1 (0, Nat.succ_pos k) = head₀.1))
    = pmfExp headLaw (fun head => if head = head₀ then (1 : ℝ) else 0) := by
      refine pmfExp_congr headLaw ?_
      intro head
      exact hinner head
    _ = pmfProb headLaw (fun head => head = head₀) := rfl
    _ = (headLaw head₀).toReal := pmfProb_singleton headLaw head₀
    _ = (finiteWeightedPMFAvailable
      baseWeight forbidden hbase_nonneg
      (havailable forbidden (by omega))) head₀).toReal := rfl

```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.measureProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} → [inst : MeasurableSpace α] → (μ : MeasureTheory.Measure α) → (p : α → Prop) → (μ {a | p a}).toReal
```

Exact Lean library declaration

```
noncomputable def measureProb {α : Type*} [MeasurableSpace α]  
  (μ : Measure α) (p : α → Prop) : ℝ :=  
  (μ {a | p a}).toReal
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfPairIndicatorExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{β : Type u_2} →
  [inst : Fintype α] →
    [inst_1 : DecidableEq α] →
      [inst_2 : Fintype β] →
        [inst_3 : DecidableEq β] →
          (μ : PMF α) →
            (ν : PMF β) →
              (p : α × β → Prop) →
                [inst_4 : DecidablePred p] →
                  (f : α × β → ℝ) → EconCSLib.pmfPairExp μ ν fun a b => if p (a, b) then f (a, b) else 0
```

Exact Lean library declaration

```
noncomputable def pmfPairIndicatorExp {α β : Type*}
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]
  (μ : PMF α) (ν : PMF β) (p : α × β → Prop) [DecidablePred p]
  (f : α × β → ℝ) : ℝ :=
  pmfPairExp μ ν (fun a b => if p (a, b) then f (a, b) else 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProd

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] → [inst_1 : Fintype β] → (μ : PMF α) → (ν : PMF β) → PMF.ofFintype (fun p => μ p.1 * ν p.2) ...
```

Exact Lean library declaration

```
noncomputable def pmfProd {α β : Type*} [Fintype α] [Fintype β]  
  (μ : PMF α) (ν : PMF β) : PMF (α × β) :=  
  PMF.ofFintype (fun p : α × β => μ p.1 * ν p.2) (by  
    classical  
    have hμ : ∑ a : α, μ a = 1 := by  
      rw [← PMF.tsum_coe μ, tsum_fintype]  
    have hν : ∑ b : β, ν b = 1 := by  
      rw [← PMF.tsum_coe ν, tsum_fintype]  
    calc  
      ∑ p : α × β, μ p.1 * ν p.2  
      = ∑ a : α, ∑ b : β, μ a * ν b := by  
        rw [Fintype.sum_prod_type]  
      = ∑ a : α, μ a * (∑ b : β, ν b) := by  
        simp [Finset.mul_sum]  
      = 1 := by  
        simp [hμ, hν])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 133-138

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} (trueRanking : EconCSLib.SocialChoice.Ranking.Ranking n) (F : KR21Monoculture.AccuracyFamily n),
  EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy trueRanking F.value →
  Fintype.card (EconCSLib.SocialChoice.Ranking.Candidate n) = n + 2 ∧
  2 ≤ Fintype.card (EconCSLib.SocialChoice.Ranking.Candidate n) ∧
  (∀ {a b : EconCSLib.SocialChoice.Ranking.Candidate n},
    EconCSLib.SocialChoice.Ranking.rankOf trueRanking a < EconCSLib.SocialChoice.Ranking.rankOf trueRanking b →
    F.value b < F.value a) ∧
  ∀ (theta : ℝ), 0 < theta → (F.dist theta).toMeasure Set.univ = 1
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_model_candidates_values_and_ranking_law

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 139-165

Verbatim source input

Definition 1 (Noisy permutation family). A noisy permutation family F_θ is a family of distributions over permutations that satisfies the following conditions for any $\theta > 0$ and set of candidates x :

1. (Differentiability) For any permutation π , $\Pr F_\theta [\pi]$ is continuous and differentiable in θ .
2. (Asymptotic optimality) For the true ranking π^* , $\lim_{\theta \rightarrow \infty} \Pr F_\theta [\pi^*] = 1$.
3. (Monotonicity) For any (possibly empty) $S \subset x$, let $\pi(-S)$ be the partial ranking produced by removing

$(-S)$
the items in S from π . Let π_1
denote the value of the top-ranked candidate according to $\pi(-S)$. For

θ
any $\theta > 0$,

h
 i
 h
 i
 $(-S)$
 $(-S)$
 $EF_\theta \pi_1$
 $\geq EF_\theta \pi_1$

.

(1)
Moreover, for $S = \emptyset$, (1) holds with strict inequality.

θ serves as an “accuracy parameter”: for large θ , the noisy ranking converges to the true ranking over candidates. The monotonicity condition states that a higher value of θ leads to a better first choice, even if some of the candidates are removed after ranking. Removal after ranking (as opposed to before) is important because some of the ranking models we will consider later do not satisfy Independence of Irrelevant Alternatives. Examples of noisy permutation families include Random Utility Models [23] and the Mallows Model [14], both of which we will discuss in detail later.

Expanded PaperInterface specification

```
∀ {n : ℕ} (F : KR21Monoculture.AccuracyFamily n) (trueRanking : EconCSLib.SocialChoice.Ranking.Ranking n),
  EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy trueRanking F.value →
  (KR21Monoculture.SourceDefinition1 F trueRanking →
    (∀ (theta : ℝ),
      0 < theta →
        ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
          ContinuousAt (fun theta' => ((F.dist theta') pi).toReal) theta ∧
          DifferentiableAt ℝ (fun theta' => ((F.dist theta') pi).toReal) theta) ∧
        Filter.Tendsto (fun theta => ((F.dist theta) trueRanking).toReal) Filter.atTop (nhds 1) ∧
        ∀ (thetaA thetaH : ℝ),
          0 < thetaH →
            thetaH < thetaA →
              (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
                remaining.Nonempty →
                  KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value remaining ≤
                  KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value remaining) ∧
                  KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value Finset.univ <
                  KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value Finset.univ)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_definition1_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 143-159

Verbatim source input

3. (Monotonicity) For any (possibly empty) $S \subset x$, let $\pi(-S)$ be the partial ranking produced by removing $(-S)$ the items in S from π . Let π_1 denote the value of the top-ranked candidate according to $\pi(-S)$. For any $\theta > \theta$,

$$\begin{aligned} &h \\ &i \\ &h \\ &i \\ &(-S) \\ &(-S) \\ &EF\theta\theta \pi_1 \\ &\geq EF\theta \pi_1 \\ &\cdot \\ &(1) \end{aligned}$$

Moreover, for $S = \emptyset$, (1) holds with strict inequality.

Expanded PaperInterface specification

```
∀ {n : ℕ} {F : KR21Monoculture.AccuracyFamily n} {trueRanking : EconCSLib.SocialChoice.Ranking.Ranking n},
  EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy trueRanking F.value →
  KR21Monoculture.SourceDefinition1 F trueRanking →
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
      remaining.Nonempty →
      KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value remaining ≤
      KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value remaining) ∧
    KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value Finset.univ <
    KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value Finset.univ
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_equation1_removal_monotonicity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 176-192

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} (M : KR21Monoculture.Model n),
  M.firstMoverEU KR21Monoculture.Strategy.algorithm =
    EconCSLib.SocialChoice.Ranking.expectedFirstMoverUtility M.algorithmRanking M.value ∧
  M.firstMoverEU KR21Monoculture.Strategy.human =
    EconCSLib.SocialChoice.Ranking.expectedFirstMoverUtility M.humanRanking M.value ∧
  M.secondMoverEU KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm =
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared M.algorithmRanking M.value ∧
  M.secondMoverEU KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human =
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent M.humanRanking M.algorithmRanking M.value ∧
  M.secondMoverEU KR21Monoculture.Strategy.human KR21Monoculture.Strategy.algorithm =
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent M.algorithmRanking M.humanRanking M.value ∧
  M.secondMoverEU KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human =
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent M.humanRanking M.humanRanking M.value ∧
  ∀ (self other : KR21Monoculture.Strategy),
    M.payoffAgainst self other = (M.firstMoverEU self + M.secondMoverEU other self) / 2
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_two_firm_selection_game_semantics

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 190-198

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} (trueRanking : EconCSLib.SocialChoice.Ranking.Ranking n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D],
  (∀m (realizedValue : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
    EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy trueRanking realizedValue) →
  ∀ (value : KR21Monoculture.ValueProfile n),
    EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy trueRanking value →
    D Set.univ = 1 ∧
    (MeasureTheory.Measure.dirac value) Set.univ = 1 ∧
    MeasureTheory.IsProbabilityMeasure (MeasureTheory.Measure.dirac value)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_model_outer_distribution_probability_and_point_mass

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 210-216

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} (F : KR21Monoculture.DistributionalAccuracyFamily n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D] (theta : ℝ),
  0 < theta →
  ∀
    (hatom :
      ∀ (ranking : EconCSLib.SocialChoice.Ranking.Ranking n), Measurable fun value => (F.dist theta value) ranking),
    MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1)
      (F.outerIndependentPairJointLaw D theta hatom) →
      MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2)
        (F.outerIndependentPairJointLaw D theta hatom) →
      0 <
        ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
          if
            EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
              EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
              1
            else 0 ∂F.outerIndependentPairJointLaw D theta hatom →
          have J := F.outerIndependentPairJointLaw D theta hatom;
          have d2Numerator :=
            ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
              if
                EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                  EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                  x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
                    x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
                else 0 ∂J;
          have d2Denominator :=
            ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
              if
                EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                  EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                  1
                else 0 ∂J;
          have d2Shared :=
            ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
              EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1 ∂J;
          have d2Independent :=
            ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
              EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2 ∂J;
          J = D.compProd (F.independentPairKernel theta hatom) ∧
            (∀ (value : KR21Monoculture.ValueProfile n),
              (F.independentPairKernel theta hatom) value =
                (EconCSLib.pmfProd (F.dist theta value) (F.dist theta value)).toMeasure) ∧
            (KR21Monoculture.SourceDefinition2ConditionalAt F D theta hatom ↔ 0 < d2Numerator / d2Denominator) ∧
            (0 < d2Numerator / d2Denominator ↔ d2Shared < d2Independent)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_definition2_literal_outer_joint_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 217-242

Verbatim source input

Definition 3 (Preference for weaker competition.). A candidate distribution D and noisy permutation family F_θ , exhibits a preference for weaker competition if the following holds: for all $\theta_1 > \theta_2$, $\sigma \sim F_{\theta_1}$ and $\pi, \tau \sim F_{\theta_2}$,

h
 i
 h
 i
 $(-\{\sigma_1\})$
 $E \pi_1$
 $(-\{\tau_1\})$
 $< E \pi_1$
 $.$

Intuitively, suppose we have a higher accuracy parameter θ_1 and a lower accuracy parameter $\theta_2 < \theta_1$; we draw a permutation π from F_{θ_2} ; and we then derive two permutations from π : $\pi(-\{\sigma_1\})$ obtained by deleting the first-ranked element of a permutation σ drawn from the more accurate distribution F_{θ_1} , and $\pi(-\{\tau_1\})$ obtained by deleting the first-ranked element of a permutation τ drawn from the less accurate distribution F_{θ_2} .

Then the expected value of the first-ranked candidate in $\pi(-\{\tau_1\})$ is strictly greater than the expected value of the first-ranked candidate in $\pi(-\{\sigma_1\})$ – that is, when a random candidate is removed from π , the best remaining candidate is better in expectation when the randomly removed candidate is chosen based on a noisier ranking.

Expanded PaperInterface specification

```
∀ {n : ℕ} (F : KR21Monoculture.DistributionalAccuracyFamily n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D]
  (thetaA thetaH : ℝ),
  0 < thetaH →
  thetaH < thetaA →
  (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    MeasureTheory.AEStronglyMeasurable (fun value => ((F.dist thetaH value) pi).toReal) D) →
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    MeasureTheory.AEStronglyMeasurable (fun value => ((F.dist thetaA value) pi).toReal) D) →
  have d3Algorithm :=
    ∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value) fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma ∂D;
  have d3Human :=
    ∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaH value) fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma ∂D;
  MeasureTheory.Integrable
    (fun value =>
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value) fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma)
  D ∧
  MeasureTheory.Integrable
    (fun value =>
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaH value) fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma)
  D ∧
  (KR21Monoculture.SourceDefinition3At F D thetaA thetaH ↔ d3Algorithm < d3Human)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.source_definition3_literal_outer_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

Using these two conditions, we can state our theorem.

Theorem 1. Suppose that a given candidate distribution D and noisy permutation family F_θ satisfy Definition 2 (preference for the $\hookrightarrow$ first position) and Definition 3 (preference for weaker competition).

Then, for any θ_H , there exists $\theta_A > \theta_H$ such that using the algorithmic ranking is a strictly dominant strategy for both firms, but social welfare would be higher if both firms used human evaluators.

[Next verbatim source excerpt]

Proving Theorem 1

With this intuition, we are ready to prove Theorem 1.

Proof of Theorem 1. For given values of θ_A and θ_H , using the algorithmic ranking is a strictly dominant strategy as long as

$$UA(\theta_A, \theta_H) + UAA(\theta_A, \theta_H) > UH(\theta_A, \theta_H) + UAH(\theta_A, \theta_H)$$

(4)

$$UA(\theta_A, \theta_H) + UHA(\theta_A, \theta_H) > UH(\theta_A, \theta_H) + UHH(\theta_A, \theta_H)$$

(5)

Note that (5) is always true for $\theta_A > \theta_H$ by the monotonicity assumption on F_θ : $UA(\theta_A, \theta_H) \geq UH(\theta_A, \theta_H)$ because a more accurate ranking produces a top-ranked candidate with higher expected value, and $UHA(\theta_A, \theta_H) \geq UHH(\theta_A, \theta_H)$ because this holds even conditioned on removing any candidate from the pool (in this case, the

[form-feed in source extraction]

candidate randomly selected by the firm that hires first). Crucially, in (5), the first firm's random selection is independent from the second firm's selection; the same logic could not be used to argue that (4) always holds for $\theta_A \geq \theta_H$. Moreover, when $\theta_A > \theta_H$, $UA(\theta_A, \theta_H) > UH(\theta_A, \theta_H)$ by the monotonicity assumption, meaning (5) holds.

Let $W_{s_1 s_2}(\theta_A, \theta_H)$ denote social welfare when the two firms employ strategies $s_1, s_2 \in \{A, H\}$. Then, when both firms use the algorithmic ranking, social welfare is

$$WAA(\theta_A, \theta_H) = UA(\theta_A, \theta_H) + UAA(\theta_A, \theta_H).$$

By (2), Definition 2 implies that for any θ , $UAA(\theta, \theta) < UAH(\theta, \theta)$, implying

$$UA(\theta_H, \theta_H) + UAA(\theta_H, \theta_H) < UH(\theta_H, \theta_H) + UAH(\theta_H, \theta_H).$$

However, by the optimality assumption on F_θ in Definition 1, for sufficiently large θ_A ,

$$UA(\theta_A, \theta_H) + UAA(\theta_A, \theta_H) > UH(\theta_A, \theta_H) + UAH(\theta_A, \theta_H).$$

Note that $U_{s_1}(\theta_A, \theta_H)$ and $U_{s_1 s_2}(\theta_A, \theta_H)$ are continuous with respect to θ_A for any $s_1, s_2 \in \{A, H\}$ since they are expectations over discrete distributions with probabilities that are by assumption differentiable with

$\ast$
 $> \theta_H$

respect to θ_A . Therefore, by the Differentiability assumption on F_θ from Definition 1, there is some θ_A such that

$\ast$
 $\ast$
 $\ast$
 $\ast$
 $\ast$

$$\begin{aligned} UA(\theta_A, \theta_H) + UAA(\theta_A, \theta_H) &= UH(\theta_A, \theta_H) + UAH(\theta_A, \theta_H), \\ (6) \end{aligned}$$

i.e., given that its competitor uses the algorithmic ranking, a firm is indifferent between the two strategies.

$\ast$
 $\ast$, using the algorithmic ranking is still a weakly dominant strategy. By definition of WAA ,
For such θ_A

$\ast$
 $\ast$
 $\ast$

$$\begin{aligned} WAA(\theta_A, \theta_H) &= UH(\theta_A, \theta_H) + UAH(\theta_A, \theta_H), \end{aligned}$$

If both firms had instead used human evaluators, social welfare would be

$\ast$
 $\ast$
 $\ast$

$$\begin{aligned} WHH(\theta_A, \theta_H) &= UH(\theta_A, \theta_H) + UHH(\theta_A, \theta_H). \end{aligned}$$

By Definition 3, for $\sigma \sim F_{\theta_A}^\ast$ and $\pi, \tau \sim F_{\theta_H}$,
 h
 i

```

h
i
(-{σ})
(-{τ})
E π1 1
< E π1 1 .
Note that
h
i
(-{σ})
*
UAH (θA
, θH ) = E π1 1
h
i
(-{τ})
*
UHH (θA
, θH ) = E π1 1
*
*
, θH ). As a result for θA* > θH , using
Thus, Definition 3 implies that for θA* > θH , UHH (θA
, θH ) > UAH (θA
the algorithmic ranking is a weakly dominant strategy, but
*
*
*
WHH (θA
, θH ) = UH (θA
, θH ) + UHH (θA
, θH )
*
*
> UH (θA
, θH ) + UAH (θA
, θH )
*
*
= UA (θA
, θH ) + UAA (θA
, θH )
*
= WAA (θA
, θH ),
meaning social welfare would have been higher had both firms used human evaluators.
θ
We can show that this effect persists for a value θA
such that using the algorithmic ranking is a strictly
*
dominant strategy. Intuitively, this is simply by slightly increasing θA
so the algorithmic ranking is strictly
dominant. For fixed θH , define
f (θA ) = UA (θA , θH ) + UAA (θA , θH )
g(θA ) = UH (θA , θH ) + UAH (θA , θH )
h(θA ) = UH (θA , θH ) + UHH (θA , θH )
6

[form-feed in source extraction]
θ
θ
θ
Because (5) always holds for θA > θH , it suffices to show that there exists θA
such that g(θA
) < f (θA
) <
) <
θ
θ
θ
θ
θ
θ
h(θA ). This is because g(θA ) < f (θA ) is equivalent to (4) and f (θA ) < h(θA ) is equivalent to WAA (θA , θH ) <
θ
WHH (θA
, θH ).
First, note that h(θA ) is a constant, and by Definition 3, g(θA ) < h(θA ) for all θA > θH . By the
optimality assumption of Definition 1, there exists sufficiently large θA such that f (θA ) > g(θA ). Recall
*
*
*
that by definition of θA
, f (θA
) = g(θA
). Both f and g are continuous by the Differentiability assumption in
θ
*
θ
θ
θ
Definition 1. Thus, there must exist some θA
> θA
such that g(θA
) < f (θA
) < h(θA
)

```

). This means that for
 θ
 θ_A , using the algorithmic ranking is a strictly dominant strategy, but social welfare would still be larger if both firms used human evaluators.

Expanded PaperInterface specification

```

∀ {n : ℕ} (F : KR21Monoculture.DistributionalAccuracyFamily n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)),
  MeasureTheory.IsProbabilityMeasure D →
    ∀ (center : EconCSLib.SocialChoice.Ranking.Ranking n) (thetaH : ℝ),
      0 < thetaH →
        (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
          (∀ (theta : ℝ) (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
            MeasureTheory.AEStronglyMeasurable (fun value => ((F.dist theta value) pi).toReal) D) →
            (∀ (value : KR21Monoculture.ValueProfile n) (theta : ℝ),
              0 < theta →
                ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                  ContinuousAt (fun theta' => ((F.dist theta' value) pi).toReal) theta) →
                (∀ (value : KR21Monoculture.ValueProfile n) (theta : ℝ),
                  0 < theta →
                    ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                      DifferentiableAt ℝ (fun theta' => ((F.dist theta' value) pi).toReal) theta) →
                    (∀ (value : KR21Monoculture.ValueProfile n),
                      Filter.Tendsto (fun theta => ((F.dist theta value) center).toReal) Filter.atTop (nhds 1)) →
                    (∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
                      EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value) →
                    ∀
                      (hatom_measurable :
                        ∀ (theta : ℝ) (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                          Measurable fun value => (F.dist theta value) pi),
                      (∀ (theta : ℝ),
                        0 < theta →
                          MeasureTheory.Integrable
                            (fun value =>
                              EconCSLib.pmfPairExp (F.dist theta value) (F.dist theta value) fun pi sigma =>
                                if
                                  EconCSLib.SocialChoice.Ranking.firstChoice pi ≠
                                  EconCSLib.SocialChoice.Ranking.firstChoice sigma then
                                    1
                                  else 0)
                            D) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              MeasureTheory.Integrable
                                (fun value =>
                                  EconCSLib.pmfExp (F.dist theta value) fun pi =>
                                    value (EconCSLib.SocialChoice.Ranking.secondChoice pi))
                                D) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              MeasureTheory.Integrable
                                (fun value =>
                                  EconCSLib.pmfPairExp (F.dist theta value) (F.dist theta value) fun pi sigma =>
                                    EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma)
                                D) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              MeasureTheory.Integrable
                                (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1)
                                (F.outerIndependentPairJointLaw D theta →)) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              MeasureTheory.Integrable
                                (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2)
                                (F.outerIndependentPairJointLaw D theta →)) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              0 <
                                ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                                  if
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                                      1
                                    else 0 ∂F.outerIndependentPairJointLaw D theta →)) →
                          (∀ (theta : ℝ),
                            0 < theta →
                              0 <
                                (∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                                  if
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                                      x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
                                      x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
                                    else 0 ∂F.outerIndependentPairJointLaw D theta →) /
                                ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                                  if
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                                      1
                                    else 0 ∂F.outerIndependentPairJointLaw D theta →)) →
                          (∀ (thetaA thetaH : ℝ),
                            0 < thetaH →
                              thetaH < thetaA →
                                MeasureTheory.Integrable

```

```

      (fun value =>
        EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value)
      fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma)
    D) →
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    MeasureTheory.Integrable
      (fun value =>
        EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value)
      fun pi sigma =>
        EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma)
    D) →
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    ∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value)
    fun pi sigma =>
      EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi
      sigma ∂D <
    ∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp (F.dist thetaH value) (F.dist thetaA value)
    fun pi sigma =>
      EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi
      sigma ∂D) →
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    ∀ (value : KR21Monoculture.ValueProfile n)
      (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
      remaining.Nonempty →
      KR21Monoculture.expectedBestInSet (F.dist thetaH value) value
      remaining ≤
      KR21Monoculture.expectedBestInSet (F.dist thetaA value) value
      remaining) →
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    ∀ (value : KR21Monoculture.ValueProfile n),
      KR21Monoculture.expectedBestInSet (F.dist thetaH value) value
      Finset.univ <
      KR21Monoculture.expectedBestInSet (F.dist thetaA value) value
      Finset.univ) →
  ∃ thetaA,
    thetaH < thetaA ∧
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.human ∂D +
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
      KR21Monoculture.Strategy.algorithm
      KR21Monoculture.Strategy.human ∂D <
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.algorithm ∂D +
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
      KR21Monoculture.Strategy.algorithm
      KR21Monoculture.Strategy.algorithm ∂D ∧
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.human ∂D +
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
      KR21Monoculture.Strategy.human
      KR21Monoculture.Strategy.algorithm ∂D <
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.algorithm ∂D +
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
      KR21Monoculture.Strategy.algorithm
      KR21Monoculture.Strategy.algorithm ∂D <
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.human ∂D +
    ∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
      KR21Monoculture.Strategy.human
      KR21Monoculture.Strategy.human ∂D
  )

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem1_raw_outer_literal_payoff_terminal_conclusion

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 9

Source locator: cited publication, lines 263-276

Verbatim source input

To do so, we need to introduce some notation. Recall that the two firms hire in a random order. Given a candidate distribution D , let $U_s(\theta_A, \theta_H)$ denote the expected utility of the first firm to hire a candidate when using ranking s , where $s \in \{A, H\}$ is either the algorithmic ranking or the ranking generated by a human evaluator respectively. Similarly, let $U_{s2}(\theta_A, \theta_H)$ be the expected utility of the second firm to hire given that the first firm used strategy $s1$ and the second firm uses strategy $s2$, where again $s1, s2 \in \{A, H\}$. Finally, let $\pi, \sigma \sim F\theta$.

In what follows, we will show that for any θ ,
 $E[\pi_1 - \pi_2 \mid \pi_1 \neq \sigma_1] > 0 \iff U_{AH}(\theta, \theta) > U_{AA}(\theta, \theta)$.

(2)

In other words, whenever a ranking model meets Definition 2, the firm chosen to select second will prefer to use an independent ranking mechanism from its competitor, given that the ranking mechanisms are equally accurate.

Expanded PaperInterface specification

```

∀ {n : ℕ} (F : KR21Monoculture.DistributionalAccuracyFamily n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D] (theta : ℝ),
  0 < theta →
  ∀
    (hatom :
      ∀ (ranking : EconCSLib.SocialChoice.Ranking.Ranking n), Measurable fun value => (F.dist theta value) ranking),
    MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1)
      (F.outerIndependentPairJointLaw D theta hatom) →
      MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2)
        (F.outerIndependentPairJointLaw D theta hatom) →
      0 <
        ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
          if
            EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
              EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
              1
            else 0 ∂F.outerIndependentPairJointLaw D theta hatom →
          have J := F.outerIndependentPairJointLaw D theta hatom;
          have numerator :=
            ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
              if
                EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                  EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                  x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
                    x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
                else 0 ∂J;
            have denominator :=
              ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                if
                  EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
                    EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
                    1
                  else 0 ∂J;
            have UAA :=
              ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1 ∂J;
            have UAH :=
              ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2 ∂J;
            J = D.compProd (F.independentPairKernel theta hatom) ∧
              (∀ (value : KR21Monoculture.ValueProfile n),
                (F.independentPairKernel theta hatom) value =
                  (EconCSLib.pmfProd (F.dist theta value) (F.dist theta value)).toMeasure) ∧
                (0 < numerator / denominator ↔ UAA < UAH))

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation2_outer_joint_payoff_equivalence_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 277-290

Verbatim source input

First, we can write

$$UAH(\theta_A, \theta_H) = E[\pi_1 \cdot 1_{\pi_1 \neq \sigma_1} + \pi_2 \cdot 1_{\pi_1 = \sigma_1}]$$

$$UAA(\theta_A, \theta_H) = E[\sigma_2]$$

$$= E[\sigma_2 \cdot 1_{\pi_1 \neq \sigma_1} + \sigma_2 \cdot 1_{\pi_1 = \sigma_1}]$$
Thus,

$$UAH(\theta_A, \theta_H) - UAA(\theta_A, \theta_H)$$

$$= E[(\pi_1 - \sigma_2) \cdot 1_{\pi_1 \neq \sigma_1} + (\pi_2 - \sigma_2) \cdot 1_{\pi_1 = \sigma_1}]$$
Conditioned on either $\pi_1 = \sigma_1$ or $\pi_1 \neq \sigma_1$, π_2 and σ_2 are identically distributed and therefore have equal expectations. As a result,

$$UAH(\theta_A, \theta_H) - UAA(\theta_A, \theta_H) = E[(\pi_1 - \pi_2) \cdot 1_{\pi_1 \neq \sigma_1}],$$
(3)

which implies (2). Thus, whenever a ranking model meets Definition 2, firms rationally prefer independent

Expanded PaperInterface specification

```

∀ {n : ℕ} (F : KR21Monoculture.DistributionalAccuracyFamily n)
  (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D] (theta : ℝ)
  (hatom :
    ∀ (ranking : EconCSLib.SocialChoice.Ranking.Ranking n), Measurable fun value => (F.dist theta value) ranking),
  MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1)
    (F.outerIndependentPairJointLaw D theta hatom) →
  MeasureTheory.Integrable (fun x => EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2)
    (F.outerIndependentPairJointLaw D theta hatom) →
  have J := F.outerIndependentPairJointLaw D theta hatom;
  have numerator :=
    ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
      if EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠ EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
        x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
        x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
      else 0 ∂J;
  have UAA :=
    ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
      EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.1 ∂J;
  have UAH :=
    ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
      EconCSLib.SocialChoice.Ranking.secondMoverUtility x.1 x.2.1 x.2.2 ∂J;
  J = D.compProd (F.independentPairKernel theta hatom) ∧
    (∀ (value : KR21Monoculture.ValueProfile n),
      (F.independentPairKernel theta hatom) value =
        (EconCSLib.pmfProd (F.dist theta value) (F.dist theta value)).toMeasure) ∧
    UAH - UAA = numerator

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation3_outer_joint_payoff_identity_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

With this intuition, we are ready to prove Theorem 1.
Proof of Theorem 1. For given values of θ_A and θ_H , using the algorithmic ranking is a strictly dominant strategy as long as
$$U_A(\theta_A, \theta_H) + U_{AA}(\theta_A, \theta_H) > U_H(\theta_A, \theta_H) + U_{AH}(\theta_A, \theta_H)$$

(4)

Expanded PaperInterface specification

```
∀ {n : ℕ} (M : KR21Monoculture.Model n),
  M.firstMoverEU KR21Monoculture.Strategy.algorithm +
    M.secondMoverEU KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm >
    M.firstMoverEU KR21Monoculture.Strategy.human +
    M.secondMoverEU KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human ↔
  M.payoffAgainst KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm >
    M.payoffAgainst KR21Monoculture.Strategy.human KR21Monoculture.Strategy.algorithm
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation4_algorithm_best_response_against_algorithm_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

$$U_A(\theta_A, \theta_H) + U_H(\theta_A, \theta_H) > U_H(\theta_A, \theta_H) + U_{HH}(\theta_A, \theta_H)$$

(5)

Note that (5) is always true for $\theta_A > \theta_H$ by the monotonicity assumption on F_θ : $U_A(\theta_A, \theta_H) \geq U_H(\theta_A, \theta_H)$ because a more accurate ranking produces a top-ranked candidate with higher expected value, and $U_H(\theta_A, \theta_H) \geq U_{HH}(\theta_A, \theta_H)$ because this holds even conditioned on removing any candidate from the pool (in this case, the

[form-feed in source extraction]
candidate randomly selected by the firm that hires first). Crucially, in (5), the first firm's random selection is independent from the second firm's selection; the same logic could not be used to argue that (4) always holds for $\theta_A \geq \theta_H$. Moreover, when $\theta_A > \theta_H$, $U_A(\theta_A, \theta_H) > U_H(\theta_A, \theta_H)$ by the monotonicity assumption, meaning (5) holds.

Expanded PaperInterface specification

```
∀ {n : ℕ} (F : KR21Monoculture.AccuracyFamily n) (thetaA thetaH : ℝ),
  0 < thetaH →
  thetaH < thetaA →
  (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
    remaining.Nonempty →
    KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value remaining ≤
    KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value remaining) →
  KR21Monoculture.expectedBestInSet (F.dist thetaH) F.value Finset.univ <
  KR21Monoculture.expectedBestInSet (F.dist thetaA) F.value Finset.univ →
  (F.modelAt thetaA thetaH).firstMoverEU KR21Monoculture.Strategy.algorithm +
  (F.modelAt thetaA thetaH).secondMoverEU KR21Monoculture.Strategy.human
  >
  (F.modelAt thetaA thetaH).firstMoverEU KR21Monoculture.Strategy.human +
  (F.modelAt thetaA thetaH).secondMoverEU KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation5_from_literal_definition1_removal

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

The choice of which ranking mechanism to use leads to a game-theoretic setting: both firms know the accuracy parameters of the human evaluators (θ_H) and the algorithm (θ_A), and they must decide whether to

[form-feed in source extraction]

use a human evaluator or the algorithm. This choice introduces a subtlety: for many ranking models, a firm's rational behavior depends not only on the accuracy of the ranking mechanism, but also on the underlying candidate values $x_1, \dots, x_n$. Thus, to fully specify a firm's behavior, we assume that $x_1, \dots, x_n$ are drawn from a known joint distribution D . Our main results will hold for any D , meaning they apply even when the candidate values (but not their identities) are deterministically known.

[Next verbatim source excerpt]

Definition 2 (Preference for the first position.). A candidate distribution D and noisy permutation family F_θ exhibits a preference for the first position if for all $\theta > 0$, if $\pi, \sigma \sim F_\theta$, $E[\pi_1 - \pi_2 \mid \pi_1 \neq \sigma_1] > 0$.

In other words, for any $\theta > 0$, suppose we draw two permutations π and σ independently from F_θ , and suppose that the first-ranked candidates differ in π and σ . Then the expected value of the first-ranked candidate in π is strictly greater than the expected value of the second-ranked candidate in π .

[Next verbatim source excerpt]

To do so, we need to introduce some notation. Recall that the two firms hire in a random order. Given a candidate distribution D , let $U_s(\theta_A, \theta_H)$ denote the expected utility of the first firm to hire a candidate when using ranking s , where $s \in \{A, H\}$ is either the algorithmic ranking or the ranking generated by a human evaluator respectively. Similarly, let $U_{s2}(\theta_A, \theta_H)$ be the expected utility of the second firm to hire given that the first firm used strategy $s1$ and the second firm uses strategy $s2$, where again $s1, s2 \in \{A, H\}$. Finally, let $\pi, \sigma \sim F_\theta$. In what follows, we will show that for any θ , $E[\pi_1 - \pi_2 \mid \pi_1 \neq \sigma_1] > 0 \iff U_{AH}(\theta, \theta) > U_{AA}(\theta, \theta)$.

(2)

In other words, whenever a ranking model meets Definition 2, the firm chosen to select second will prefer to

[Next verbatim source excerpt]

Let $W_{s1s2}(\theta_A, \theta_H)$ denote social welfare when the two firms employ strategies $s1, s2 \in \{A, H\}$. Then, when both firms use the algorithmic ranking, social welfare is $W_{AA}(\theta_A, \theta_H) = U_A(\theta_A, \theta_H) + U_{AA}(\theta_A, \theta_H)$. By (2), Definition 2 implies that for any θ , $U_{AA}(\theta, \theta) < U_{AH}(\theta, \theta)$, implying $U_A(\theta_H, \theta_H) + U_{AA}(\theta_H, \theta_H) < U_H(\theta_H, \theta_H) + U_{AH}(\theta_H, \theta_H)$. However, by the optimality assumption on F_θ in Definition 1, for sufficiently large θ_A , $U_A(\theta_A, \theta_H) + U_{AA}(\theta_A, \theta_H) > U_H(\theta_A, \theta_H) + U_{AH}(\theta_A, \theta_H)$. Note that $U_{s1}(\theta_A, \theta_H)$ and $U_{s1s2}(\theta_A, \theta_H)$ are continuous with respect to θ_A for any $s1, s2 \in \{A, H\}$ since they are expectations over discrete distributions with probabilities that are by assumption differentiable with

$> \theta_H$ respect to θ_A . Therefore, by the Differentiability assumption on F_θ from Definition 1, there is some θ_A such that

*

*

*

*

$U_A(\theta_A, \theta_H) + U_{AA}(\theta_A, \theta_H) = U_H(\theta_A, \theta_H) + U_{AH}(\theta_A, \theta_H)$,

(6)

i.e., given that its competitor uses the algorithmic ranking, a firm is indifferent between the two strategies.

*

, using the algorithmic ranking is still a weakly dominant strategy. By definition of W_{AA} , For such θ_A

*

*

$W_{AA}(\theta_A, \theta_H) = U_H(\theta_A, \theta_H) + U_{AH}(\theta_A, \theta_H)$.

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} (F : \text{KR21Monoculture.DistributionalAccuracyFamily } n)$
 $(D : \text{MeasureTheory.Measure } (\text{KR21Monoculture.ValueProfile } n)),$

```

MeasureTheory.IsProbabilityMeasure D →
  ∀ (center : EconCSLib.SocialChoice.Ranking.Ranking n) (thetaH : ℝ),
    0 < thetaH →
      (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
        (∀ (theta : ℝ) (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
          MeasureTheory.AEStronglyMeasurable (fun value => ((F.dist theta value) pi).toReal) D) →
          (∀ (value : KR21Monoculture.ValueProfile n) (theta : ℝ),
            0 < theta →
              ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                ContinuousAt (fun theta' => ((F.dist theta' value) pi).toReal) theta) →
              (∀ (value : KR21Monoculture.ValueProfile n) (theta : ℝ),
                0 < theta →
                  ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                    DifferentiableAt ℝ (fun theta' => ((F.dist theta' value) pi).toReal) theta) →
              (∀ (value : KR21Monoculture.ValueProfile n),
                Filter.Tendsto (fun theta => ((F.dist theta value) center).toReal) Filter.atTop (nhds 1)) →
              (∀m (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
                EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value) →
              ∀
                (hatom_measurable :
                  ∀ (theta : ℝ) (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                    Measurable fun value => (F.dist theta value) pi),
                (∀ (theta : ℝ),
                  0 < theta →
                    MeasureTheory.Integrable
                      (fun value => KR21Monoculture.disagreementProb (F.dist theta value)) D) →
                  (∀ (theta : ℝ),
                    0 < theta →
                      MeasureTheory.Integrable
                        (fun value =>
                          EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared (F.dist theta value) value)
                        D) →
                    (∀ (theta : ℝ),
                      0 < theta →
                        MeasureTheory.Integrable
                          (fun value =>
                            EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent (F.dist theta value)
                              (F.dist theta value) value)
                          D) →
                      (∀ (theta : ℝ),
                        0 < theta →
                          MeasureTheory.Integrable
                            KR21Monoculture.DistributionalAccuracyFamily.jointSharedSecondMoverPayoff
                              (F.outerIndependentPairJointLaw D theta -)) →
                      (∀ (theta : ℝ),
                        0 < theta →
                          MeasureTheory.Integrable
                            KR21Monoculture.DistributionalAccuracyFamily.jointIndependentSecondMoverPayoff
                              (F.outerIndependentPairJointLaw D theta -)) →
                      (∀ (theta : ℝ),
                        0 < theta →
                          ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
                            if KR21Monoculture.disagreementEvent x.2 then 1
                            else 0 ∂F.outerIndependentPairJointLaw D theta -) →
                      (∀ (theta : ℝ), 0 < theta → 0 < F.jointLawDisagreementConditionalGain D theta -) →
                      ∃ thetaA,
                        thetaH < thetaA ∧
                        ∫ (value : KR21Monoculture.ValueProfile n),
                          ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
                            KR21Monoculture.Strategy.algorithm ∂D +
                          ∫ (value : KR21Monoculture.ValueProfile n),
                            ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
                              KR21Monoculture.Strategy.algorithm
                                KR21Monoculture.Strategy.algorithm ∂D =
                          ∫ (value : KR21Monoculture.ValueProfile n),
                            ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
                              KR21Monoculture.Strategy.human ∂D +
                          ∫ (value : KR21Monoculture.ValueProfile n),
                            ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
                              KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human ∂D
                    )
              )
          )
        )

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation6_outer_payoff_indifference_from_source_conditions

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 493-504

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} (E : MeasureTheory.Measure ℝ) [inst : MeasureTheory.IsProbabilityMeasure E],
  MeasureTheory.MemLp id 2 E →
    0 < ProbabilityTheory.variance id E →
      ∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) (theta : ℝ),
        0 < theta →
          ProbabilityTheory.iIndepFun (fun i epsilon => epsilon i)
            (KR21Monoculture.sourceRUMNormalizedIIDNoiseLaw E √(ProbabilityTheory.variance id E)) ∧
            MeasureTheory.MemLp id 2
              (KR21Monoculture.sourceRUMNormalizedScalarNoiseLaw E √(ProbabilityTheory.variance id E)) ∧
              (∀ (i : EconCSLib.SocialChoice.Ranking.Candidate n),
                ProbabilityTheory.variance (fun epsilon => epsilon i)
                  (KR21Monoculture.sourceRUMNormalizedIIDNoiseLaw E √(ProbabilityTheory.variance id E)) =
                    1) ∧
                KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (MeasureTheory.Measure.pi fun x => E) value
                  theta =
                    KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF
                      (KR21Monoculture.sourceRUMNormalizedIIDNoiseLaw E √(ProbabilityTheory.variance id E)) value
                      (theta / √(ProbabilityTheory.variance id E))
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.section31_iidFinitePositiveVariance_normalization

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

Because RUMs are notoriously difficult to work with analytically, we restrict ourselves to the case where $n = 3$, i.e., there are 3 candidates. Under this restriction, we can show that for Gaussian and Laplacian noise

[form-feed in source extraction]
Figure 3: Regions for different equilibria. When human evaluators are more accurate than the algorithm, both firms decide to employ humans (HH). When the algorithm is significantly more accurate, both firms use the algorithm (AA). When the algorithm is slightly more accurate than human evaluators, two possible equilibria exist: (1) one firm uses the algorithm and the other employs a human (AH), or (2) both decide whether to use the algorithm with some probability p . The shaded portion of the green AA region depicts where social welfare is smaller at the AA equilibrium than it would be if both firms used human evaluators. distributions, Definition 2 and 3 – the two conditions of Theorem 1 – are met, regardless of the candidate distribution D . We defer the proof to Appendix C.
Theorem 2. Let F_0 be the family of RUMs with either Gaussian or Laplacian noise with standard deviation $1/\theta$. Then, for any candidate distribution D over 3 candidates, the conditions of Theorem 1 are satisfied. It might be tempting to generalize Theorem 2 to other distributions and more candidates; however, certain noise and candidate distributions violate the conditions of Theorem 1. Even for 3-candidate RUMs, there exist distributions for which each of the conditions is violated; we provide such examples in Appendix B.

Expanded PaperInterface specification

```
(V {theta x1 x2 x3 : ℝ},
  0 < theta →
  MeasureTheory.MeasurePreserving (↑(KR21Monoculture.rightTripleToCandidateMeasurableEquiv ℝ))
    ((ProbabilityTheory.gaussianReal 0 1).prod
      ((ProbabilityTheory.gaussianReal 0 1).prod (ProbabilityTheory.gaussianReal 0 1)))
    (MeasureTheory.Measure.pi fun x => ProbabilityTheory.gaussianReal 0 1) ∧
    ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist theta).toMeasure =
      MeasureTheory.Measure.map
        (fun epsilon =>
          EconCSLib.SocialChoice.Ranking.rankByScore fun i =>
            KR21Monoculture.threeCandidateValueProfile x1 x2 x3 i +
            KR21Monoculture.rightTripleToCandidateFunction epsilon i / theta)
          ((ProbabilityTheory.gaussianReal 0 1).prod
            ((ProbabilityTheory.gaussianReal 0 1).prod (ProbabilityTheory.gaussianReal 0 1))) ∧
            (KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist theta =
              KR21Monoculture.gaussianThreeCandidateRankingLaw theta x1 x2 x3) ∧
  ProbabilityTheory.variance id (ProbabilityTheory.gaussianReal 0 1) = 1 ∧
  ProbabilityTheory.variance id
    (KR21Monoculture.w11BaseNoiseLaw KR21Monoculture.sourceUnitVarianceLaplaceBaseDensity) =
    1 ∧
  (V {theta x1 x2 x3 : ℝ},
    0 < theta →
    MeasureTheory.MeasurePreserving (↑(KR21Monoculture.rightTripleToCandidateMeasurableEquiv ℝ))
      ((KR21Monoculture.theorem7LaplaceMeasure (√2) 0).prod
        ((KR21Monoculture.theorem7LaplaceMeasure (√2) 0).prod
          (KR21Monoculture.theorem7LaplaceMeasure (√2) 0)))
      (MeasureTheory.Measure.pi fun x => KR21Monoculture.theorem7LaplaceMeasure (√2) 0) ∧
      ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist theta).toMeasure =
        MeasureTheory.Measure.map
          (fun epsilon =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun i =>
              KR21Monoculture.threeCandidateValueProfile x1 x2 x3 i +
              KR21Monoculture.rightTripleToCandidateFunction epsilon i / theta)
            ((KR21Monoculture.theorem7LaplaceMeasure (√2) 0).prod
              ((KR21Monoculture.theorem7LaplaceMeasure (√2) 0).prod
                (KR21Monoculture.theorem7LaplaceMeasure (√2) 0))) ∧
              (KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist theta =
                KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateRankingLaw theta x1 x2 x3) ∧
  (V {x1 x2 x3 : ℝ},
    x2 < x1 →
    x3 < x2 →
    (V (theta : ℝ),
      0 < theta →
      V (pi : EconCSLib.SocialChoice.Ranking.Ranking 1),
        ContinuousAt
          (fun theta' =>
            ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist theta') pi).toReal)
          theta ∧
        DifferentiableAt ℝ
          (fun theta' =>
            ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist theta') pi).toReal)
          theta) ∧
    Filter.Tendsto
      (fun theta =>
        ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist theta)
```



```

    KR21Monoculture.rum3Ranking012).toReal)
  Filter.atTop (nhds 1) ∧
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
      thetaH < thetaA →
        (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate 1)),
          remaining.Nonempty →
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist thetaH)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) remaining ≤
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist thetaA)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) remaining) ∧
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist thetaH)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) Finset.univ <
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.theorem2GaussianScaledNoiseFamily x1 x2 x3).dist thetaA)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) Finset.univ) ∧
  (∀ {x1 x2 x3 : ℝ},
    x2 < x1 →
      x3 < x2 →
        (∀ (theta : ℝ),
          0 < theta →
            ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking 1),
              ContinuousAt
                (fun theta' =>
                  ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist theta')
                  pi).toReal)
                theta ∧
              DifferentiableAt ℝ
                (fun theta' =>
                  ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist theta')
                  pi).toReal)
                theta) ∧
  Filter.Tendsto
    (fun theta =>
      ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist theta)
      KR21Monoculture.rum3Ranking012).toReal)
  Filter.atTop (nhds 1) ∧
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
      thetaH < thetaA →
        (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate 1)),
          remaining.Nonempty →
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist
                thetaH)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) remaining ≤
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist
                thetaA)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) remaining) ∧
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist thetaH)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) Finset.univ <
            KR21Monoculture.expectedBestInSet
              ((KR21Monoculture.sourceUnitVarianceLaplaceScaledNoiseFamily x1 x2 x3).dist thetaA)
              (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) Finset.univ) ∧
  (∀ (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile 1)) [MeasureTheory.IsProbabilityMeasure D],
    (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate 1),
      MeasureTheory.Integrable (fun value => value c) D) →
      (∀ (value : KR21Monoculture.ValueProfile 1) ∂D, value 1 < value 0 ∧ value 2 < value 1) →
        (∀ (theta : ℝ),
          0 < theta →
            (regularity :
              KR21Monoculture.gaussianThreeCandidateDistributionalFamily.OuterIndependentRerankingJointRegularity
                D theta),
              0 <
                ∫ (x : KR21Monoculture.ValueProfile 1 × KR21Monoculture.RankingPair 1),
                  if KR21Monoculture.disagreementEvent x.2 then 1
                  else
                    0 ∂KR21Monoculture.gaussianThreeCandidateDistributionalFamily.outerIndependentPairJointLaw
                      D theta -- ∧
                0 <
                  KR21Monoculture.gaussianThreeCandidateDistributionalFamily.jointLawDisagreementConditionalGain
                    D theta --) ∧
            ∀ (thetaA thetaH : ℝ),
              0 < thetaH →
                thetaH < thetaA →
                  ∫ (value : KR21Monoculture.ValueProfile 1),
                    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
                      (KR21Monoculture.gaussianThreeCandidateDistributionalFamily.dist thetaH value)
                      (KR21Monoculture.gaussianThreeCandidateDistributionalFamily.dist thetaA value)
                      value ∂D <
                  ∫ (value : KR21Monoculture.ValueProfile 1),
                    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
                      (KR21Monoculture.gaussianThreeCandidateDistributionalFamily.dist thetaH value)
                      (KR21Monoculture.gaussianThreeCandidateDistributionalFamily.dist thetaH value)
                      value ∂D) ∧
  (∀ (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile 1)) [MeasureTheory.IsProbabilityMeasure D],
    (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate 1),
      MeasureTheory.Integrable (fun value => value c) D) →
      (∀ (value : KR21Monoculture.ValueProfile 1) ∂D, value 1 < value 0 ∧ value 2 < value 1) →
        (∀ (theta : ℝ),

```

```

0 < theta →
  ∀
    (regularity :
      KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.OuterIndependentReran
      ↪ kingJointRegularity
      D theta),
    0 <
      ∫ (x : KR21Monoculture.ValueProfile 1 × KR21Monoculture.RankingPair 1),
        if KR21Monoculture.disagreementEvent x.2 then 1
        else
          0 ∂KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.outerIndepen
          ↪ dentPairJointLaw
          D theta ... ∧
    0 <
      KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.jointLawDisagreemen
      ↪ tConditionalGain
      D theta ...) ∧
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
      thetaH < thetaA →
        ∫ (value : KR21Monoculture.ValueProfile 1),
          EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
          (KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.dist
            thetaH value)
          (KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.dist
            thetaA value)
          value ∂D <
        ∫ (value : KR21Monoculture.ValueProfile 1),
          EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
          (KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.dist
            thetaH value)
          (KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily.dist
            thetaH value)
          value ∂D) ∧
  (∀ (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile 1)) [MeasureTheory.IsProbabilityMeasure D]
    (thetaH : ℝ),
      0 < thetaH →
        (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate 1),
          MeasureTheory.Integrable (fun value => value c) D) →
          (∀m (value : KR21Monoculture.ValueProfile 1) ∂D, value 1 < value 0 ∧ value 2 < value 1) →
            KR21Monoculture.PaperInterface.literal_outer_payoff_paradox
            KR21Monoculture.gaussianThreeCandidateDistributionalFamily D thetaH) ∧
  (∀ (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile 1)) [MeasureTheory.IsProbabilityMeasure D]
    (thetaH : ℝ),
      0 < thetaH →
        (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate 1),
          MeasureTheory.Integrable (fun value => value c) D) →
          (∀m (value : KR21Monoculture.ValueProfile 1) ∂D, value 1 < value 0 ∧ value 2 < value 1) →
            KR21Monoculture.PaperInterface.literal_outer_payoff_paradox
            KR21Monoculture.sourceUnitVarianceLaplaceThreeCandidateDistributionalFamily D thetaH
  )

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem2_gaussian_laplace_source_semantic_complete_literal_payoff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 530-540

Verbatim source input

Finally, there exist noise distributions that violate Definition 2 for any candidate distribution D. In particular, the RUM family defined by the Gumbel distribution is well-known to be equivalent to the PlackettLuce model of $\hookrightarrow$ ranking, which is generated by sequentially selecting candidate i with probability $\frac{\exp(\theta x_i)}{\sum_{j \in S} \exp(\theta x_j)}$

P

(7)

where S is the set of remaining candidates $[12, 4]$. Under the Plackett-Luce model, for any θ , $\text{UAH}(\theta, \theta) =$

Expanded PaperInterface specification

```
∀ {n : ℕ} (theta : ℝ) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
  0 < theta →
  ∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n))
    {i : EconCSLib.SocialChoice.Ranking.Candidate n},
    i ∈ remaining →
      KR21Monoculture.plackettLuceChoiceProb theta value remaining i =
        Real.exp (theta * value i) / ∑ j ∈ remaining, Real.exp (theta * value j)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation7_plackettLuce_choice_probability

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 530-548

Verbatim source input

Finally, there exist noise distributions that violate Definition 2 for any candidate distribution D . In particular, the RUM family defined by the Gumbel distribution is well-known to be equivalent to the Plackett-Luce model of $\rightarrow$ ranking, which is generated by sequentially selecting candidate i with probability $\exp(\theta x_i)$

$\sum_{j \in S} \exp(\theta x_j)$

P

(7)

where S is the set of remaining candidates [12, 4]. Under the Plackett-Luce model, for any θ , $U_A(\theta, \theta) = U_A(\theta, \theta)$. To see this, suppose the firm that hires first selects candidate i^* . Then, the firm that hires second

[form-feed in source extraction]
gets each candidate i with probability given by (7) with $S = \{1, \dots, n\} \setminus i^*$. As a result, by (3), if $\pi, \sigma \sim F_\theta$, $E[\pi_1 - \pi_2 \mid \pi_1 \neq \sigma_1] = 0$
for any candidate distribution D , meaning the Plackett-Luce model never meets Definition 2. Thus, under the Plackett-Luce model, monoculture has no effect—the optimal strategy is always to use the best available ranking, regardless of competitors' strategies.

Expanded PaperInterface specification

```

V {n : ℕ} (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D]
{location theta : ℝ},
0 < theta →
(V (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
(KR21Monoculture.sourceUnitVarianceGumbelNoiseLaw location =
  MeasureTheory.Measure.pi fun x =>
    MeasureTheory.Measure.map (fun arrival => location + -(√6 / Real.pi) * Real.log arrival)
      (ProbabilityTheory.expMeasure 1)) ∧
(V (profile epsilon : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
  (i : EconCSLib.SocialChoice.Ranking.Candidate n),
  KR21Monoculture.sourceUnitVarianceGumbelScores theta profile epsilon i = profile i + epsilon i / theta) ∧
(V (profile : KR21Monoculture.ValueProfile n),
  (KR21Monoculture.sourceUnitVarianceGumbelRUMRankingPMF location theta profile).toMeasure =
    MeasureTheory.Measure.map
      (fun epsilon => EconCSLib.SocialChoice.Ranking.rankByScore fun i => profile i + epsilon i / theta)
      (MeasureTheory.Measure.pi fun x =>
        MeasureTheory.Measure.map (fun arrival => location + -(√6 / Real.pi) * Real.log arrival)
          (ProbabilityTheory.expMeasure 1))) ∧
(V (profile : KR21Monoculture.ValueProfile n),
  KR21Monoculture.sourceUnitVarianceGumbelRUMRankingPMF location theta profile =
    KR21Monoculture.plackettLuceRankingPMF (theta / KR21Monoculture.unitVarianceGumbelScale) profile) ∧
have F :=
  KR21Monoculture.DistributionalAccuracyFamily.sourceUnitVarianceGumbelRUMDistributionalFamily location;
∃ (hatom :
  V (ranking : EconCSLib.SocialChoice.Ranking.Ranking n),
  Measurable fun value => (F.dist theta value) ranking),
have J := F.outerIndependentPairJointLaw D theta hatom;
have numerator :=
  ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
  if
    EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
      x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
        x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
    else 0 ∂J;
have denominator :=
  ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
  if
    EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
      1
    else 0 ∂J;
J = D.compProd (F.independentPairKernel theta hatom) ∧
(V (profile : KR21Monoculture.ValueProfile n),
  (F.independentPairKernel theta hatom) profile =
    (EconCSLib.pmfProd (F.dist theta profile) (F.dist theta profile)).toMeasure) ∧
MeasureTheory.Integrable
  (fun value =>
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent (F.dist theta value)
      (F.dist theta value) value)
D ∧
MeasureTheory.Integrable
  (fun value =>
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared (F.dist theta value) value)
D ∧
0 < denominator ∧
numerator / denominator = 0 ∧
  ∫ (value : KR21Monoculture.ValueProfile n),
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent (F.dist theta value)
      (F.dist theta value) value ∂D =
  ∫ (value : KR21Monoculture.ValueProfile n),
    EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared (F.dist theta value) value ∂D

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.section31_literalUnitVarianceGumbel_outer_zero_effect_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 18

Source locator: cited publication, lines 574-587

Verbatim source input

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π :

$$\Pr[\pi] = \frac{1}{Z} \phi^{-d(\pi, \pi^*)},$$

where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking r . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
  theta = phi - 1 →
  ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
  have M := KR21Monoculture.concreteMallowsSpec center theta;
  M.q = phi⁻¹ ∧
  (M.law pi).toReal =
    phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
    ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equation8_source_concrete_mallows_probability

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

$$\Pr[\pi] = \frac{1}{Z} \phi^{-d(\pi, \pi^*)},$$
where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking π^* . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

Theorem 3. Let F_θ be the family of Mallows Model distributions with parameter $\theta = \phi - 1$. Then, for any candidate distribution D , the conditions of Theorem 1 are satisfied.

[Next verbatim source excerpt]

E

Proof of Theorem 3

E.1

Verifying Definition 2

We must show that when $\pi, \tau \sim F_\theta$,
 $E[\pi_1 - \pi_2 \mid \pi_1 \neq \tau_1] > 0$.

(E.1)

We begin by expanding:
 $E[\pi_1 - \pi_2 \mid \pi_1 \neq \tau_1] =$

$$\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j) \Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1]$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j>i}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j>i}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j>i}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j>i}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j>i}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1])$$

Since $x_i > x_j$ for $i < j$, it suffices to show that for all $i < j$,
 $\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] \geq \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1]$,

(E.2)

and that this holds strictly for some $i < j$. We simplify (E.2) as follows:

$$\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] > \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1]$$

$$\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] > \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1]$$

$$\Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1] > \Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1]$$

$$>$$

$$\Pr[\pi_1 \neq \tau_1]$$

$$\Leftrightarrow \Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] > \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1]$$

$$\Leftrightarrow$$

$$\Leftrightarrow \Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \tau_1] > \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \tau_1]$$

$$\Leftrightarrow \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] \Pr[\tau_1 \neq x_i] > \Pr[\pi_1 = x_j \wedge \pi_2 = x_i] \Pr[\tau_1 \neq x_j]$$

(E.3)

We can simplify (E.3) using Lemmas 5 and 6. Let $|i - j|$ denote the difference in rank between x_i and x_j .

$$\begin{aligned} & \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] \Pr[\tau_1 \neq x_i] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i] \Pr[\tau_1 \neq x_j] \\ &= \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] (1 - \Pr[\tau_1 = x_i]) - \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] (1 - \Pr[\tau_1 = x_j]) \\ &= \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] (1 - \Pr[\tau_1 = x_i]) - \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] (1 - \Pr[\tau_1 = x_j]) \\ &= \Pr[\pi_1 = x_i \wedge \pi_2 = x_j] (1 - \Pr[\tau_1 = x_i] - \Pr[\tau_1 = x_j] + \Pr[\tau_1 = x_i] \Pr[\tau_1 = x_j]) \end{aligned}$$

29

[form-feed in source extraction]

This is positive if and only if

$$1 - \Pr[\tau_1 = x_i] - \Pr[\tau_1 = x_j] + \Pr[\tau_1 = x_i] \Pr[\tau_1 = x_j] > 0$$

[U+0011 control character]

[U+0010 control character]

$$\Leftrightarrow \Pr[\tau_1 = x_i] (1 - \Pr[\tau_1 = x_j]) > 1 - \Pr[\tau_1 = x_j]$$

$$1 - \Pr[\tau_1 = x_j]$$

$$1 - \Pr[\tau_1 = x_j]$$

$$1 - \Pr[\tau_1 = x_j]$$

$$1 - \Pr[\tau_1 = x_j]$$

$$\Leftrightarrow 1 - \Pr[\tau_1 = x_j]$$

$$<$$

$$-n$$

$$\Pr(1 - \Pr[\tau_1 = x_j])$$

$$1 - \Pr[\tau_1 = x_j]$$

$$\Leftrightarrow \Pr[\tau_1 = x_i] <$$

$$\Leftrightarrow \Pr[\tau_1 = x_i] (1 - \Pr[\tau_1 = x_j]) > 1 - \Pr[\tau_1 = x_j]$$

This is weakly true for any $i < j$ because $\Pr[\tau_1 = x_i] \geq 1$ and $|i - j| + 1 \leq n$, and it is strictly true for any i, j other than 1 and n . Thus, $E[\pi_1 - \pi_2 | \pi_1 \neq \tau_1] > 0$.

E.2

Verifying Definition 3

Recall that Definition 3 is equivalent to $UAH(\theta_A, \theta_H) < UHH(\theta_A, \theta_H)$ for $\theta_A > \theta_H$. Let τ be the algorithmic ranking, and let π be a ranking from a human evaluator. Recall that $UH(\theta_A, \theta_H) = E[\pi_1]$. Throughout this proof, we will drop the (θ_A, θ_H) notation and simply write UH , UAH , and UHH .

UAH =

$$n$$

$$X$$

$$(\Pr[\pi_1 = x_i \wedge \tau_1 \neq x_i] + \Pr[\pi_2 = x_i \wedge \pi_1 = \tau_1]) x_i$$

$$i=1$$

$$=$$

$$=$$

$$n$$

$$X$$

$$i=1$$

$$n$$

$$X$$

$$\Pr[\pi_1 = x_i \wedge \tau_1 \neq x_i] x_i +$$

$$n$$

$$X$$

$$\Pr[\pi_2 = x_i \wedge \pi_1 = \tau_1] x_i$$

$$i=1$$

$$(P[\pi_1 = x_i] - \Pr[\pi_1 = x_i \wedge \tau_1 = x_i]) x_i +$$

$$i=1$$

$$= UH -$$

$$= UH +$$

$$n X$$

$$X$$

$$\Pr[\pi_1 = x_j \wedge \tau_1 = x_j \wedge \pi_2 = x_i] x_i$$

$$i=1 \quad j=1$$

$$n$$

$$X$$

$$i=1$$

$$n$$

$$X$$

$$\Pr[\pi_1 = x_i \wedge \tau_1 = x_i] x_i +$$

$$n$$

$$X$$

$$\Pr[\pi_1 = x_i \wedge \tau_1 = x_i] E[\pi_2 | \pi_1 = x_i \wedge \tau_1 = x_i]$$

$$i=1$$

$$\Pr[\pi_1 = x_i] \Pr[\tau_1 = x_i] (E[\pi_2 | \pi_1 = x_i] - x_i)$$

$$i=1$$

Similarly, because two human evaluators are independent,
 $U_{HH} = U_H +$

$\sum_{i=1}^n$
 X

$\Pr[\pi_1 = x_i] (E[\pi_2 | \pi_1 = x_i] - x_i) .$

$i=1$

Let $V_i = E[\pi_2 | \pi_1 = x_i]$. Note that conditioned on $\pi_1 = x_i$, the remaining elements of π_1 follow a Mallows model distribution over $n - 1$ candidates. Because the Mallows model is value-independent, increasing any item value increases the expected value of the top-ranked item (and in fact, the item ranked at any position). Thus, V_i increases as i increases (since x_i , the value of the unavailable candidate, decreases). Moreover, x_i is strictly decreasing in i , so $V_i - x_i$ is strictly increasing in i . With this, we have

$U_{AH} - U_H =$
 $U_{HH} - U_H =$

$\sum_{i=1}^n$
 X
 $i=1$
 $\sum_{i=1}^n$
 X

$\Pr[\pi_1 = x_i] \Pr[\tau_1 = x_i] (V_i - x_i)$
 $\Pr[\pi_1 = x_i]^2 (V_i - x_i)$

$i=1$

P_n

P_n
Let $CA = \Pr[\pi_1 = \tau_1] = \sum_{i=1}^n \Pr[\pi_1 = x_i] \Pr[\tau_1 = x_i]$, and similarly let $CH = \sum_{i=1}^n \Pr[\pi_1 = x_i]^2$. $CA > CH$ by Lemma 4 with $y_{i0} = \Pr[\pi_1 = x_{n-i+1}]$, $p_{0i} = \Pr[\pi_1 = x_{n-i+1}]$ and $q_{i0} = \Pr[\tau_1 = x_{n-i+1}]$.

3θ

[form-feed in source extraction]

Let $p_i = \Pr[\pi_{10} = i] \Pr[\pi_1 = i]/CA$, $q_i = \Pr[\pi_{10} = i]^2 / CH$, and $y_i = V_i - x_i$. Then, we have

$\sum_{i=1}^n$
 X
 $U_{AH} - U_H$
 $=$
 $\sum_{i=1}^n p_i y_i$
 CA
 $i=1$
 $\sum_{i=1}^n$
 X
 $U_{HH} - U_H$
 $=$
 $\sum_{i=1}^n q_i y_i$
 CH
 $i=1$

With $\phi_A = \theta_A + 1$ and $\phi_H = \theta_H + 1$,

$\sum_{i=1}^n$
 CH
 p_i
 $=$
 $\sum_{i=1}^n$
 q_i
 CA

$1-\phi-1$
 A
 $i-1$
 $\phi_A (1-\phi-n$
 $A)$
 $1-\phi-1$
 H
 $-n$
 $\phi i-1$
 $H (1-\phi H)$

P_n

which is decreasing in i since $\phi_H < \phi_A$. By Lemma 4,
 θ by Lemma 7, so

$\sum_{i=1}^n$
 X

$p_i y_i <$

$\sum_{i=1}^n$
 X

$i=1$

α

$\phi i-1$
 H

$\phi i-1$
 A

$i=1 p_i y_i <$

Pn

i=1 qi yi . Finally, note that UHH -UH <

qi yi

i=1

UHH - UH

UAH - UH

<

CA

CH

CH (UAH - UH)

< UHH - UH

CA

UAH - UH < UHH - UH

(CA > CH , and UHH - UH < 0)

UAH < UHH

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For Candidate n = Fin(n+2) with n > 0, a fixed rank-labelled Mallows center, and a probability law D on value profiles with
→ coordinatewise finite first moments and D-a.e. strict center order: for every phi > 1 and theta = phi - 1, the fixed-center
→ source Mallows law has q = phi^(-1) and, for every value profile v and ranking pi, mass phi^(-KendallTau(center, pi)) /
→ sum_tau phi^(-KendallTau(center, tau)). For every theta > 0 the actual outer-law conditionally-iid-ranking top-disagreement
→ gain is positive, and for every thetaA > thetaH > 0 the Definition 3 weaker-competition conclusion holds.

Archival source anchor: cited publication, lines 590-591

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```
∀ {n : ℕ} (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D]
(center : EconCSLib.SocialChoice.Ranking.Ranking n),
0 < n →
  (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
  (∀m (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
    EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value) →
  (∀ (phi theta : ℝ),
    1 < phi →
      theta = phi - 1 →
        (KR21Monoculture.concreteMallowsSpec center theta).q = phi-1 ∧
          ∀ (value : KR21Monoculture.ValueProfile n) (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
            (((KR21Monoculture.fixedCenterMallowsDistributionalFamily center).dist theta value) pi).toReal =
              phi-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                ∑ tau, phi-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
  (∀ (value : KR21Monoculture.ValueProfile n),
    EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value →
      (∀ (theta : ℝ),
        0 < theta →
          ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
            ContinuousAt
              (fun theta' =>
                (((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist theta') pi).toReal)
                theta) ∧
            DifferentiableAt ℝ
              (fun theta' =>
                (((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist theta') pi).toReal)
                theta) ∧
            Filter.Tendsto
              (fun theta =>
                (((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist theta) center).toReal)
              Filter.atTop (nhds 1) ∧
          ∀ (thetaA thetaH : ℝ),
            0 < thetaH →
              thetaH < thetaA →
                (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
                  remaining.Nonempty →
                    KR21Monoculture.expectedBestInSet
                      ((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist thetaH) value
                      remaining ≤
                    KR21Monoculture.expectedBestInSet
                      ((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist thetaA) value
                      remaining) ∧
                    KR21Monoculture.expectedBestInSet
                      ((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist thetaH) value
                      Finset.univ <
                    KR21Monoculture.expectedBestInSet
                      ((KR21Monoculture.fixedCenterMallowsPointFamily center value).dist thetaA) value
                      Finset.univ) ∧
                (∀ (theta : ℝ),
                  0 < theta →
                    have J :=
                      (KR21Monoculture.fixedCenterMallowsDistributionalFamily center).outerIndependentPairJointLaw D theta
                    ...;
                  0 <
                    ∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
```

```

    if
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
    1
  else 0 ∂J ∧
0 <
  (∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
    if
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
      x.1 (EconCSLib.SocialChoice.Ranking.firstChoice x.2.1) -
      x.1 (EconCSLib.SocialChoice.Ranking.secondChoice x.2.1)
    else 0 ∂J) /
  (∫ (x : KR21Monoculture.ValueProfile n × KR21Monoculture.RankingPair n),
    if
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.1 ≠
      EconCSLib.SocialChoice.Ranking.firstChoice x.2.2 then
    1
    else 0 ∂J) ∧
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    (∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp
      ((KR21Monoculture.fixedCenterMallowsDistributionalFamily center).dist thetaH value)
      ((KR21Monoculture.fixedCenterMallowsDistributionalFamily center).dist thetaA value)
      fun pi sigma => EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma ∂D <
    (∫ (value : KR21Monoculture.ValueProfile n),
      EconCSLib.pmfPairExp
      ((KR21Monoculture.fixedCenterMallowsDistributionalFamily center).dist thetaH value)
      ((KR21Monoculture.fixedCenterMallowsDistributionalFamily center).dist thetaA value)
      fun pi sigma => EconCSLib.SocialChoice.Ranking.secondMoverUtility value pi sigma ∂D) ∧
    ∀ (thetaH : ℝ),
    0 < thetaH →
    have F := KR21Monoculture.fixedCenterMallowsDistributionalFamily center;
    ∃ thetaA,
    thetaH < thetaA ∧
    (∫ (value : KR21Monoculture.ValueProfile n),
      ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
      KR21Monoculture.Strategy.human ∂D +
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
        KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human ∂D <
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
        KR21Monoculture.Strategy.algorithm ∂D +
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
        KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm ∂D ∧
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
        KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human ∂D <
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
        KR21Monoculture.Strategy.algorithm ∂D +
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
        KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm ∂D <
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).firstMoverEU
        KR21Monoculture.Strategy.human ∂D +
      (∫ (value : KR21Monoculture.ValueProfile n),
        ((F.pointFamily value).modelAt thetaA thetaH).secondMoverEU
        KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human ∂D

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem3_source_complete_semantic_complete_literal_payoff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 133-138

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n k : ℕ} (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D]
(center : EconCSLib.SocialChoice.Ranking.Ranking n) (phiA thetaA phiH thetaH : ℝ),
1 < phiA →
1 < phiH →
  thetaA = phiA - 1 →
  thetaH = phiH - 1 →
  k < Fintype.card (EconCSLib.SocialChoice.Ranking.Candidate n) →
  (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
  have algorithm := KR21Monoculture.concreteMallowsSpec center thetaA;
  have human := KR21Monoculture.concreteMallowsSpec center thetaH;
  algorithm.q = phiA-1 ∧
  human.q = phiH-1 ∧
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    (algorithm.law pi).toReal =
      phiA-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
      ∑ tau, phiA-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
    (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
      (human.law pi).toReal =
        phiH-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
        ∑ tau, phiH-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
    (phiA ≤ phiH →
      (∀n (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
        EconCSLib.SocialChoice.Ranking.WeaklyOrderedBy center value) →
        ∀ (i : Fin k) (hired : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
          hired.card = ↑i →
            ∀ (strategy : KR21Monoculture.Strategy),
              ∫ (value : KR21Monoculture.ValueProfile n),
                ∑ pi,
                  ((match strategy with
                     | KR21Monoculture.Strategy.algorithm => algorithm.law
                     | KR21Monoculture.Strategy.human => human.law)
                   pi).toReal *
                  value (KR21Monoculture.bestInSet pi (Finset.univ \ hired)) ∂D ≤
                ∫ (value : KR21Monoculture.ValueProfile n),
                  ∑ pi,
                    (human.law pi).toReal *
                    value (KR21Monoculture.bestInSet pi (Finset.univ \ hired)) ∂D) ∧
            (phiA < phiH →
              (∀n (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
                EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value) →
                ∀ (i : Fin k) (hired : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
                  hired.card = ↑i →
                    ∀ (strategy : KR21Monoculture.Strategy),
                      (∀ (alternative : KR21Monoculture.Strategy),
                        ∫ (value : KR21Monoculture.ValueProfile n),
                          ∑ pi,
                            ((match alternative with
                               | KR21Monoculture.Strategy.algorithm => algorithm.law
                               | KR21Monoculture.Strategy.human => human.law)
                             pi).toReal *
                            value (KR21Monoculture.bestInSet pi (Finset.univ \ hired)) ∂D ≤
                          ∫ (value : KR21Monoculture.ValueProfile n),
                            ∑ pi,
                              ((match strategy with
                                 | KR21Monoculture.Strategy.algorithm => algorithm.law
                                 | KR21Monoculture.Strategy.human => human.law)
                                pi).toReal *
                              value (KR21Monoculture.bestInSet pi (Finset.univ \ hired)) ∂D) →
                        strategy = KR21Monoculture.Strategy.human)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem4_source_mallows_outer_horizon_lt_literal_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 782-951

Verbatim source input

Random Utility Models satisfying Definition 1

Theorem 5. Let f be the pdf of E . The family of RUMs F_θ given by ranking $x_i + \varepsilon_i \theta$ with $\varepsilon_i \sim E$ satisfies the conditions of Definition 1 if:

- f is differentiable
- f has positive support on $(-\infty, \infty)$

Proof. We need to show that F_θ satisfies the differentiability, asymptotic optimality, and monotonicity conditions in Definition 1.

Qn

Differentiability: The probability density of any realization of the n noise samples ε_i / θ is $i=1 f(\varepsilon_i / \theta)$.

Let $\varepsilon = [\varepsilon_1 / \theta, \dots, \varepsilon_n / \theta]$ be the vector of noise values and let $M(\pi) \subseteq \mathbb{R}^n$ be the region such that any $\varepsilon \in M(\pi)$ will produce the ranking π . The probability of any permutation π is

$\int_{M(\pi)} f(\varepsilon) d\varepsilon$

$\Pr[\pi] =$

$\int_{M(\pi)} f(\varepsilon) d\varepsilon$

$\Pr[\pi] =$

$\int_{M(\pi)} f(\varepsilon) d\varepsilon$

$\Pr[\pi] =$

$\int_{M(\pi)} f(\varepsilon) d\varepsilon$

$\Pr[\pi] =$

$\Pr[\pi] =$

Because f is differentiable,

$\frac{d}{d\theta} \Pr[\pi] = \int_{M(\pi)} \frac{d}{d\theta} f(\varepsilon) d\varepsilon$

$\frac{d}{d\theta} \Pr[\pi] = \int_{M(\pi)} \frac{d}{d\theta} f(\varepsilon) d\varepsilon$

$\frac{d}{d\theta} \Pr[\pi] =$

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2
Thus, for sufficiently large θ , the probability that x_i and x_{i+1} are incorrectly ordered is at most δ . Repeating this analysis for all $n - 1$ pairs of adjacent elements, taking the maximum of all the θ 's, and taking a union bound yields that the probability of incorrectly ordering any pair of elements is at most $(n - 1)\delta$, meaning the probability of outputting the correct ranking is at least $1 - (n - 1)\delta$. Since δ is arbitrary, this probability can be made arbitrarily close to 1, satisfying the asymptotic optimality condition.
14

[form-feed in source extraction]

Monotonicity: The removal of any elements does not alter the distribution of the remaining elements, meaning that the distribution of $\pi(-S)$ is equivalent to a RUM with $n - |S|$ elements. Thus, it suffices to show that for a RUM with positive support on $(-\infty, \infty)$, the probability of ranking the best candidate first strictly increases with θ .

Recall that by definition, the candidates are ranked according to $x_i + \epsilon\theta_i$. The probability that x_1 is ranked first is

```
[U+0014 control character]
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[U+0015 control character]
 $\epsilon_i$ 
 $\epsilon_i$ 
 $\epsilon_1$ 
 $\epsilon_1$ 
 $\Pr x_1 +$ 
 $> \max_{2 \leq i \leq n} x_i +$ 
 $= \Pr$ 
 $> \max_{2 \leq i \leq n} x_i - x_1 +$ 
 $2 \leq i \leq n$ 
 $\theta$ 
 $\theta$ 
 $\theta$ 
 $\theta$ 
[U+0014 control character]
[U+0015 control character]
 $= \Pr \epsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i$ 
 $2 \leq i \leq n$ 
[U+0014 control character]
[U+0015 control character]
 $= E \epsilon_2, \dots, \epsilon_n \Pr \epsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i \mid \epsilon_2, \dots, \epsilon_n$ 
(A.1)
 $2 \leq i \leq n$ 
```

We want to show that (A.1) is increasing in θ . Intuitively, this is because as θ increases, the right hand side of the inequality inside the probability decreases. To prove this formally, it suffices to show that the subderivative of (A.1) with respect to θ only includes strictly positive numbers. First, we have

```
[U+0014 control character]
[U+0015 control character]
 $\partial$ 
 $E \epsilon_2, \dots, \epsilon_n \Pr \epsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i \mid \epsilon_2, \dots, \epsilon_n \in \mathbb{R}_{>0}$ 
 $2 \leq i \leq n$ 
 $\partial \theta$ 
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[U+0015 control character]
 $\partial$ 
 $\Pr \epsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i \mid \epsilon_2, \dots, \epsilon_n \in \mathbb{R}_{>0}$ 
 $\iff$ 
 $2 \leq i \leq n$ 
 $\partial \theta$ 
Let  $F$  and  $f$  be the cumulative density function and probability density function of  $E$  respectively. Then,
[U+0014 control character]
[U+0015 control character]
[U+0012 control character]
[U+0013 control character]
 $\Pr \epsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i \mid \epsilon_2, \dots, \epsilon_n = 1 - F \max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i$ 
 $2 \leq i \leq n$ 
 $2 \leq i \leq n$ 
```

Note that $F(\cdot)$ is strictly increasing (since f is assumed to have positive support on $(-\infty, \infty)$), so it suffices to show that

```
 $\partial$ 
 $\max_{2 \leq i \leq n} \theta(x_i - x_1) + \epsilon_i \in \mathbb{R}_{<0}$ 
 $\partial \theta$ 
For any  $i$ ,
 $d$ 
 $\theta(x_i - x_1) + \epsilon_i = x_i - x_1 < 0$ .
 $d \theta$ 
```

Thus, the subderivative of the max of such functions includes only strictly negative numbers, which completes the proof.

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

Let Candidate $n = \text{Fin}(n+2)$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be measurable, integrable, everywhere strictly positive, normalized by integral $\hookrightarrow \text{ENNReal.ofReal}(f) = 1$, and absolutely continuous on every real interval; let f_{prime} be integrable and agree almost everywhere with $\text{deriv } f$. For a value profile strictly descending along a center ranking, the iid product density law is a probability law, and there is a ranking law F_{θ} such that for every $\theta > 0$ it is exactly the pushforward of that product law under $\text{epsilon} \mapsto \text{rank}(\text{value}_i + \text{epsilon}_i/\theta)$. Every ranking atom of F_{θ} is continuous and differentiable at every positive θ , every atom tends to the pure center-ranking mass as θ tends to infinity, expected value of the best candidate in every nonempty remaining set is weakly increasing with θ , and full-set expected value is strictly increasing when $\theta_A > \theta_H > 0$.

Archival source anchor: cited publication, lines 784-787

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```

∀ {n : ℕ} (f derivative : ℝ → ℝ),
  MeasureTheory.Integrable f MeasureTheory.volume →
  MeasureTheory.Integrable derivative MeasureTheory.volume →
  Measurable f →
  (∀ (x : ℝ), 0 < f x) →
  (∀ (a b : ℝ), AbsolutelyContinuousOnInterval f a b) →
  derivative = "[MeasureTheory.volume] deriv f →
  ⌊- (x : ℝ), ENNReal.ofReal (f x) = 1 →
  ∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
    (center : EconCSLib.SocialChoice.Ranking.Ranking n),
    (∀ (i : Fin (n + 1)), value (center i.succ) < value (center i.castSucc)) →
    Fintype.card (EconCSLib.SocialChoice.Ranking.Candidate n) = n + 2 ∧
    MeasureTheory.IsProbabilityMeasure
      (MeasureTheory.Measure.pi fun x =>
        MeasureTheory.volume.withDensity fun x => ENNReal.ofReal (f x)) ∧
  ∃ rankingLaw,
    (∀ (theta : ℝ),
      0 < theta →
      (rankingLaw theta).toMeasure =
        MeasureTheory.Measure.map
          (fun epsilon =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun i => value i + epsilon i / theta)
          (MeasureTheory.Measure.pi fun x =>
            MeasureTheory.volume.withDensity fun x => ENNReal.ofReal (f x))) ∧
  (∀ (theta : ℝ),
    0 < theta →
    ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
      ContinuousAt (fun theta' => ((rankingLaw theta') pi).toReal) theta ∧
      DifferentiableAt ℝ (fun theta' => ((rankingLaw theta') pi).toReal) theta) ∧
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    Filter.Tendsto (fun theta => ((rankingLaw theta) pi).toReal) Filter.atTop
      (nhds ((PMF.pure center) pi).toReal)) ∧
  (∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    ∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
      remaining.Nonempty →
      KR21Monoculture.expectedBestInSet (rankingLaw thetaH) value remaining ≤
      KR21Monoculture.expectedBestInSet (rankingLaw thetaA) value remaining) ∧
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
    thetaH < thetaA →
    KR21Monoculture.expectedBestInSet (rankingLaw thetaH) value Finset.univ <
    KR21Monoculture.expectedBestInSet (rankingLaw thetaA) value Finset.univ

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixA_theorem5_corrected_source_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 493-504

Verbatim source input

In Random Utility Models, the underlying candidate values x_i are perturbed by independent and identically distributed noise $\varepsilon_i \sim E$, and the perturbed values are ranked to produce π . Originally conceived in the psychology literature [23], this model has been well-studied over nearly a century, [7, 4, 9, 24, 16, 22], including more recently in the computer science and machine learning literature [2, 1, 19, 25, 13]. First, we must define a family of RUMs that satisfies the conditions of Definition 1. Assume without loss of generality that the noise distribution E has unit variance. Then, consider the family of RUMs parameterized by θ in which candidates are ranked according to $x_i + \varepsilon_i \theta$. By this definition, the standard deviation of the noise for a particular value of θ is simply $1/\theta$. Intuitively, larger values of θ reduce the effect of the noise, making the ranking more accurate. In Theorem 5 in Appendix A, we show as long as the noise distribution E has positive support on $(-\infty, \infty)$, this definition of F_θ meets the differentiability, asymptotic optimality, and monotonicity conditions in Definition 1. For distributions with finite support, many of our results can be generalized by relaxing strict inequalities in Definition 1 and Theorem 1 to weak inequalities.

[Next verbatim source excerpt]

[form-feed in source extraction]

Monotonicity: The removal of any elements does not alter the distribution of the remaining elements, meaning that the distribution of $\pi(-S)$ is equivalent to a RUM with $n - |S|$ elements. Thus, it suffices to show that for a RUM with positive support on $(-\infty, \infty)$, the probability of ranking the best candidate first strictly increases with θ .

Recall that by definition, the candidates are ranked according to $x_i + \varepsilon_i \theta$. The probability that x_1 is ranked first is

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 $\varepsilon_i$ 
 $\varepsilon_i$ 
 $\varepsilon_1$ 
 $\varepsilon_1$ 
 $\Pr x_1 +$ 
 $> \max_{2 \leq i \leq n} x_i +$ 
 $= \Pr$ 
 $> \max_{2 \leq i \leq n} x_i - x_1 +$ 
 $2 \leq i \leq n$ 
 $2 \leq i \leq n$ 
 $\theta$ 
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[U+0014 control character]
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 $= \Pr \varepsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i$ 
 $2 \leq i \leq n$ 
[U+0014 control character]
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 $= E \varepsilon_2, \dots, \varepsilon_n \Pr \varepsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i \mid \varepsilon_2, \dots, \varepsilon_n$ 
(A.1)
 $2 \leq i \leq n$ 
```

We want to show that (A.1) is increasing in θ . Intuitively, this is because as θ increases, the right hand side of the inequality inside the probability decreases. To prove this formally, it suffices to show that the subderivative of (A.1) with respect to θ only includes strictly positive numbers. First, we have

```
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 $\partial$ 
 $E \varepsilon_2, \dots, \varepsilon_n \Pr \varepsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i \mid \varepsilon_2, \dots, \varepsilon_n < R > 0$ 
 $2 \leq i \leq n$ 
 $\partial \theta$ 
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 $\partial$ 
 $\Pr \varepsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i \mid \varepsilon_2, \dots, \varepsilon_n < R > 0$ 
```

$\Leftrightarrow$
 $2 \leq i \leq n$
 $\partial \theta$
Let F and f be the cumulative density function and probability density function of E respectively. Then,

```
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 $\Pr \varepsilon_1 > \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i \mid \varepsilon_2, \dots, \varepsilon_n = 1 - F \max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i$ 
 $2 \leq i \leq n$ 
```

$2 \leq i \leq n$

Note that $F(\cdot)$ is strictly increasing (since f is assumed to have positive support on $(-\infty, \infty)$), so it suffices to show that

```
 $\partial$ 
 $\max_{2 \leq i \leq n} \theta(x_i - x_1) + \varepsilon_i < R < 0$ 
 $\partial \theta$ 
 $2 \leq i \leq n$ 
For any  $i$ ,
```

```
 $d$ 
 $\theta(x_i - x_1) + \varepsilon_i = x_i - x_1 < 0$ 
 $d \theta$ 
```

Thus, the subderivative of the max of such functions includes only strictly negative numbers, which completes

Expanded PaperInterface specification

```

∀ {n : ℕ} (f : ℝ → ℝ),
  Measurable f →
    (∀ (x : ℝ), 0 ≤ f x) →
      ∫- (x : ℝ), ENNReal.ofReal (f x) = 1 →
        ∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
          (∀ (d : Fin (n + 1)), value d.succ < value 0) →
            ∀ {theta : ℝ},
              0 < theta →
                (EconCSLib.measureProb
                  ((KR21Monoculture.sourceAppendixARestNoiseLaw n (KR21Monoculture.w11BaseNoiseLaw f)).prod
                    (KR21Monoculture.w11BaseNoiseLaw f))
                  fun z =>
                    EconCSLib.SocialChoice.Ranking.firstChoice
                      (EconCSLib.SocialChoice.Ranking.rankByScore fun i =>
                        value i + KR21Monoculture.sourceAppendixAProductNoise z i / theta) =
                      0) =
                ∫ (rest : Fin (n + 1) → ℝ),
                  EconCSLib.measureProb (KR21Monoculture.w11BaseNoiseLaw f) fun epsilon =>
                    ∀ (d : Fin (n + 1)),
                      theta * (value d.succ - value 0) + rest d <
                        epsilon → KR21Monoculture.sourceAppendixARestNoiseLaw n (KR21Monoculture.w11BaseNoiseLaw f)

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationA1_source_w11_iid_literal_event_conditionalTail_integral

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 190-198

Verbatim source input

The choice of which ranking mechanism to use leads to a game-theoretic setting: both firms know the accuracy parameters of the human evaluators (θ_H) and the algorithm (θ_A), and they must decide whether to

[form-feed in source extraction]
use a human evaluator or the algorithm. This choice introduces a subtlety: for many ranking models, a firm's rational behavior depends not only on the accuracy of the ranking mechanism, but also on the underlying candidate values $x_1, \dots, x_n$. Thus, to fully specify a firm's behavior, we assume that $x_1, \dots, x_n$ are drawn from a known joint distribution D . Our main results will hold for any D , meaning they apply even when the candidate values (but not their identities) are deterministically known.

[Next verbatim source excerpt]

B

3-candidate RUM Counterexamples

B.1

Violating Definition 2

Here, we provide a noise mode E , accuracy parameter θ , and candidate distribution D such that $UAH < UAA$. Choose the noise distribution E and accuracy parameter θ such that

[U+F8F1 delimiter glyph]
[U+F8F4 delimiter glyph]1
w.p. 2δ
 ε [U+F8F2 delimiter glyph]
 $= \theta$
w.p. $1 - \delta$
 θ [U+F8F4 delimiter glyph]
[U+F8F3 delimiter glyph]
-1 w.p. 2δ

Note that this distribution does not satisfy Definition 1 because it is neither differentiable nor supported on $(-\infty, \infty)$; however, we can provide a "smooth" approximation to this distribution by expressing it as the sum of arbitrarily tightly concentrated Gaussians with the same results.

15

[form-feed in source extraction]

We choose the candidate distribution D such that $x_1 - 1 > x_2 > x_3 > x_1 - 2$. For example,

7
4
1
 $x_2 =$
2
 $x_3 = \theta$
 $x_1 =$

Under this condition, assuming $x_3 = \theta$ without loss of generality,
 $UAH(\theta, \theta) - UAA(\theta, \theta) =$

[U+0001 control character]
62 3
 $\delta x_1 - 4\delta x_1 + 4\delta x_1 + 2\delta x_2 - 14\delta x_2 + 20\delta x_2 - 8x_2$
32
2

Notice that the lowest-power δ term is $-64x_2$. Therefore, for sufficiently small δ , this is negative. For example, plugging in the values given above with $\delta = .1$, $UAH(\theta, \theta) - UAA(\theta, \theta) \approx -0.00076$.

B.2

Expanded PaperInterface specification

```
have componentLaw := KR21Monoculture.appendixB1NoisePMF;
have rawNoiseLaw := EconCSLib.pmfProd (EconCSLib.pmfProd componentLaw componentLaw) componentLaw;
have rawRank := fun noise =>
  EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
    KR21Monoculture.appendixB1Value c +
    KR21Monoculture.appendixB1NoiseValue (KR21Monoculture.appendixB1NoiseTripleFunction noise c);
have rawLaw := PMF.map rawRank rawNoiseLaw;
(KR21Monoculture.appendixB1Value 0 = 7 / 4  $\wedge$ 
 KR21Monoculture.appendixB1Value 1 = 1 / 2  $\wedge$  KR21Monoculture.appendixB1Value 2 = 0)  $\wedge$ 
(KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.plusOne = 1  $\wedge$ 
 KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.zero = 0  $\wedge$ 
 KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.minusOne = -1)  $\wedge$ 
(componentLaw KR21Monoculture.AppendixB1NoiseAtom.plusOne).toReal = 1 / 20  $\wedge$ 
(componentLaw KR21Monoculture.AppendixB1NoiseAtom.zero).toReal = 9 / 10  $\wedge$ 
(componentLaw KR21Monoculture.AppendixB1NoiseAtom.minusOne).toReal = 1 / 20  $\wedge$ 
( $\forall$  (noise : KR21Monoculture.AppendixB1NoiseTriple),
  rawRank noise = KR21Monoculture.appendixB1DiscreteTripleRank noise)  $\wedge$ 
rawLaw = KR21Monoculture.appendixB1RankingPMF  $\wedge$ 
EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent rawLaw rawLaw
KR21Monoculture.appendixB1Value -
EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared rawLaw KR21Monoculture.appendixB1Value =
-(9749 / 12800000)  $\wedge$ 
~KR21Monoculture.Model.PrefersIndependentReranking rawLaw KR21Monoculture.appendixB1Value
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixB1_source_discrete_definition2_counterexample_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Review row 24

Source locator: cited publication, lines 190-198

Verbatim source input

The choice of which ranking mechanism to use leads to a game-theoretic setting: both firms know the accuracy parameters of the human evaluators (θ_H) and the algorithm (θ_A), and they must decide whether to

3

[form-feed in source extraction]

use a human evaluator or the algorithm. This choice introduces a subtlety: for many ranking models, a firm's rational behavior depends not only on the accuracy of the ranking mechanism, but also on the underlying candidate values $x_1, \dots, x_n$. Thus, to fully specify a firm's behavior, we assume that $x_1, \dots, x_n$ are drawn from a known joint distribution D . Our main results will hold for any D , meaning they apply even when the candidate values (but not their identities) are deterministically known.

[Next verbatim source excerpt]

Violating Definition 3

Next, we'll give a 3-candidate RUM for which $UAH < UHH$ does not hold in general. Consider the following 3-candidate example.

$x_1 = 3$
 $x_2 = 2$
 $x_3 = 0$
Choose ϵ and θ such that

[U+F8F1 delimiter glyph]
1
[U+F8F4 delimiter glyph]
[U+F8F4 delimiter glyph]
[U+F8F4 delimiter glyph]
[U+F8F2 delimiter glyph]
-1
 ϵ
=
 θ [U+F8F4 delimiter glyph]
10
[U+F8F4 delimiter glyph]
[U+F8F4 delimiter glyph]
[U+F8F3 delimiter glyph]
-10

w.p. 1- δ
2
w.p. 1- δ
2
w.p. 2δ
w.p. 2δ

Again, while this noise model doesn't satisfy Definition 1, we can approximate it arbitrarily closely with the sum of tightly concentrated Gaussians. Let the $\theta_A = 1.1\theta$ and $\theta_H = 0.9\theta$.

We will show that for these parameters, $UAH(\theta_A, \theta_H) > UHH(\theta_A, \theta_H)$, i.e., it is somehow better to choose after a better opponent than after a worse opponent. At a high level, the reasoning for this is as follows:

1. When choosing first, the only difference between the algorithm and the human evaluator is that the algorithm is more likely to choose x_2 than x_3 . Both strategies have identical probabilities of selecting x_1 .

2. When choosing second, the human evaluator's utility is higher when x_2 has already been chosen than when x_3 has already been chosen. This is because when x_2 is unavailable, the human evaluator is almost guaranteed to get x_1 ; when x_3 is unavailable, the human evaluator will choose x_2 with probability $\approx 1/4$.

Let τ and π be rankings generated by the algorithm and human evaluator respectively. First, we will show that

$\Pr[\tau_1 = x_1] = \Pr[\pi_1 = x_1]$

(B.1)

$\Pr[\tau_1 = x_2] > \Pr[\pi_1 = x_2]$

(B.2)

To do so, consider the realizations of $\epsilon_1, \epsilon_2, \epsilon_3$ that result in different rankings under θ_A and θ_H . In fact, the only set of realizations that result in different rankings are when $\epsilon_2/\theta = -1$ and $\epsilon_3/\theta = 1$. Thus, the algorithm and human evaluator always rank x_1 in the same position, conditioned on a realization, which proves (B.1); the only difference is that the algorithm sometimes ranks x_2 above x_3 when the human

16

[form-feed in source extraction]

evaluator does not. Moreover, whenever $\epsilon_1/\theta = -10$, x_2 is more strictly more likely to be ranked first under the algorithm than the human evaluator, which proves (B.2).

Next, we must show that when choosing second, the human evaluator is better off when x_2 is unavailable than when x_3 is unavailable. This is clearly true because for the human evaluator,

[U+0014 control character]
[U+0015 control character]
 $\epsilon_1 \theta_H$
 $\epsilon_3 \theta_H$
 $\Pr x_1 +$
 $> x_3 +$
 $\approx 1 - O(\delta)$
 θ
 θ
[U+0014 control character]
[U+0015 control character]

```

ε1 0H
ε3 0H
3
Pr x1 +
> x2 +
≈
0
0
4
Thus, conditioned on x2 being unavailable, the human evaluator gets utility ≈ 3, whereas when x3 is
unavailable, the human evaluator gets utility ≈ 2.75. Let u-i be the expected utility for the human evaluator
when xi is unavailable. Putting this together, we get
UAH (0A , 0H ) - UHH (0A , 0H )
=

3
X

(Pr[τ1 = xi ] - Pr[π1 = xi ])u-i

i=1

= (Pr[τ1 = x1 ] - Pr[π1 = x1 ])u-1 + (Pr[τ1 = x2 ] - Pr[π1 = x2 ])u-2 + (Pr[τ1 = x3 ] - Pr[π1 = x3 ])u-3
= (Pr[τ1 = x2 ] - Pr[π1 = x2 ])u-2 + (Pr[τ1 = x3 ] - Pr[π1 = x3 ])u-3
(Pr[τ1 = x1 ] = Pr[π1 = x1 ])
P3
P3
= (Pr[τ1 = x2 ] - Pr[π1 = x2 ])(u-2 - u-3 )
( i=1 Pr[τ1 = xi ] = i=1 Pr[π1 = xi ])
>0
The last step follows from (B.2) and because u-2 > u-3 .

```

Expanded PaperInterface specification

```

have componentLaw := KR21Monoculture.appendixB2NoisePMF;
have rawNoiseLaw := EconCSLib.pmfProd (EconCSLib.pmfProd componentLaw componentLaw) componentLaw;
have algorithmRank := fun noise =>
  EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
    KR21Monoculture.appendixB2Value c +
    10 / 11 * KR21Monoculture.appendixB2NoiseValue (KR21Monoculture.appendixB2NoiseTripleFunction noise c);
have humanRank := fun noise =>
  EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
    KR21Monoculture.appendixB2Value c +
    10 / 9 * KR21Monoculture.appendixB2NoiseValue (KR21Monoculture.appendixB2NoiseTripleFunction noise c);
have algorithmLaw := PMF.map algorithmRank rawNoiseLaw;
have humanLaw := PMF.map humanRank rawNoiseLaw;
(KR21Monoculture.appendixB2Value 0 = 3 ∧
  KR21Monoculture.appendixB2Value 1 = 2 ∧ KR21Monoculture.appendixB2Value 2 = 0) ∧
(KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusOne = 1 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusOne = -1 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusTen = 10 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusTen = -10) ∧
(componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusOne).toReal = 9 / 20 ∧
(componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusOne).toReal = 9 / 20 ∧
(componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusTen).toReal = 1 / 20 ∧
(componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusTen).toReal = 1 / 20 ∧
(11 / 10 > 9 / 10 ∧ 10 / 11 * (11 / 10) = 1 ∧ 10 / 9 * (9 / 10) = 1) ∧
(∀ (noise : KR21Monoculture.AppendixB2NoiseTriple),
  algorithmRank noise = KR21Monoculture.appendixB2AlgorithmDiscreteTripleRank noise) ∧
(∀ (noise : KR21Monoculture.AppendixB2NoiseTriple),
  humanRank noise = KR21Monoculture.appendixB2HumanDiscreteTripleRank noise) ∧
algorithmLaw = KR21Monoculture.appendixB2AlgorithmRankingPMF ∧
humanLaw = KR21Monoculture.appendixB2HumanRankingPMF ∧
EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent humanLaw algorithmLaw
  KR21Monoculture.appendixB2Value -
  EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent humanLaw humanLaw
  KR21Monoculture.appendixB2Value =
  567 / 3200000 ∧
  -KR21Monoculture.Model.PrefersWeakerCompetition algorithmLaw humanLaw
  KR21Monoculture.appendixB2Value

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixB2_source_discrete_definition3_counterexample_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 1040-1057

Verbatim source input

2. When choosing second, the human evaluator's utility is higher when x_2 has already been chosen than when x_3 has already been chosen. This is because when x_2 is unavailable, the human evaluator is almost guaranteed to get x_1 ; when x_3 is unavailable, the human evaluator will choose x_2 with probability $\approx 1/4$.

Let τ and π be rankings generated by the algorithm and human evaluator respectively. First, we will show that

$$\Pr[\tau_1 = x_1] = \Pr[\pi_1 = x_1]$$

(B.1)

$$\Pr[\tau_1 = x_2] > \Pr[\pi_1 = x_2]$$

(B.2)

To do so, consider the realizations of ε_1 , ε_2 , ε_3 that result in different rankings under θ_A and θ_H . In fact, the only set of realizations that result in different rankings are when $\varepsilon_2 / \theta = -1$ and $\varepsilon_3 / \theta = 1$. Thus, the algorithm and human evaluator always rank x_1 in the same position, conditioned on a realization, which proves (B.1); the only difference is that the algorithm sometimes ranks x_2 above x_3 when the human

Expanded PaperInterface specification

```
have componentLaw := KR21Monoculture.appendixB2NoisePMF;
have rawNoiseLaw := EconCSLib.pmfProd (EconCSLib.pmfProd componentLaw componentLaw) componentLaw;
have algorithmRank := fun noise =>
  EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
    KR21Monoculture.appendixB2Value c +
    10 / 11 * KR21Monoculture.appendixB2NoiseValue (KR21Monoculture.appendixB2NoiseTripleFunction noise c);
have humanRank := fun noise =>
  EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
    KR21Monoculture.appendixB2Value c +
    10 / 9 * KR21Monoculture.appendixB2NoiseValue (KR21Monoculture.appendixB2NoiseTripleFunction noise c);
have algorithmLaw := PMF.map algorithmRank rawNoiseLaw;
have humanLaw := PMF.map humanRank rawNoiseLaw;
(KR21Monoculture.appendixB2Value 0 = 3  $\wedge$ 
  KR21Monoculture.appendixB2Value 1 = 2  $\wedge$  KR21Monoculture.appendixB2Value 2 = 0)  $\wedge$ 
  (KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusOne = 1  $\wedge$ 
    KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusOne = -1  $\wedge$ 
    KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusTen = 10  $\wedge$ 
    KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusTen = -10)  $\wedge$ 
  (componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusOne).toReal = 9 / 20  $\wedge$ 
  (componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusOne).toReal = 9 / 20  $\wedge$ 
  (componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusTen).toReal = 1 / 20  $\wedge$ 
  (componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusTen).toReal = 1 / 20  $\wedge$ 
  algorithmLaw = KR21Monoculture.appendixB2AlgorithmRankingPMF  $\wedge$ 
  humanLaw = KR21Monoculture.appendixB2HumanRankingPMF  $\wedge$ 
  KR21Monoculture.firstChoiceProb algorithmLaw 0 = KR21Monoculture.firstChoiceProb humanLaw 0
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationB1_counterexample_first_choice_x1_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 1044-1105

Verbatim source input

Let τ and π be rankings generated by the algorithm and human evaluator respectively. First, we will

show that

$$\Pr[\tau_1 = x_1] = \Pr[\pi_1 = x_1]$$

(B.1)

$$\Pr[\tau_1 = x_2] > \Pr[\pi_1 = x_2]$$

(B.2)

To do so, consider the realizations of ε_1 , ε_2 , ε_3 that result in different rankings under θ_A and θ_H . In fact, the only set of realizations that result in different rankings are when $\varepsilon_2/\theta = -1$ and $\varepsilon_3/\theta = 1$. Thus, the algorithm and human evaluator always rank x_1 in the same position, conditioned on a realization, which proves (B.1); the only difference is that the algorithm sometimes ranks x_2 above x_3 when the human

16

[form-feed in source extraction]

evaluator does not. Moreover, whenever $\varepsilon_1/\theta = -10$, x_2 is more strictly more likely to be ranked first under the algorithm than the human evaluator, which proves (B.2).

Next, we must show that when choosing second, the human evaluator is better off when x_2 is unavailable than when x_3 is unavailable. This is clearly true because for the human evaluator,

[U+0014 control character]

[U+0015 control character]

$\varepsilon_1 \theta_H$

$\varepsilon_3 \theta_H$

Pr x_1 +

> x_3 +

$\approx 1 - 0(5)$

θ

θ

[U+0014 control character]

[U+0015 control character]

$\varepsilon_1 \theta_H$

$\varepsilon_3 \theta_H$

3

Pr x_1 +

> x_2 +

$\approx$

θ

θ

4

Thus, conditioned on x_2 being unavailable, the human evaluator gets utility ≈ 3 , whereas when x_3 is unavailable, the human evaluator gets utility ≈ 2.75 . Let u_i be the expected utility for the human evaluator when x_i is unavailable. Putting this together, we get

UAH (θ_A , θ_H) - UHH (θ_A , θ_H)

=

3

X

$$(\Pr[\tau_1 = x_i] - \Pr[\pi_1 = x_i])u_i$$

$i=1$

$$= (\Pr[\tau_1 = x_1] - \Pr[\pi_1 = x_1])u_1 + (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])u_2 + (\Pr[\tau_1 = x_3] - \Pr[\pi_1 = x_3])u_3$$

$$= (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])u_2 + (\Pr[\tau_1 = x_3] - \Pr[\pi_1 = x_3])u_3$$

$$(\Pr[\tau_1 = x_1] - \Pr[\pi_1 = x_1])$$

P3

P3

$$= (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])(u_2 - u_3)$$

$$(i=1 \Pr[\tau_1 = x_i] = i=1 \Pr[\pi_1 = x_i])$$

>0

The last step follows from (B.2) and because $u_2 > u_3$.

Expanded PaperInterface specification

have componentLaw := KR21Monoculture.appendixB2NoisePMF;

have rawNoiseLaw := EconCSLib.pmfProd (EconCSLib.pmfProd componentLaw componentLaw) componentLaw;

have algorithmRank := fun noise =>

EconCSLib.SocialChoice.Ranking.rankByScore fun c =>

KR21Monoculture.appendixB2Value c +

$10 / 11 * \text{KR21Monoculture.appendixB2NoiseValue} (\text{KR21Monoculture.appendixB2NoiseTripleFunction noise } c);$

have humanRank := fun noise =>

EconCSLib.SocialChoice.Ranking.rankByScore fun c =>

KR21Monoculture.appendixB2Value c +

$10 / 9 * \text{KR21Monoculture.appendixB2NoiseValue} (\text{KR21Monoculture.appendixB2NoiseTripleFunction noise } c);$

have algorithmLaw := PMF.map algorithmRank rawNoiseLaw;

have humanLaw := PMF.map humanRank rawNoiseLaw;

(KR21Monoculture.appendixB2Value 0 = 3 $\wedge$

KR21Monoculture.appendixB2Value 1 = 2 $\wedge$ KR21Monoculture.appendixB2Value 2 = 0) $\wedge$

(KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusOne = 1 $\wedge$

KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusOne = -1 $\wedge$

KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusTen = 10 $\wedge$

KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusTen = -10) $\wedge$

(componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusOne).toReal = 9 / 20 $\wedge$

(componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusOne).toReal = 9 / 20 $\wedge$

(componentLaw KR21Monoculture.AppendixB2NoiseAtom.plusTen).toReal = 1 / 20 $\wedge$

```
(componentLaw KR21Monoculture.AppendixB2NoiseAtom.minusTen).toReal = 1 / 20 ∧  
algorithmLaw = KR21Monoculture.appendixB2AlgorithmRankingPMF ∧  
humanLaw = KR21Monoculture.appendixB2HumanRankingPMF ∧  
KR21Monoculture.firstChoiceProb humanLaw 1 < KR21Monoculture.firstChoiceProb algorithmLaw 1
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationB2_counterexample_first_choice_x2_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 27

Source locator: cited publication, lines 1111-1147

Verbatim source input

C.1

Verifying Definition 2

By (2), we can equivalently show that for any θ , $UAH(\theta, \theta) > UAA(\theta, \theta)$. Let τ and π be the algorithmic and human-generated rankings respectively. Note that they're identically distributed because $\theta_A = \theta_H$. Define

(
 $\pi_1 \pi_1 6 = \tau_1$
 $Y,$
 π_2 otherwise
Note that $UAH(\theta, \theta) = E[Y]$ and $UAA(\theta, \theta) = E[\tau_2]$. We want to show that $UAH(\theta, \theta) - UAA(\theta, \theta) = E[xY - x\tau_2] > 0$. It is sufficient to show that for any k , $E[Y - \tau_2 \mid \tau_1 = x_k] > 0$. Let $X_i = x_i + \epsilon_i / \theta$. Note that for distinct i, j, k and $x_i > x_j$,
 $E[Y - \tau_2 \mid \tau_1 = x_k] > 0 \iff$

$Pr[\tau_2 = x_i \mid \tau_1 = x_k]$
 $Pr[Y = x_i \mid \tau_1 = x_k]$
>
 $Pr[Y = x_j \mid \tau_1 = x_k]$
 $Pr[\tau_2 = x_j \mid \tau_1 = x_k]$

$\iff Pr[Y = x_i \mid \tau_1 = x_k] > Pr[\tau_2 = x_i \mid \tau_1 = x_k]$
(numerator and denominator sum to 1)
 $\iff Pr[X_i > X_j] > Pr[X_i > X_j \mid X_k > X_i \cap X_k > X_j]$
 $\iff Pr[X_i > X_j] > EX_k [Pr[X_i > X_j \mid X_k = a, X_i < a, X_j < a]]$.
Thus, it suffices to show that for any a ,
 $Pr[X_i > X_j] > Pr[X_i > X_j \mid X_i < a, X_j < a]$.

(C.1)

Since $Pr[X_i > X_j] = \lim_{a \rightarrow \infty} Pr[X_i > X_j \mid X_i < a, X_j < a]$, it suffices to show that for all a ,

d
 $Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$,
(C.2)
 da

and that it is strictly positive for some a . In other words, the higher a is, the more likely i and j are to be correctly ordered. In Theorems 7 and 8, we show that (C.2) holds for both Laplacian and Gaussian noise

[Next verbatim source excerpt]

Finally, we show that (C.2) holds for both Laplacian and Gaussian noise.

Theorem 7. For any $a \in \mathbb{R}$ and $X_i = x_i + \sigma \epsilon_i$ where ϵ_i is Laplacian with unit variance,

d
 $Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$.
 da

Moreover, it is strictly positive for some a .

Proof. First, we must derive an expression for $Pr[X_i > X_j \mid X_i < a, X_j < a]$. Recall that the Laplace distribution parameterized by μ and λ has pdf

$f(x; \mu, \lambda) =$
and cdf

λ
 $\exp(-\lambda|x - \mu|)$
2

(
 $F(x; \mu, \lambda) =$

1
2 $\exp(-\lambda(\mu - x))$
1 - 1/2 $\exp(-\lambda(x - \mu))$

21

$x < \mu$
 $x \geq \mu$

[form-feed in source extraction]

Note that x_i and x_j be the respective means of X_i and X_j , with $x_i > x_j$. Because the Laplace distribution is piecewise defined, we must consider 3 cases and show that in all 3 cases, (C.2) holds. Note that

Ra
 $\int_a^\infty f(x; x_i, \lambda) F(x; x_j, \lambda) dx$
(C.3)

$Pr[X_i > X_j \mid X_i < a, X_j < a] = -\infty$

$\int_a^\infty f(a; x_i, \lambda) F(a; x_j, \lambda)$

Case 1: $a \leq x_j$.

Then, the numerator of (C.3) is

$\int_a^\infty$
 $\int_a^\infty$
 λ
1
 λa
 $\exp(-\lambda(x_i + x_j - 2x)) dx$
 $\exp(-\lambda(x_i - x)) \cdot \exp(-\lambda(x_j - x)) dx =$
2
4 $-\infty$
 $-\infty$ 2
 $\int_a^\infty$
 $\lambda \exp(-\lambda(x_i + x_j)) a$

```

exp(2λx) dx
=
4
-∞
λ exp(-λ(xi + xj )) 1
=
exp(2λa)
4
2λ
exp(-λ(xi + xj - 2a))
=
8
The denominator is
1
1
1
exp(-λ(xi - a)) · exp(-λ(xj - a)) = exp(-λ(xi + xj - 2a)).
2
2
4
Thus,
Pr[Xi > Xj | Xi < a, Xj < a] =

1
,
2

so its derivative is trivially nonnegative.
Case 2: xj < a ≤ xi .
Then, the numerator of (C.3) is
[U+0012 control character]
[U+0013 control character]
Z xj
Z a
λ
1
λ
1
exp (-λ(xi - x)) · exp(-λ(xj - x)) dx +
exp (-λ(xi - x)) 1 - exp(-λ(x - xj )) dx
2
2
-∞ 2
xj 2
Z a
Z a
exp(-λ(xi - xj )) λ
λ
+
exp(-λ(xi - x)) dx -
exp(-λ(xi - xj )) dx
=
8
2 xj
4 xj
exp(-λ(xi - xj )) λ 1
λ
+
(exp(-λ(xi - a)) - exp(-λ(xi - xj ))) - (a - xj ) exp(-λ(xi - xj ))
8
2[U+0012 control character]λ
4
[U+0013 control character]
1
3 λ
= exp(-λ(xi - a)) -
+ (a - xj ) exp(-λ(xi - xj ))
2
8
4
=

The denominator is
[U+0012 control character]
[U+0013 control character]
1
1
1
1
1 - exp(-λ(a - xj )) · exp(λ(xj - a)) = exp(-λ(xi - a)) - exp(-λ(xi - xj ))
2
2
2
4
We can factor out 41 exp(-λ(xi - xj )) from both, so
2 exp(λ(a - xj )) - 23 + λ(a - xj )
Pr[Xi > Xj | Xi < a, Xj < a] =
2 exp(λ(a - xj )) - 1
=

[U+0001 control character]

[U+0001 control character]
2 exp(λ(a - xj )) - 1 - 21 + λ(a - xj )
2 exp(λ(a - xj )) - 1
=1-

```

```

1
2 + λ(a - xj )
2 exp(λ(a - xj )) - 1
22

[form-feed in source extraction]
Thus,
d
Pr[Xi > Xj | Xi < a, Xj < a] > 0
da
1
d
2 + λ(a - xj )

$$\frac{2 \exp(\lambda(a - x_j)) - 1}{2 \exp(\lambda(a - x_j)) - 1} > 0$$

da 2 exp(λ(a - xj )) - 1
[U+0012 control character]
[U+0013 control character]
1

$$\frac{(2 \exp(\lambda(a - x_j)) - 1)\lambda}{2 \exp(\lambda(a - x_j)) - 1} < 2$$

+ λ(a - xj ) 2λ exp(λ(a - xj ))
2
[U+0012 control character]
[U+0013 control character]
1

$$2 - \exp(-\lambda(a - x_j)) < 2$$

+ λ(a - xj )
2

$$1 - \exp(-\lambda(a - x_j)) < 2\lambda(a - x_j)$$


$$\exp(-\lambda(a - x_j)) > 1 - 2\lambda(a - x_j)$$

This is true because λ(a - xj ) > 0, and for z > 0,
exp(-z) > 1 - z > 1 - 2z.
Case 3: a > xi .
Then, the numerator of (C.3) is
[U+0012 control character]
[U+0013 control character]
Z xj
Z xi
λ
1
λ
1
exp (-λ(xi - x)) · exp(-λ(xj - x)) dx +
exp (-λ(xi - x)) 1 - exp(-λ(x - xj )) dx
2
2
-∞ 2
xj 2
[U+0012 control character]
[U+0013 control character]
Z a
1
λ
exp(-λ(x - xi )) 1 - exp(-λ(x - xj )) dx
+
2
2
xi
[U+0012 control character]
[U+0013 control character]
Z
1
3 λ
1
λ a
= -
+ (xi - xj ) exp(-λ(xi - xj )) + (1 - exp(-λ(a - xi ))) -
exp(-λ(2x - xi - xj )) dx
2
8
4
2
4 xi
[U+0012 control character]
[U+0013 control character]
3 λ
1
=1-
+ (xi - xj ) exp(-λ(xi - xj )) - exp(-λ(a - xi ))
8
4
2
1
+ exp(λ(xi + xj ))(exp(-2λa) - exp(-2λxi ))
8 [U+0012 control character]
[U+0013 control character]
1
1
1 λ
+ (xi - xj ) exp(-λ(xi - xj )) - exp(-λ(a - xi )) + exp(-λ(2a - xi - xj ))
=1-
2
4

```

```

2
8
The denominator is
[U+0012 control character]
[U+0013 control character][U+0012 control character]
[U+0013 control character]
1
1
1 - exp(-λ(t - xi ))
1 - exp(-λ(t - xj ))
2
2
1
1
1
= 1 - exp(-λ(a - xi )) - exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
2
2
4
Thus,
Pr[Xi > Xj | Xi < a, Xj < a]
[U+0001 control character]
1 - 21 + 14λ(xi - xj ) exp(-λ(xi - xj )) - 12 exp(-λ(a - xi )) + 81 exp(-λ(2a - xi - xj ))
=
1 - 12 exp(-λ(a - xi )) - 21 exp(-λ(a - xj )) + 14 exp(-λ(2a - xi - xj ))
8 - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )) - 4 exp(-λ(a - xi )) + exp(-λ(2a - xi - xj ))
α
4 - 2 exp(-λ(a - xi )) - 2 exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
23

[form-feed in source extraction]
We're interested in
d
Pr[Xi > Xj | Xi < a, Xj < a] > 0
da
⇔ (4 - 2 exp(-λ(a - xi )) - 2 exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
· (4λ exp(-λ(a - xi )) - 2λ exp(-λ(2a - xi - xj )))
> (8 - 4 exp(-λ(a - xi )) + exp(-λ(2a - xi - xj )) - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )))
· (2λ exp(-λ(a - xi )) + 2λ exp(-λ(a - xj )) - 2λ exp(-λ(2a - xi - xj )))
⇔ 16 exp(-λ(a - xi )) - 8 exp(-λ(2a - xi - xj )) - 8 exp(-2λ(a - xi )) + 4 exp(-λ(3a - 2xi - xj ))
- 8 exp(-λ(2a - xi - xj )) + 4 exp(-λ(3a - xi - 2xj )) + 4 exp(-λ(3a - 2xi - xj ))
- 2 exp(-2λ(2a - xi - xj ))
> 16 exp(-λ(a - xi )) + 16 exp(-λ(a - xj )) - 16 exp(-λ(2a - xi - xj ))
- 8 exp(-2λ(a - xi )) - 8 exp(-λ(2a - xi - xj )) + 8 exp(-λ(3a - 2xi - xj ))
+ 2 exp(-λ(3a - 2xi - xj )) + 2 exp(-λ(3a - xi - 2xj )) - 2 exp(-2λ(2a - xi - xj ))
- 2(4 + 2λ(xi - xj )) exp(-λ(a - xj )) - 2(4 + 2λ(xi - xj )) exp(-λ(a + xi - 2xj ))
+ 2(4 + 2λ(xi - xj )) exp(-2λ(a - xj ))
⇔ exp(-λ(3a - xi - 2xj ))
> 8 exp(-λ(a - xj )) - 4 exp(-λ(2a - xi - xj )) + exp(-λ(3a - 2xi - xj ))
- (4 + 2λ(xi - xj )) exp(-λ(a - xj )) - (4 + 2λ(xi - xj )) exp(-λ(a + xi - 2xj ))
+ (4 + 2λ(xi - xj )) exp(-2λ(a - xj ))
⇔ exp(-λ(2a - xi - xj ))
> 8 - 4 exp(-λ(a - xi )) + exp(-λ(2a - 2xi ))
- (4 + 2λ(xi - xj )) - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )) + (4 + 2λ(xi - xj )) exp(-λ(a - xj ))
⇔ exp(-λ(2a - xi - xj )) - 8 + 4 exp(-λ(a - xi )) - exp(-2λ(a - xi ))
+ (4 + 2λ(xi - xj ))(1 + exp(-λ(xi - xj ))) - (4 + 2λ(xi - xj )) exp(-λ(a - xj ))
> 0

(C.4)

Note that for any z ≥ 0, we have
(4 + 2z)(1 + e-z) - 8 ≥ 0 ⇔ (2 + z)(1 + e-z) ≥ 4
⇔ z + 2e-z + ze-z ≥ 2
For z = 0, this holds with equality, and the left hand side is increasing since
d
z + 2e-z + ze-z ≥ 0 ⇔ 1 - 2e-z + e-z - ze-z ≥ 0
dx
⇔ 1 ≥ e-z + ze-z
1
⇔
≥ e-z
1+z
⇔ 1 + z ≤ ez
Therefore, choosing z = λ(xi - xj ) and plugging back to (C.4), we have
exp(-λ(2a - xi - xj )) - 8 + 4 exp(-λ(a - xi )) - exp(-2λ(a - xi ))
+ (4 + 2λ(xi - xj ))(1 + exp(-λ(xi - xj ))) - (4 + 2λ(xi - xj )) exp(-λ(a - xj )) > 0
⇔ exp(-λ(2a - xi - xj )) + 4 exp(-λ(a - xi )) - exp(-2λ(a - xi )) - (4 + 2λ(xi - xj )) exp(-λ(a - xj )) > 0
⇔ exp(-λ(a - xj )) + 4 - exp(-λ(a - xi )) - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )) > 0
⇔ 4(1 - exp(-λ(xi - xj ))) + exp(-λ(a - xi ))(exp(-λ(xi - xj )) - 1) - 2λ(xi - xj ) exp(-λ(xi - xj )) > 0
⇔ (4 - exp(-λ(a - xi )))(1 - exp(-λ(xi - xj ))) - 2λ(xi - xj ) exp(-λ(xi - xj )) > 0
⇔ 3(1 - exp(-λ(xi - xj ))) - 2λ(xi - xj ) exp(-λ(xi - xj )) > 0
24
(exp(-λ(a - xi )) < 1)

[form-feed in source extraction]
Again letting z = λ(xi - xj ), this is true if and only if
3(1 - e-z) > 2ze-z ⇔ 3(ez - 1) > 2z
⇔ 3ez > 3 + 2z
which is true because ez > 1 + z for z > 0. This completes the proof for Case 3.
As a result, we have that
d
Pr[Xi > Xj | Xi < a, Xj < a] ≥ 0
da

```

for all a , with strict inequality for some a , which proves the theorem.

[Next verbatim source excerpt]

Theorem 8. For any $a \in \mathbb{R}$ and $X_i = x_i + \sigma \varepsilon_i$ where $\varepsilon_i \sim N(0, 1)$,

$\Pr[X_i > X_j \mid X_i < a, X_j < a] > 0$.

$\frac{d}{da}$

Proof. Assume $\sigma = 1/2$. This

$\sqrt{\cdot}$ is without loss of generality because for any instance with arbitrary σ , there is an instance with $\sigma = 1/2$ that yields the same distribution over rankings (simply by scaling all item values by σ/σ_0). First, we have

$\Pr[X_i = x] \Pr[X_j < x] dx$
 $\Pr[X_i > X_j \mid X_i < a, X_j < a] = -\infty$
 $\Pr[X_i < a] \Pr[X_j < a]$

$\frac{d}{da}$
 $\exp(-(x - x_i)^2)/\pi \cdot (1 + \operatorname{erf}(x - x_j))/2 dx$
 $=$

$(1 + \operatorname{erf}(a - x_i))/2 \cdot (1 + \operatorname{erf}(a - x_j))/2$

$\frac{d}{da}$
 $\exp(-(x - x_i)^2)(1 + \operatorname{erf}(x - x_j)) dx$
 $\frac{d}{da}$

$= \frac{d}{da} \int_{-\infty}^{\infty} (1 + \operatorname{erf}(a - x_i)) \cdot (1 + \operatorname{erf}(a - x_j))$

π
The derivative with respect to a is positive if and only if
 $(1 + \operatorname{erf}(a - x_i))(1 + \operatorname{erf}(a - x_j)) \exp(-(a - x_i)^2)(1 + \operatorname{erf}(a - x_j))$

$\frac{d}{da}$
 $>$
 $\exp(-(x - x_i)^2)(1 + \operatorname{erf}(x - x_j)) dx$
 $=$

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$\frac{d}{da}$
 $\cdot \sqrt{(1 + \operatorname{erf}(a - x_i)) \exp(-(a - x_j)^2) + (1 + \operatorname{erf}(a - x_j)) \exp(-(a - x_i)^2)}$
 π

(C.5)

Let $t = a - x_i$ and $\delta = x_i - x_j$. Then, using the fact that

$\frac{d}{da}$

$\frac{d}{da} a - x_i$

$\frac{d}{da}$

$\exp(-(x - x_i)^2)(1 + \operatorname{erf}(x - x_j)) dx =$
 $=$

$\frac{d}{da}$

$\frac{d}{da} a - x_i$

$\exp(-x^2) dx +$
 $=$

$\frac{d}{da}$
 $=$

π
 $(1 + \operatorname{erf}(a - x_i)) +$
 $\frac{d}{da}$

$\exp(-x^2) \operatorname{erf}(x + \delta) dx$

$=$

$\frac{d}{da} a - x_i$

$\exp(-x^2) \operatorname{erf}(x + \delta) dx,$

$=$

(C.5) becomes

$\frac{d}{da}$

$\frac{d}{da}$

$\frac{d}{da} t$

π

$(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)$

π

$\cdot$

$>$

$(1$

$+ \operatorname{erf}(t))$

$+ \exp(-x^2) \operatorname{erf}(x + \delta) dx$

$\frac{d}{da} (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2)$

$\frac{d}{da}$

$\frac{d}{da} t$

$(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)$

$\frac{d}{da}$

$\sqrt{\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)}} \exp(-t^2) > 0$
 $\lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) = 0$
(C.6)
To show that this is true, we will use the fact that $f(t) > 0$ whenever the following conditions are met:
1. $f(t)$ is continuous and differentiable everywhere
25

[form-feed in source extraction]

2. $\lim_{t \rightarrow -\infty} f(t) = 0$

$\frac{d}{dt} f(t) > 0$
3. $\frac{d}{dt} f(t) > 0$

We'll show that these conditions hold for the LHS of (C.6).

$\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) = 0$

$\frac{d}{dt} f(t)$

$\lim$

$\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) = 0$

$= \lim$

$\exp(-x^2) \operatorname{erf}(x + \delta) dx$

$-\infty$

(C.7)

Observe that both the numerator and denominator of (C.7) are positive, so this limit must be at least 0.

We can upper bound it by

$\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\leq$
 $\lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $= \lim_{t \rightarrow -\infty} \frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)}$
 $\lim_{t \rightarrow -\infty}$

$t \rightarrow -\infty$

$= 0$

Thus, the limit is 0. Now, we must show that the derivative is positive. The derivative is

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[U+0014 control character]

$\frac{d}{dt} f(t)$

$\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

(C.8)
Taking this derivative and factoring out

$\sqrt{\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)}} \exp(4t^2)$

$\frac{2(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta)) \exp(4t^2)}{2}$

$\pi (\operatorname{erf}(t) + 1) e^{t^2} + (\operatorname{erf}(\delta + t) + 1) e^{(\delta + t)^2}$

[U+0001 control character]2 ,

we get that (C.8) is positive if and only if

$\sqrt{\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)}} \exp(4t^2)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$
 $\frac{d}{dt} \left(\frac{1 + \operatorname{erf}(t)}{1 + \operatorname{erf}(t + \delta)} \exp(-t^2) \right)$

$$1 + \operatorname{erf}(t)$$

$$\Leftrightarrow \delta \pi \exp((t + \delta)^2)(1 + \operatorname{erf}(t)) +$$

$$- \exp(2\delta t + t^2) > 0$$

$$1 + \operatorname{erf}(t + \delta)$$

$$\sqrt{1 + \operatorname{erf}(t)}$$

$$\Leftrightarrow \delta \pi \exp(t^2)(1 + \operatorname{erf}(t)) + \exp(-2\delta t - t^2)$$

$$- 1 > 0$$

$$1 + \operatorname{erf}(t + \delta)$$

[U+0014 control character]

[U+0015 control character]

$$\sqrt{\exp(-2\delta t - \delta^2)}$$

$$\Leftrightarrow (1 + \operatorname{erf}(t)) \delta \pi \exp(t^2) +$$

$$- 1 > 0$$

$$1 + \operatorname{erf}(t + \delta)$$

[U+0015 control character]

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$$1 + \operatorname{erf}(t) \sqrt{\exp(-(t + \delta)^2)}$$

$$\Leftrightarrow$$

$$\delta$$

$$\pi$$

$$+$$

$$- 1 > 0$$

(C.9)

$$\exp(-t^2)$$

$$1 + \operatorname{erf}(t + \delta)$$

Define

$$g(t),$$

$$1 + \operatorname{erf}(t)$$

$$\cdot \exp(-t^2)$$

Then, (C.9) is

[U+0014 control character]

$$\sqrt{g(t) \delta \pi +}$$

[U+0015 control character]

$$1$$

$$- 1 > 0$$

$$g(t + \delta)$$

$$\sqrt{1}$$

$$1$$

$$\Leftrightarrow$$

$$-$$

$$< \delta \pi$$

$$g(t) g(t + \delta)$$

$$2\delta$$

[form-feed in source extraction]

By the Mean Value Theorem,

$$\frac{1}{d} \frac{1}{dt}$$

$$= -\delta$$

$$g(t) g(t + \delta)$$

$$\frac{d}{dt} g(t) \big|_{t=t^*}$$

for some $t \leq t^* \leq t + \delta$. Thus, it suffices to show that

$$\sqrt{\frac{d}{dt} \frac{1}{dt} g(t)}$$

$$> -\pi$$

(C.10)

for all t . To do this, consider Mills Ratio [17]

$$Z \infty$$

$$2$$

$$R(t), \exp(t/2)$$

$$\exp(-x^2/2) dx.$$

Note that this is quite similar in functional form to $g(t)$, and with some manipulation, we can relate the two:

$$Z \infty$$

$$2$$

$$R(t) = \exp(t/2)$$

$$\exp(-x^2/2) dx$$

$$t$$

$$Z \infty$$

$$\sqrt{R(2t) = \exp(t^2) \sqrt{\exp(-x^2/2) dx}}$$

$$2t$$

$$Z \infty$$

$$\sqrt{2}$$

$$= 2 \exp(t)$$

$$\exp(-x^2) dx$$


```

0 < gaussianDenominator ∧
((gaussian.prod gaussian)
  {epsilon |
    xi + epsilon.1 / theta < a ∧
    xj + epsilon.2 / theta < a ∧ xj + epsilon.2 / theta < xi + epsilon.1 / theta}).toReal /
  gaussianDenominator <
((gaussian.prod gaussian) {epsilon | xj + epsilon.2 / theta < xi + epsilon.1 / theta}).toReal

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixC1_source_pairwise_full_events

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 1139-1147

Verbatim source input

(C.1)
Since $\Pr[X_i > X_j] = \lim_{a \rightarrow -\infty} \Pr[X_i > X_j \mid X_i < a, X_j < a]$, it suffices to show that for all a ,
d
 $\Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$,
(C.2)
da
and that it is strictly positive for some a . In other words, the higher a is, the more likely i and j are to be correctly ordered. In Theorems 7 and 8, we show that (C.2) holds for both Laplacian and Gaussian noise

[Next verbatim source excerpt]

Finally, we show that (C.2) holds for both Laplacian and Gaussian noise.
Theorem 7. For any $a \in \mathbb{R}$ and $X_i = x_i + \sigma \varepsilon_i$ where ε_i is Laplacian with unit variance,
d
 $\Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$.
da
Moreover, it is strictly positive for some a .
Proof. First, we must derive an expression for $\Pr[X_i > X_j \mid X_i < a, X_j < a]$. Recall that the Laplace distribution parameterized by μ and λ has pdf
f (x; μ , λ) =
and cdf

$$\frac{\lambda}{2} \exp(-\lambda|x - \mu|)$$
$$F(x; \mu, \lambda) = \begin{cases} \frac{1}{2} \exp(-\lambda(\mu - x)) & x < \mu \\ 1 - \frac{1}{2} \exp(-\lambda(x - \mu)) & x \geq \mu \end{cases}$$

[form-feed in source extraction]
Note that x_i and x_j be the respective means of X_i and X_j , with $x_i > x_j$. Because the Laplace distribution is piecewise defined, we must consider 3 cases and show that in all 3 cases, (C.2) holds. Note that

Ra
f (x; x_i , λ)F (x; x_j , λ) dx
(C.3)
 $\Pr[X_i > X_j \mid X_i < a, X_j < a] = \frac{\int_{-\infty}^a f(a; x_i , λ)F(a; x_j , λ) dx}{\int_{-\infty}^a f(x; x_i , λ)F(x; x_j , λ) dx}$
Case 1: $a \leq x_j$.
Then, the numerator of (C.3) is
$$\int_{-\infty}^a \frac{\lambda}{2} \exp(-\lambda(x_i + x_j - 2x)) \cdot \exp(-\lambda(x_j - x)) dx = \frac{\lambda}{2} \int_{-\infty}^a \exp(-\lambda(x_i + x_j - x)) dx = \frac{\lambda}{2} \exp(-\lambda(x_i + x_j - 2a))$$

The denominator is
$$\int_{-\infty}^a \frac{\lambda}{2} \exp(-\lambda(x_i - a)) \cdot \exp(-\lambda(x_j - a)) dx = \exp(-\lambda(x_i + x_j - 2a)).$$

Thus,
$$\Pr[X_i > X_j \mid X_i < a, X_j < a] = \frac{1}{2}$$

so its derivative is trivially nonnegative.

$$4 - 2 \exp(-\lambda(a - x_i)) - 2 \exp(-\lambda(a - x_j)) + \exp(-\lambda(2a - x_i - x_j))$$

23

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We're interested in

d

$$\Pr[X_i > X_j \mid X_i < a, X_j < a] > 0$$

da

$$\begin{aligned} & \Leftrightarrow (4 - 2 \exp(-\lambda(a - x_i)) - 2 \exp(-\lambda(a - x_j)) + \exp(-\lambda(2a - x_i - x_j))) \\ & \cdot (4 \exp(-\lambda(a - x_i)) - 2 \exp(-\lambda(2a - x_i - x_j))) \\ & > (8 - 4 \exp(-\lambda(a - x_i)) + \exp(-\lambda(2a - x_i - x_j))) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(x_i - x_j))) \\ & \cdot (2 \exp(-\lambda(a - x_i)) + 2 \exp(-\lambda(a - x_j)) - 2 \exp(-\lambda(2a - x_i - x_j))) \\ & \Leftrightarrow 16 \exp(-\lambda(a - x_i)) - 8 \exp(-\lambda(2a - x_i - x_j)) - 8 \exp(-2 \exp(-\lambda(a - x_i))) + 4 \exp(-\lambda(3a - 2x_i - x_j)) \\ & - 8 \exp(-\lambda(2a - x_i - x_j)) + 4 \exp(-\lambda(3a - x_i - 2x_j)) + 4 \exp(-\lambda(3a - 2x_i - x_j)) \\ & - 2 \exp(-2 \exp(-\lambda(a - x_i))) \\ & > 16 \exp(-\lambda(a - x_i)) + 16 \exp(-\lambda(a - x_j)) - 16 \exp(-\lambda(2a - x_i - x_j)) \\ & - 8 \exp(-2 \exp(-\lambda(a - x_i))) - 8 \exp(-\lambda(2a - x_i - x_j)) + 8 \exp(-\lambda(3a - 2x_i - x_j)) \\ & + 2 \exp(-\lambda(3a - 2x_i - x_j)) + 2 \exp(-\lambda(3a - x_i - 2x_j)) - 2 \exp(-2 \exp(-\lambda(a - x_i))) \\ & - 2(4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) - 2(4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a + x_i - 2x_j)) \\ & + 2(4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-2 \exp(-\lambda(a - x_j))) \\ & \Leftrightarrow \exp(-\lambda(3a - x_i - 2x_j)) \\ & > 8 \exp(-\lambda(a - x_j)) - 4 \exp(-\lambda(2a - x_i - x_j)) + \exp(-\lambda(3a - 2x_i - x_j)) \\ & - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a + x_i - 2x_j)) \\ & + (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-2 \exp(-\lambda(a - x_j))) \\ & \Leftrightarrow \exp(-\lambda(2a - x_i - x_j)) \\ & > 8 - 4 \exp(-\lambda(a - x_i)) + \exp(-\lambda(2a - 2x_i)) \\ & - (4 + 2 \exp(-\lambda(x_i - x_j))) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(x_i - x_j)) + (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) \\ & \Leftrightarrow \exp(-\lambda(2a - x_i - x_j)) - 8 + 4 \exp(-\lambda(a - x_i)) - \exp(-2 \exp(-\lambda(a - x_i))) \\ & + (4 + 2 \exp(-\lambda(x_i - x_j)))(1 + \exp(-\lambda(x_i - x_j))) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) \\ & > 0 \end{aligned}$$

(C.4)

Note that for any $z \geq 0$, we have

$$(4 + 2z)(1 + e^{-z}) - 8 \geq 0 \Leftrightarrow (2 + z)(1 + e^{-z}) \geq 4$$

$$\Leftrightarrow z + 2e^{-z} + ze^{-z} \geq 2$$

For $z = 0$, this holds with equality, and the left hand side is increasing since

d

$$z + 2e^{-z} + ze^{-z} \geq 0 \Leftrightarrow 1 - 2e^{-z} + e^{-z} - ze^{-z} \geq 0$$

dx

$$\Leftrightarrow 1 \geq e^{-z} + ze^{-z}$$

1

$\Leftrightarrow$

$$\geq e^{-z}$$

1+z

$$\Leftrightarrow 1 + z \leq e^z$$

Therefore, choosing $z = \lambda(x_i - x_j)$ and plugging back to (C.4), we have

$$\begin{aligned} & \exp(-\lambda(2a - x_i - x_j)) - 8 + 4 \exp(-\lambda(a - x_i)) - \exp(-2 \exp(-\lambda(a - x_i))) \\ & + (4 + 2 \exp(-\lambda(x_i - x_j)))(1 + \exp(-\lambda(x_i - x_j))) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) > 0 \\ & \Leftrightarrow \exp(-\lambda(2a - x_i - x_j)) + 4 \exp(-\lambda(a - x_i)) - \exp(-2 \exp(-\lambda(a - x_i))) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(a - x_j)) > 0 \\ & \Leftrightarrow \exp(-\lambda(a - x_j)) + 4 - \exp(-\lambda(a - x_i)) - (4 + 2 \exp(-\lambda(x_i - x_j))) \exp(-\lambda(x_i - x_j)) > 0 \\ & \Leftrightarrow 4(1 - \exp(-\lambda(x_i - x_j))) + \exp(-\lambda(a - x_i))(\exp(-\lambda(x_i - x_j)) - 1) - 2 \exp(-\lambda(x_i - x_j)) \exp(-\lambda(x_i - x_j)) > 0 \\ & \Leftrightarrow (4 - \exp(-\lambda(a - x_i)))(1 - \exp(-\lambda(x_i - x_j))) - 2 \exp(-\lambda(x_i - x_j)) \exp(-\lambda(x_i - x_j)) > 0 \\ & \Leftrightarrow 3(1 - \exp(-\lambda(x_i - x_j))) - 2 \exp(-\lambda(x_i - x_j)) \exp(-\lambda(x_i - x_j)) > 0 \end{aligned}$$

24

$$(\exp(-\lambda(a - x_i))) < 1)$$

[form-feed in source extraction]

Again letting $z = \lambda(x_i - x_j)$, this is true if and only if

$$3(1 - e^{-z}) > 2ze^{-z} \Leftrightarrow 3(e^z - 1) > 2z$$

$$\Leftrightarrow 3ez > 3 + 2z$$

which is true because $ez > 1 + z$ for $z > 0$. This completes the proof for Case 3.

As a result, we have that

d

$$\Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$$

da

for all a , with strict inequality for some a , which proves the theorem.

[Next verbatim source excerpt]

Theorem 8. For any $a \in \mathbb{R}$ and $X_i = x_i + \sigma \epsilon_i$ where $\epsilon_i \sim N(0, 1)$,

d

$$\Pr[X_i > X_j \mid X_i < a, X_j < a] > 0.$$

da

$\sqrt{\cdot}$

0

Proof. Assume $\sigma = 1/2$. This

$\sqrt{\cdot}$ is without loss of generality because for any instance with arbitrary σ , there is an instance with $\sigma = 1/2$ that yields the same distribution over rankings (simply by scaling all item values by σ/σ_0). First, we have

Ra

$$\Pr[X_i = x] \Pr[X_j < x] dx$$

$$\Pr[X_i > X_j \mid X_i < a, X_j < a] = -\infty$$

$$\Pr[X_i < a] \Pr[X_j < a]$$

Ra

$\sqrt{\cdot}$

$$\exp(-(x - x_i)^2) / \pi \cdot (1 + \operatorname{erf}(x - x_j)) / 2 dx$$

$-\infty$

$=$

$$(1 + \operatorname{erf}(a - x_i)) / 2 \cdot (1 + \operatorname{erf}(a - x_j)) / 2$$

Ra

$$\exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) dx$$

2

$$\begin{aligned}
&= \sqrt{-\infty} \\
&(1 + \operatorname{erf}(a - x_i)) \cdot (1 + \operatorname{erf}(a - x_j)) \\
&\pi \\
&\text{The derivative with respect to } a \text{ is positive if and only if} \\
&(1 + \operatorname{erf}(a - x_i))(1 + \operatorname{erf}(a - x_j)) \exp(-(a - x_i)^2)(1 + \operatorname{erf}(a - x_j)) \\
&\int_a^\infty \\
&> \\
&\exp(-(x - x_i)^2)(1 + \operatorname{erf}(x - x_j)) \, dx \\
&-\infty \\
&[\text{U+0001 control character}] \\
&2 \\
&\cdot \sqrt{(1 + \operatorname{erf}(a - x_i)) \exp(-(a - x_j)^2) + (1 + \operatorname{erf}(a - x_j)) \exp(-(a - x_i)^2)} \\
&\pi \\
&(\text{C.5})
\end{aligned}$$

Let $t = a - x_i$ and $\delta = x_i - x_j$. Then, using the fact that

$$\begin{aligned}
&\int_a^\infty \\
&2 \\
&\exp(-(x - x_i)^2)(1 + \operatorname{erf}(x - x_j)) \, dx = \\
&-\infty \\
&2 \\
&\int_a^\infty \\
&\exp(-x^2) \, dx + \\
&-\infty \\
&\sqrt{ \\
&= \\
&\pi \\
&(1 + \operatorname{erf}(a - x_i)) + \\
&2
\end{aligned}$$

$$\begin{aligned}
&\exp(-x^2) \operatorname{erf}(x + \delta) \, dx \\
&-\infty \\
&\int_a^\infty \\
&\exp(-x^2) \operatorname{erf}(x + \delta) \, dx, \\
&-\infty
\end{aligned}$$

(C.5) becomes

$$\begin{aligned}
&\sqrt{ \\
&\int_t^\infty \\
&\pi \\
&(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \\
&\pi \\
&\cdot \\
&> \\
&(1 \\
&+ \\
&\operatorname{erf}(t)) \\
&+ \\
&\exp(-x^2) \operatorname{erf}(x + \delta) \, dx \\
&2 (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) \\
&2 \\
&-\infty \\
&\int_t^\infty \\
&(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \\
&2 \\
&\sqrt{ \\
&\Leftrightarrow \\
&- \\
&(1 \\
&+ \\
&\operatorname{erf}(t)) \\
&- \\
&\exp(-x^2) \operatorname{erf}(x + \delta) \, dx > 0 \\
&(1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) \\
&\pi \quad -\infty \\
&(\text{C.6})
\end{aligned}$$

To show that this is true, we will use the fact that $f(t) > 0$ whenever the following conditions are met:

1. $f(t)$ is continuous and differentiable everywhere

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2. $\lim_{t \rightarrow -\infty} f(t) = 0$
- d
3. $\frac{dt}{f(t)} > 0$

We'll show that these conditions hold for the LHS of (C.6).

$$\begin{aligned}
&(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \\
&2 \\
&- (1 + \operatorname{erf}(t)) - \sqrt{ \\
&t \rightarrow -\infty (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2)
\end{aligned}$$

$1 + \operatorname{erf}(t + \delta)$
 Define
 $g(t)$,
 $1 + \operatorname{erf}(t)$
 $\cdot$
 $\exp(-t^2)$
 Then, (C.9) is
 [U+0014 control character]
 $\sqrt{g(t) \delta \pi +}$
 [U+0015 control character]
 $\frac{1}{-1 > 0}$
 $g(t + \delta)$
 $\sqrt{\frac{1}{1}}$
 $\Leftrightarrow$
 $-\frac{< \delta \pi}{g(t) g(t + \delta)}$
 $\frac{26}{26}$

[form-feed in source extraction]
 By the Mean Value Theorem,
 $\frac{1}{d} \frac{1}{d} \frac{1}{d}$
 $= -\delta$
 $\frac{g(t) g(t + \delta)}{dt g(t) t=t*}$
 for some $t \leq t^* \leq t + \delta$. Thus, it suffices to show that
 $\sqrt{\frac{d}{dt} \frac{1}{g(t)}}$
 $> -\pi$
 (C.10)

for all t . To do this, consider Mills Ratio [17]
 $Z \infty$

2
 $R(t)$, $\exp(t/2)$
 $\exp(-x^2/2) dx$.
 t

Note that this is quite similar in functional form to $g(t)$, and with some manipulation, we can relate the two:

$Z \infty$
 2
 $R(t) = \exp(t/2)$
 $\exp(-x^2/2) dx$
 t
 $Z \infty$
 $\sqrt{R(2t) = \exp(t^2) \sqrt{\exp(-x^2/2) dx}}$
 $2t$
 $Z \infty$
 $\sqrt{2}$
 $= 2 \exp(t)$
 $\exp(-x^2) dx$
 t
 $Z -t$
 $\sqrt{2}$
 $= 2 \exp(t)$
 $\exp(-x^2) dx$
 ($\exp(-x^2)$ is symmetric)
 $-\infty$

$\sqrt{R(-2t) =}$
 $\sqrt{2 \exp(t^2)}$
 $Z t$
 $\exp(-x^2) dx$
 $-\infty$
 $\sqrt{}$
 $\sqrt{}$


```

π
2 exp(t) .
(1 + erf(t))
2
r [U+0012 control character]
[U+0013 control character]
π 1 + erf(t)
=
2 exp(-t2)
r
√
π
R(- 2t) =
g(t)
2
=
2

d 1
Sampford [21, Eq. (3)] proved that dt
R(t) < 1 for any t. Thus,

```

```

d 1
d
1
q
=
=
dt g(t)
dt 2 R(-√2t)
π
r
π d
1
√
>
2 dt R(- 2t)
r
√
√
π
· 1 · - 2 = - π,
2

```

which proves (C.10) and completes the proof.

Expanded PaperInterface specification

```

∀ {theta xi xj : ℝ},
  0 < theta →
    xj < xi →
      have laplace := MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (√2 / 2 * Real.exp (-√2 * |z|));
      have gaussian := ProbabilityTheory.gaussianReal 0 1;
      have laplaceRatio := fun u =>
        ((laplace.prod laplace)
          {epsilon |
            xi + epsilon.1 / theta < u ∧
              xj + epsilon.2 / theta < u ∧ xj + epsilon.2 / theta < xi + epsilon.1 / theta}).toReal /
          ((laplace.prod laplace) {epsilon | xi + epsilon.1 / theta < u ∧ xj + epsilon.2 / theta < u}).toReal;
      have gaussianRatio := fun u =>
        ((gaussian.prod gaussian)
          {epsilon |
            xi + epsilon.1 / theta < u ∧
              xj + epsilon.2 / theta < u ∧ xj + epsilon.2 / theta < xi + epsilon.1 / theta}).toReal /
          ((gaussian.prod gaussian) {epsilon | xi + epsilon.1 / theta < u ∧ xj + epsilon.2 / theta < u}).toReal;
      ((∀ (a : ℝ),
        0 < ((laplace.prod laplace) {epsilon | xi + epsilon.1 / theta < a ∧ xj + epsilon.2 / theta < a}).toReal ∧
          ∃ d, HasDerivAt laplaceRatio d a ∧ 0 ≤ d) ∧
        ∃ a d, HasDerivAt laplaceRatio d a ∧ 0 < d) ∧
        ∀ (a : ℝ),
          0 < ((gaussian.prod gaussian) {epsilon | xi + epsilon.1 / theta < a ∧ xj + epsilon.2 / theta < a}).toReal ∧
            ∃ d, HasDerivAt gaussianRatio d a ∧ 0 < d

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixC2_source_pairwise_full_events

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 1151-1162

Verbatim source input

C.2

Verifying Definition 3

Next, we show that for both Laplacian and Gaussian distributions, $UAH(\theta_A, \theta_H) < UHH(\theta_A, \theta_H)$ for all $\theta_A > \theta_H$. In fact, for 3-candidate RUM families, we will show that this is always true for any well-ordered distribution, defined as follows.

Definition 4. A noise model with density $f(\cdot)$ is well-ordered if for any $a > b$ and $c > d$, $f(a - c)f(b - d) > f(a - d)f(b - c)$.

In other words, for a well-ordered noise model, given two numbers, two candidates are more likely to be correctly ordered than inverted conditioned on realizing those two numbers in some order. Lemma 1 shows that both Gaussian and Laplacian distributions are well-ordered.

Expanded PaperInterface specification

```
∀ (f : ℝ → ℝ),
  KR21Monoculture.PaperInterface.strictlyWellOrderedNoise f ↔
    ∀ {{a b c d : ℝ}}, b < a → d < c → f (a - c) * f (b - d) > f (a - d) * f (b - c)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.definition4_strictlyWellOrderedNoise_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 217-242

Verbatim source input

Definition 3 (Preference for weaker competition.). A candidate distribution D and noisy permutation family F_θ , exhibits a preference for weaker competition if the following holds: for all $\theta_1 > \theta_2$, $\sigma \sim F_{\theta_1}$ and $\pi, \tau \sim F_{\theta_2}$,

h
 i
 h
 i
 $(-\{\sigma_1\})$

$E \pi_1$

$(-\{\tau_1\})$

$< E \pi_1$

.

Intuitively, suppose we have a higher accuracy parameter θ_1 and a lower accuracy parameter $\theta_2 < \theta_1$; we draw a permutation π from F_{θ_2} ; and we then derive two permutations from π : $\pi(-\{\sigma_1\})$ obtained by deleting the first-ranked element of a permutation σ drawn from the more accurate distribution F_{θ_1} , and $\pi(-\{\tau_1\})$ obtained by deleting the first-ranked element of a permutation τ drawn from the less accurate distribution F_{θ_2} .

Then the expected value of the first-ranked candidate in $\pi(-\{\tau_1\})$ is strictly greater than the expected value of the first-ranked candidate in $\pi(-\{\sigma_1\})$ – that is, when a random candidate is removed from π , the best remaining candidate is better in expectation when the randomly removed candidate is chosen based on a noisier ranking.

[Next verbatim source excerpt]

In Random Utility Models, the underlying candidate values x_i are perturbed by independent and identically distributed noise $\varepsilon_i \sim E$, and the perturbed values are ranked to produce π . Originally conceived in the psychology literature [23], this model has been well-studied over nearly a century, [7, 4, 9, 24, 16, 22], including more recently in the computer science and machine learning literature [2, 1, 19, 25, 13]. First, we must define a family of RUMs that satisfies the conditions of Definition 1. Assume without loss of generality that the noise distribution E has unit variance. Then, consider the family of RUMs parameterized by θ in which candidates are ranked according to $x_i + \varepsilon_{\theta i}$. By this definition, the standard deviation of the noise for a particular value of θ is simply $1/\theta$. Intuitively, larger values of θ reduce the effect of the noise, making the ranking more accurate. In Theorem 5 in Appendix A, we show as long as the noise distribution E has positive support on $(-\infty, \infty)$, this definition of F_θ meets the differentiability, asymptotic optimality, and monotonicity conditions in Definition 1. For distributions with finite support, many of our results can

[Next verbatim source excerpt]

Definition 4. A noise model with density $f(\cdot)$ is well-ordered if for any $a > b$ and $c > d$, $f(a - c)f(b - d) > f(a - d)f(b - c)$.

In other words, for a well-ordered noise model, given two numbers, two candidates are more likely to be correctly ordered than inverted conditioned on realizing those two numbers in some order. Lemma 1 shows that both Gaussian and Laplacian distributions are well-ordered.

Thus, it suffices to show that for any 3-candidate RUM with a well-ordered noise model, $UAH(\theta_A, \theta_H) < UHH(\theta_A, \theta_H)$ when $\theta_A > \theta_H$.

Theorem 6. For 3 candidates with unique values $x_1 > x_2 > x_3$ and well-ordered i.i.d. noise with support $(-\infty, \infty)$, if $\theta_A > \theta_H$, then $UAH(\theta_A, \theta_H) < UHH(\theta_A, \theta_H)$.

Proof. Define u_i to be the expected utility of the maximum element of the human-generated ranking when i is not available. Because we're in the 3-candidate setting, we have

$u_1 = \lambda_1 x_2 + (1 - \lambda_1)x_3$

$u_2 = \lambda_2 x_1 + (1 - \lambda_2)x_3$

$u_3 = \lambda_3 x_1 + (1 - \lambda_3)x_2$

where $1/2 < \lambda_i < 1$. This is because the noise has support everywhere, so it is impossible to correctly rank any two candidates with probability 1, and any two candidates are more likely than not to be correctly ordered:

[U+0014 control character]

[U+0015 control character]

i

$h\varepsilon$

ε_j

$\varepsilon_i - \varepsilon_j$

1

i

$-$

$> -\delta = \Pr[\varepsilon_i - \varepsilon_j \geq 0] + \Pr 0 >$

$> -\delta >$

$\Pr$

θ

θ

θ

2

Note that $\lambda_2 > \lambda_1$ and $\lambda_2 > \lambda_3$, since

$\lambda_2 = \Pr[\varepsilon_1 - \varepsilon_3 > -\theta(x_1 - x_3)] > \max\{\Pr[\varepsilon_1 - \varepsilon_3 > -\theta(x_2 - x_3)], \Pr[\varepsilon_1 - \varepsilon_3 > -\theta(x_1 - x_2)]\} = \max\{\lambda_1, \lambda_3\}$.

Let $\tau \sim F_{\theta_A}$ and $\pi \sim F_{\theta_H}$. With this, we can write

$UAH(\theta_A, \theta_H) =$

3

X

$\Pr[\tau_1 = i]u_i$

$i=1$

$UHH(\theta_A, \theta_H) =$

```

3
X

Pr[ $\pi_1 = i$ ] $u_i$ 

i=1

Define
 $\Delta p_1 = \Pr[\tau_1 = i] - \Pr[\pi_1 = i]$ 
Using Lemmas 2, and 3, we have
 $\Delta p_1 > 0$ 

(By monotonicity of RUM families, see Appendix A)

 $\Delta p_1 \geq \Delta p_2$ 
 $\Delta p_3 \leq 0$ 
Also,  $\Delta p_1 + \Delta p_2 + \Delta p_3 = 0$ . We must show that
 $UAH(\theta_A, \theta_H) - UHH(\theta_A, \theta_H) =$ 

3
X
i=1

18

 $\Delta p_i u_i < 0$ .

[form-feed in source extraction]
We consider 2 cases.
Case 1:  $\Delta p_2 \leq 0$ .
Then,  $\Delta p_1 = -(\Delta p_2 + \Delta p_3)$ . This yields
3
X

 $\Delta p_i u_i = \Delta p_1 u_{-1} + \Delta p_2 u_{-2} + \Delta p_3 u_{-3}$ 

i=1

 $\leq \Delta p_1 u_{-1} - \Delta p_1 \min(u_{-2}, u_{-3})$ 
 $= \Delta p_1 (\lambda_1 x_2 + (1 - \lambda_1)x_3 - \min\{\lambda_2 x_1 + (1 - \lambda_2)x_3, \lambda_3 x_1 + (1 - \lambda_3)x_2\})$ 
 $\leq \Delta p_1 (\lambda_1 x_2 + (1 - \lambda_1)x_3 - \min\{\lambda_2 x_1 + (1 - \lambda_2)x_3, x_2\})$ 
We can show that this is at most 0 regardless of which term attains the minimum. Because  $\lambda_2 > \lambda_1$ ,
 $\lambda_1 x_2 + (1 - \lambda_1)x_3 - \lambda_2 x_1 - (1 - \lambda_2)x_3 = \lambda_1 x_2 + x_3 - \lambda_1 x_3 - \lambda_2 x_1 - x_3 + \lambda_2 x_3$ 
 $= \lambda_1 x_2 - \lambda_1 x_3 - \lambda_2 x_1 + \lambda_2 x_3$ 
 $= \lambda_1 (x_2 - x_3) + \lambda_2 (x_3 - x_1)$ 
 $< \lambda_1 (x_2 - x_3) + \lambda_1 (x_3 - x_1)$ 
 $= \lambda_1 (x_2 - x_1)$ 
 $< 0$ 
For the second term, we have
 $\lambda_1 x_2 + (1 - \lambda_1)x_3 - x_2 = (1 - \lambda_1)(x_3 - x_2) < 0$ .
Thus,
3
X

 $\Delta p_i u_i < 0$ .

i=1

Case 2:  $\Delta p_2 > 0$ . Note that  $u_{-1} < x_2 < u_{-3}$ . Then, using  $\Delta p_3 = -(\Delta p_1 + \Delta p_2)$ ,
3
X

 $\Delta p_i u_i = \Delta p_1 u_{-1} + \Delta p_2 u_{-2} + \Delta p_3 u_{-3}$ 

i=1

 $= \Delta p_1 (u_{-1} - u_{-3}) + \Delta p_2 (u_{-2} - u_{-3})$ 
 $\leq \Delta p_2 (u_{-1} - u_{-3}) + \Delta p_2 (u_{-2} - u_{-3})$ 

( $\Delta p_1 \geq \Delta p_2$  and  $u_{-1} < u_{-3}$ )

 $= \Delta p_2 (u_{-1} + u_{-2} - 2u_{-3})$ 
 $\leq \Delta p_2 (x_2 + x_1 - 2(\lambda_3 x_1 + (1 - \lambda_3)x_2))$ 
[U+0013 control character][U+0013 control character]
[U+0012 control character]
[U+0012 control character]
1
1
 $x_1 + x_2$ 
 $< \Delta p_2 x_2 + x_1 - 2$ 
2
2
 $= 0$ 
Thus,  $UAH(\theta_A, \theta_H) < UHH(\theta_A, \theta_H)$ .

[Next verbatim source excerpt]

Next, we show a few basic facts. Let  $f_A(r)$  be the density function of the joint realization  $R = [X_1, \dots, X_n] = [x_1 + \varepsilon_1/\theta_A, \dots, x_n + \varepsilon_n/\theta_A]$  under the algorithmic ranking and  $f_H(r)$  be the similarly defined density function under the human-generated ranking. Consider the “contraction” operation  $r_0 = \text{cont}(r)$  such that  $r_{i0} = x_i + (r_i - x_i) \cdot \theta_H$ . Essentially, the contraction defines a coupling between  $f_A(\cdot)$  and  $f_H(\cdot)$ , since
A
for  $r_0 = \text{cont}(r)$ ,  $f_A(r_0) dr_0 = f_H(r) dr$ . Let  $\pi(r)$  be the ranking induced by  $r$ . Note that contraction cannot introduce any new inversions in  $\pi(r)$ —that is, if  $i$  is ranked above  $j$  in  $\pi(r)$  for  $i < j$ , then  $i$  is ranked above  $j$  in  $\pi(\text{cont}(r))$ . Intuitively, this is because contraction pulls values closer to their means, and can therefore

```

only correct existing inversions, not introduce new ones. This fact will allow us to prove some useful lemmas.

Expanded PaperInterface specification

```

∀ (f : ℝ → ℝ) {thetaA thetaH x1 x2 x3 : ℝ},
  Measurable f →
    (∀ {a b c d : ℝ}, b < a → d < c → f (a - c) * f (b - d) > f (a - d) * f (b - c)) →
      (∀ (z : ℝ), 0 ≤ f z) →
        ∫- (z : ℝ), ENNReal.ofReal (f z) = 1 →
          0 < thetaH →
            thetaH < thetaA →
              x2 < x1 →
                x3 < x2 →
                  MeasureTheory.IsProbabilityMeasure
                    (MeasureTheory.Measure.pi fun x =>
                      MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z)) ∧
                    ∃ rankingLaw,
                      (∀ (theta : ℝ),
                        0 < theta →
                          (rankingLaw theta).toMeasure =
                            MeasureTheory.Measure.map
                              (fun epsilon =>
                                EconCSLib.SocialChoice.Ranking.rankByScore fun i =>
                                  KR21Monoculture.threeCandidateValueProfile x1 x2 x3 i + epsilon i / theta)
                              (MeasureTheory.Measure.pi fun x =>
                                MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z))) ∧
                                EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent (rankingLaw thetaH)
                                  (rankingLaw thetaA) (KR21Monoculture.threeCandidateValueProfile x1 x2 x3) <
                                EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent (rankingLaw thetaH)
                                  (rankingLaw thetaH) (KR21Monoculture.threeCandidateValueProfile x1 x2 x3)

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem6_source_raw_iid_density_literal_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 1293-1329

Verbatim source input

Lemma 1. Both Gaussian and Laplacian distributions are well-ordered.
Proof. The Gaussian noise model is well-ordered:

$$\begin{aligned} & \frac{1}{2\sigma^2\pi} \exp\left(-\frac{(a-c)^2 - (b-d)^2}{2\sigma^2}\right) \\ &= \frac{1}{2\sigma^2\pi} \exp\left(-\frac{(a-d)^2 - (b-c)^2 - 2(ac + bd - ad - bc)}{2\sigma^2}\right) \\ &= f(a-d)f(b-c) \exp(-2((a-b)(c-d))) \\ f(a-c)f(b-d) &= \\ &< f(a-d)f(b-c) \\ 19 \\ (\lambda 3 > 21) \end{aligned}$$

[form-feed in source extraction]
Laplacian noise is as well:

$$\begin{aligned} & \frac{1}{4} \exp(-|a-c| - |b-d|) \\ & \frac{1}{4} f(a-d)f(b-c) = \exp(-|a-d| - |b-c|) \\ & f(a-c)f(b-d) = \end{aligned}$$

It suffices to show that for $a > b$ and $c > d$, $|a-c| + |b-d| < |a-d| + |b-c|$. To show this, plot (a, b) and (c, d) in the (x, y) plane. Note that they're both below the $y = x$ line, and that the ℓ_1 distance between them is $|a-c| + |b-d|$. Moreover, the ℓ_1 distance between any two points must be realized by some Manhattan path, which is a combination of horizontal and vertical line segments. Consider the point (b, a) , which is above the $y = x$ line. Any Manhattan path from (b, a) to (c, d) must cross the $y = x$ line at some point (w, w) . Since (b, a) and (a, b) are equidistant from (w, w) , for any Manhattan path from (b, a) to (c, d) , there exists a Manhattan path from (a, b) to (c, d) passing through (w, w) of the same length, meaning the ℓ_1 distance from (a, b) to (c, d) is smaller than the ℓ_1 distance from (b, a) to (c, d) . As a result, $|a-c| + |b-d| < |a-d| + |b-c|$.

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For every $\kappa > 0$ and $\lambda > 0$, the Gaussian kernel $G_\kappa(r) = \exp(-\kappa r^2)$ satisfies $G_\kappa(a-c)G_\kappa(b-d) > G_\kappa(a-d)G_\kappa(b-c)$ for all $a > b$ and $c > d$. The Laplace kernel $L_\lambda(r) = \exp(-\lambda|r|)$ satisfies the corresponding $\geq$ inequality globally, does not satisfy it strictly for every such quadruple, and satisfies the strict inequality on the explicit overlap domain $b < a$, $d < c$, $b < c$, and $d < a$.

Archival source anchor: cited publication, lines 1293-1329

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```
∀ {κ lam : ℝ},
  0 < κ →
  0 < lam →
    KR21Monoculture.StrictlyWellOrderedNoise (KR21Monoculture.gaussianNoiseKernel κ) ∧
    KR21Monoculture.WeaklyWellOrderedNoise (KR21Monoculture.laplacianNoiseKernel lam) ∧
    ¬KR21Monoculture.StrictlyWellOrderedNoise (KR21Monoculture.laplacianNoiseKernel lam) ∧
      ∀ {a b c d : ℝ},
        b < a →
        d < c →
        b < c →
        d < a →
          KR21Monoculture.laplacianNoiseKernel lam (a - c) *
            KR21Monoculture.laplacianNoiseKernel lam (b - d) >
              KR21Monoculture.laplacianNoiseKernel lam (a - d) *
                KR21Monoculture.laplacianNoiseKernel lam (b - c)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemmal_corrected_gaussian_laplace_kernel_target

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 1340-1364

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} {f : ℝ → ℝ},
  Measurable f →
  (∀ (z : ℝ), 0 ≤ f z) →
  ∀ (hnormalized : ∫- (z : ℝ), ENNReal.ofReal (f z) = 1) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
  StrictAnti value →
  ∀ {thetaA thetaH : ℝ},
    0 < thetaH →
    thetaH < thetaA →
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f) value
      thetaA).toMeasure =
      MeasureTheory.Measure.map
        (fun epsilon => EconCSLib.SocialChoice.Ranking.rankByScore fun c => value c + epsilon c / thetaA)
        (MeasureTheory.Measure.pi fun x =>
          MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z)) ∧
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f) value
      thetaH).toMeasure =
      MeasureTheory.Measure.map
        (fun epsilon =>
          EconCSLib.SocialChoice.Ranking.rankByScore fun c => value c + epsilon c / thetaH)
        (MeasureTheory.Measure.pi fun x =>
          MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z)) ∧
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f)
      value thetaA)
    (Fin.last (n + 1)) ≤
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f)
      value thetaH)
    (Fin.last (n + 1))
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma2_source_arbitraryFinite_iid_bottom_first_probability_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 1366-1464

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {n : ℕ} {f : ℝ → ℝ},
  KR21Monoculture.StrictlyWellOrderedNoise f →
  Measurable f →
  (∀ (z : ℝ), 0 ≤ f z) →
  ∀ (hnormalized : ∫- (z : ℝ), ENNReal.ofReal (f z) = 1) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
  StrictAnti value →
  ∀ {thetaA thetaH : ℝ},
    0 < thetaH →
    thetaH ≤ thetaA →
    ∀ {i : EconCSLib.SocialChoice.Ranking.Candidate n},
      i ≠ 0 →
      (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f)
        value thetaA).toMeasure =
        MeasureTheory.Measure.map
          (fun epsilon =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun c => value c + epsilon c / thetaA)
          (MeasureTheory.Measure.pi fun x =>
            MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z)) ∧
      (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF (KR21Monoculture.w11CandidateNoiseLaw f)
        value thetaH).toMeasure =
        MeasureTheory.Measure.map
          (fun epsilon =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun c => value c + epsilon c / thetaH)
          (MeasureTheory.Measure.pi fun x =>
            MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (f z)) ∧
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF
      (KR21Monoculture.w11CandidateNoiseLaw f) value thetaA)
  i -
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF
      (KR21Monoculture.w11CandidateNoiseLaw f) value thetaH)
  i ≤
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF
      (KR21Monoculture.w11CandidateNoiseLaw f) value thetaA)
  0 -
  KR21Monoculture.firstChoiceProb
    (KR21Monoculture.paper_appendixA_scaledNoiseRankingPMF
      (KR21Monoculture.w11CandidateNoiseLaw f) value thetaH)
  0
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma3_source_arbitraryFinite_iid_delta_le_top_delta_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 1467-1800

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```
∀ {sigma xi xj : ℝ},
  0 < sigma →
  xj < xi →
  have innovation := MeasureTheory.volume.withDensity fun z => ENNReal.ofReal (√2 / 2 * Real.exp (-√2 * |z|));
  have score := fun epsilon c => if c = 0 then xi + sigma * epsilon.1 else xj + sigma * epsilon.2;
  have rank := fun epsilon => EconCSLib.SocialChoice.Ranking.rankByScore (score epsilon);
  have rankLaw := MeasureTheory.Measure.map rank (innovation.prod innovation);
  Measurable rank ∧
  rankLaw Set.univ = 1 ∧
  (∀ (epsilon : ℝ × ℝ),
    xj + sigma * epsilon.2 < xi + sigma * epsilon.1 →
    EconCSLib.SocialChoice.Ranking.firstChoice (rank epsilon) = 0) ∧
  (∀ (a : ℝ),
    0 <
      ((innovation.prod innovation)
        {epsilon | xi + sigma * epsilon.1 < a ∧ xj + sigma * epsilon.2 < a}).toReal ∧
    ∃ d,
      HasDerivAt
        (fun u =>
          ((innovation.prod innovation)
            {epsilon |
              xi + sigma * epsilon.1 < u ∧
              xj + sigma * epsilon.2 < u ∧
              xj + sigma * epsilon.2 < xi + sigma * epsilon.1}).toReal /
          ((innovation.prod innovation)
            {epsilon | xi + sigma * epsilon.1 < u ∧ xj + sigma * epsilon.2 < u}).toReal)
        d a ∧
        0 ≤ d) ∧
  ∃ a d,
    HasDerivAt
      (fun u =>
        ((innovation.prod innovation)
          {epsilon |
            xi + sigma * epsilon.1 < u ∧
            xj + sigma * epsilon.2 < u ∧ xj + sigma * epsilon.2 < xi + sigma * epsilon.1}).toReal /
        ((innovation.prod innovation)
          {epsilon | xi + sigma * epsilon.1 < u ∧ xj + sigma * epsilon.2 < u}).toReal)
      d a ∧
      0 < d
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem7_source_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation


```

=
The denominator is
[U+0012 control character]
[U+0013 control character]
1
1
1
1
1 - exp(-λ(a - xj )) · exp(λ(xj - a)) = exp(-λ(xi - a)) - exp(-λ(xi - xj ))
2
2
2
4
We can factor out 41 exp(-λ(xi - xj )) from both, so
2 exp(λ(a - xj )) - 23 + λ(a - xj )
Pr[Xi > Xj | Xi < a, Xj < a] =
2 exp(λ(a - xj )) - 1
=
[U+0001 control character]
[U+0001 control character]
2 exp(λ(a - xj )) - 1 - 21 + λ(a - xj )
2 exp(λ(a - xj )) - 1
=1-
1
2 + λ(a - xj )
2 exp(λ(a - xj )) - 1
22
[form-feed in source extraction]
Thus,
d
Pr[Xi > Xj | Xi < a, Xj < a] > 0
da
1
d
2 + λ(a - xj )
=>
<0
da 2 exp(λ(a - xj )) - 1
[U+0012 control character]
[U+0013 control character]
1
=>(2 exp(λ(a - xj )) - 1)λ <
+ λ(a - xj ) 2λ exp(λ(a - xj ))
2
[U+0012 control character]
[U+0013 control character]
1
=>2 - exp(-λ(a - xj )) < 2
+ λ(a - xj )
2
=>1 - exp(-λ(a - xj )) < 2λ(a - xj )
=> exp(-λ(a - xj )) > 1 - 2λ(a - xj )
This is true because λ(a - xj ) > 0, and for z > 0,
exp(-z) > 1 - z > 1 - 2z.
Case 3: a > xi .
Then, the numerator of (C.3) is
[U+0012 control character]
[U+0013 control character]
Z xj
Z xi
λ
1
λ
1
exp (-λ(xi - x)) · exp(-λ(xj - x)) dx +
exp (-λ(xi - x)) 1 - exp(-λ(x - xj )) dx
2
2
-∞ 2
xj 2
[U+0012 control character]
[U+0013 control character]
Z a
1
λ
exp(-λ(x - xi )) 1 - exp(-λ(x - xj )) dx
+
2
2
xi
[U+0012 control character]
[U+0013 control character]
Z
1
3 λ
1
λ a

```

```

= -
+ (xi - xj ) exp(-λ(xi - xj )) + (1 - exp(-λ(a - xi ))) -
exp(-λ(2x - xi - xj )) dx
2
8
4
2
4 xi
[U+0012 control character]
[U+0013 control character]
3 λ
1
=1-
+ (xi - xj ) exp(-λ(xi - xj )) - exp(-λ(a - xi ))
8
4
2
1
+ exp(λ(xi + xj ))(exp(-2λa) - exp(-2λxi ))
8 [U+0012 control character]
[U+0013 control character]
1
1
1 λ
+ (xi - xj ) exp(-λ(xi - xj )) - exp(-λ(a - xi )) + exp(-λ(2a - xi - xj ))
=1-
2
4
2
8
The denominator is
[U+0012 control character]
[U+0013 control character][U+0012 control character]
[U+0013 control character]
1
1
1 - exp(-λ(t - xi ))
1 - exp(-λ(t - xj ))
2
2
1
1
1
= 1 - exp(-λ(a - xi )) - exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
2
2
4
Thus,
Pr[Xi > Xj | Xi < a, Xj < a]
[U+0001 control character]
1 - 21 + λ4 (xi - xj ) exp(-λ(xi - xj )) - 12 exp(-λ(a - xi )) + 81 exp(-λ(2a - xi - xj ))
=
1 - 12 exp(-λ(a - xi )) - 21 exp(-λ(a - xj )) + 14 exp(-λ(2a - xi - xj ))
8 - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )) - 4 exp(-λ(a - xi )) + exp(-λ(2a - xi - xj ))
α
4 - 2 exp(-λ(a - xi )) - 2 exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
23

[form-feed in source extraction]
We're interested in
d
Pr[Xi > Xj | Xi < a, Xj < a] > 0
da
⇒ (4 - 2 exp(-λ(a - xi )) - 2 exp(-λ(a - xj )) + exp(-λ(2a - xi - xj ))
· (4λ exp(-λ(a - xi )) - 2λ exp(-λ(2a - xi - xj )))
> (8 - 4 exp(-λ(a - xi )) + exp(-λ(2a - xi - xj )) - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )))
· (2λ exp(-λ(a - xi )) + 2λ exp(-λ(a - xj )) - 2λ exp(-λ(2a - xi - xj )))
⇒ 16 exp(-λ(a - xi )) - 8 exp(-λ(2a - xi - xj )) - 8 exp(-2λ(a - xi )) + 4 exp(-λ(3a - 2xi - xj ))
- 8 exp(-λ(2a - xi - xj )) + 4 exp(-λ(3a - xi - 2xj )) + 4 exp(-λ(3a - 2xi - xj ))
- 2 exp(-2λ(2a - xi - xj ))
> 16 exp(-λ(a - xi )) + 16 exp(-λ(a - xj )) - 16 exp(-λ(2a - xi - xj ))
- 8 exp(-2λ(a - xi )) - 8 exp(-λ(2a - xi - xj )) + 8 exp(-λ(3a - 2xi - xj ))
+ 2 exp(-λ(3a - 2xi - xj )) + 2 exp(-λ(3a - xi - 2xj )) - 2 exp(-2λ(2a - xi - xj ))
- 2(4 + 2λ(xi - xj )) exp(-λ(a - xj )) - 2(4 + 2λ(xi - xj )) exp(-λ(a + xi - 2xj ))
+ 2(4 + 2λ(xi - xj )) exp(-2λ(a - xj ))
⇒ exp(-λ(3a - xi - 2xj ))
> 8 exp(-λ(a - xj )) - 4 exp(-λ(2a - xi - xj )) + exp(-λ(3a - 2xi - xj ))
- (4 + 2λ(xi - xj )) exp(-λ(a - xj )) - (4 + 2λ(xi - xj )) exp(-λ(a + xi - 2xj ))
+ (4 + 2λ(xi - xj )) exp(-2λ(a - xj ))
⇒ exp(-λ(2a - xi - xj ))
> 8 - 4 exp(-λ(a - xi )) + exp(-λ(2a - 2xi ))
- (4 + 2λ(xi - xj )) - (4 + 2λ(xi - xj )) exp(-λ(xi - xj )) + (4 + 2λ(xi - xj )) exp(-λ(a - xj ))
⇒ exp(-λ(2a - xi - xj )) - 8 + 4 exp(-λ(a - xi )) - exp(-2λ(a - xi ))
+ (4 + 2λ(xi - xj ))(1 + exp(-λ(xi - xj ))) - (4 + 2λ(xi - xj )) exp(-λ(a - xj ))
>0

```

Expanded PaperInterface specification

```

∀ {lam xi xj a : ℝ},
0 < lam →
have density := fun location x => lam / 2 * Real.exp (-lam * |x - location|);
have cdf := fun location x =>
if x < location then 1 / 2 * Real.exp (-lam * (location - x)) else 1 - 1 / 2 * Real.exp (-lam * (x - location));
have scoreI := MeasureTheory.volume.withDensity fun x => ENNReal.ofReal (density xi x);

```

```

have scoreJ := MeasureTheory.volume.withDensity fun x => ENNReal.ofReal (density xj x);
have pairLaw := scoreI.prod scoreJ;
0 < (pairLaw {p | p.1 < a ∧ p.2 < a}).toReal ∧
(pairLaw {p | p.1 < a ∧ p.2 < p.1}).toReal / (pairLaw {p | p.1 < a ∧ p.2 < a}).toReal =
  (∫ (x : ℝ) in Set.Iic a, density xi x * cdf xj x) / (cdf xi a * cdf xj a)

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC3_laplace_strict_conditional_ratio_eq_density_cdf_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 1765-1800

Verbatim source input

(C.4)

Note that for any $z \geq 0$, we have

$$(4 + 2z)(1 + e^{-z}) - 8 \geq 0 \iff (2 + z)(1 + e^{-z}) \geq 4$$

$$\iff z + 2e^{-z} + ze^{-z} \geq 2$$

For $z = 0$, this holds with equality, and the left hand side is increasing since

$$d \quad z + 2e^{-z} + ze^{-z} \geq 0 \iff 1 - 2e^{-z} + e^{-z} - ze^{-z} \geq 0$$

$$dx \quad \iff 1 \geq e^{-z} + ze^{-z}$$

$$1$$

$$\iff$$

$$\geq e^{-z}$$

$$1+z$$

$$\iff 1 + z \leq ez$$

Therefore, choosing $z = \lambda(x_i - x_j)$ and plugging back to (C.4), we have

$$\begin{aligned} & \exp(-\lambda(2a - x_i - x_j)) - 8 + 4 \exp(-\lambda(a - x_i)) - \exp(-2\lambda(a - x_i)) \\ & + (4 + 2\lambda(x_i - x_j))(1 + \exp(-\lambda(x_i - x_j))) - (4 + 2\lambda(x_i - x_j)) \exp(-\lambda(a - x_j)) > 0 \\ \iff & \exp(-\lambda(2a - x_i - x_j)) + 4 \exp(-\lambda(a - x_i)) - \exp(-2\lambda(a - x_i)) - (4 + 2\lambda(x_i - x_j)) \exp(-\lambda(a - x_j)) > 0 \\ \iff & \exp(-\lambda(a - x_j)) + 4 - \exp(-\lambda(a - x_i)) - (4 + 2\lambda(x_i - x_j)) \exp(-\lambda(x_i - x_j)) > 0 \\ \iff & 4(1 - \exp(-\lambda(x_i - x_j))) + \exp(-\lambda(a - x_i))(\exp(-\lambda(x_i - x_j)) - 1) - 2\lambda(x_i - x_j) \exp(-\lambda(x_i - x_j)) > 0 \\ \iff & (4 - \exp(-\lambda(a - x_i)))(1 - \exp(-\lambda(x_i - x_j))) - 2\lambda(x_i - x_j) \exp(-\lambda(x_i - x_j)) > 0 \\ \iff & 3(1 - \exp(-\lambda(x_i - x_j))) - 2\lambda(x_i - x_j) \exp(-\lambda(x_i - x_j)) > 0 \end{aligned}$$

$$24$$

$$(\exp(-\lambda(a - x_i)) < 1)$$

[form-feed in source extraction]

Again letting $z = \lambda(x_i - x_j)$, this is true if and only if

$$3(1 - e^{-z}) > 2ze^{-z} \iff 3(ez - 1) > 2z$$

$$\iff 3ez > 3 + 2z$$

which is true because $ez > 1 + z$ for $z > 0$. This completes the proof for Case 3.

As a result, we have that

d

$$\Pr[X_i > X_j \mid X_i < a, X_j < a] \geq 0$$

da

for all a , with strict inequality for some a , which proves the theorem.

Expanded PaperInterface specification

```
∀ {lam xi xj a : ℝ},
  0 < lam →
    xj < xi →
      xi < a →
        0 <
          Real.exp (-lam * (2 * a - xi - xj)) - 8 + 4 * Real.exp (-lam * (a - xi)) - Real.exp (-2 * lam * (a - xi)) +
            (4 + 2 * lam * (xi - xj)) * (1 + Real.exp (-lam * (xi - xj))) -
              (4 + 2 * lam * (xi - xj)) * Real.exp (-lam * (a - xj))
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC4_laplace_case3_expression_pos

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 1801-2160

Verbatim source input

[No record available.]

Expanded PaperInterface specification

```

∀ {sigma xi xj : ℝ},
  0 < sigma →
    xj < xi →
      have innovation := ProbabilityTheory.gaussianReal 0 1;
      have score := fun epsilon c => if c = 0 then xi + sigma * epsilon.1 else xj + sigma * epsilon.2;
      have rank := fun epsilon => EconCSLib.SocialChoice.Ranking.rankByScore (score epsilon);
      have rankLaw := MeasureTheory.Measure.map rank (innovation.prod innovation);
      Measurable rank ∧
        rankLaw Set.univ = 1 ∧
        (∀ (epsilon : ℝ × ℝ),
          xj + sigma * epsilon.2 < xi + sigma * epsilon.1 →
            EconCSLib.SocialChoice.Ranking.firstChoice (rank epsilon) = 0) ∧
        ∀ (a : ℝ),
          0 <
            ((innovation.prod innovation)
              {epsilon | xi + sigma * epsilon.1 < a ∧ xj + sigma * epsilon.2 < a}).toReal ∧
        ∃ d,
          HasDerivAt
            (fun u =>
              ((innovation.prod innovation)
                {epsilon |
                  xi + sigma * epsilon.1 < u ∧
                  xj + sigma * epsilon.2 < u ∧
                  xj + sigma * epsilon.2 < xi + sigma * epsilon.1}).toReal /
                ((innovation.prod innovation)
                  {epsilon | xi + sigma * epsilon.1 < u ∧ xj + sigma * epsilon.2 < u}).toReal)
              d a ∧
            0 < d

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem8_source_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 38

Source locator: cited publication, lines 1839-1903

Verbatim source input

```
(C.5)
Let t = a - xi and δ = xi - xj . Then, using the fact that

$$\int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) dx =$$


$$\int_{-\infty}^a \exp(-x^2) dx +$$


$$\frac{\sqrt{\pi}}{2} (1 + \operatorname{erf}(a - x_i)) +$$


$$\int_{-\infty}^a \exp(-x^2) \operatorname{erf}(x + \delta) dx,$$

(C.5) becomes

$$\int_{-\infty}^a \int_{-\infty}^t (1 + \operatorname{erf}(t)) (1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) dt \cdot \operatorname{erf}(t) +$$


$$\int_{-\infty}^a \int_{-\infty}^t \exp(-x^2) \operatorname{erf}(x + \delta) dx \frac{1}{2} (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) dt$$


$$\int_{-\infty}^a \int_{-\infty}^t (1 + \operatorname{erf}(t)) (1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) dt \cdot \operatorname{erf}(t) +$$


$$\int_{-\infty}^a \int_{-\infty}^t \exp(-x^2) \operatorname{erf}(x + \delta) dx \frac{1}{2} (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) dt$$


$$\int_{-\infty}^a \int_{-\infty}^t (1 + \operatorname{erf}(t)) (1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) dt \cdot \operatorname{erf}(t) +$$


$$\int_{-\infty}^a \int_{-\infty}^t \exp(-x^2) \operatorname{erf}(x + \delta) dx \frac{1}{2} (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) dt$$

```

Expanded PaperInterface specification

```
∀ {xi xj a : ℝ},
  xj < xi →
  have scoreI := ProbabilityTheory.gaussianReal xi (1 / 2);
  have scoreJ := ProbabilityTheory.gaussianReal xj (1 / 2);
  have denominator := ((scoreI.prod scoreJ) {p | p.1 < a ∧ p.2 < a}).toReal;
  0 < denominator ∧
  ((scoreI.prod scoreJ) {p | p.1 < a ∧ p.2 < a ∧ p.2 < p.1}).toReal / denominator =
    (2 / √Real.pi *
      ∫ (x : ℝ) in Set.Iic a, Real.exp (-(x - xi) ^ 2) * (1 + KR21Monoculture.theorem8Erf (x - xj))) /
      ((1 + KR21Monoculture.theorem8Erf (a - xi)) * (1 + KR21Monoculture.theorem8Erf (a - xj))) ∧
  0 <
    (1 + KR21Monoculture.theorem8Erf (a - xi)) * (1 + KR21Monoculture.theorem8Erf (a - xj)) *
      Real.exp (-(a - xi) ^ 2) *
      (1 + KR21Monoculture.theorem8Erf (a - xj)) -
      (∫ (x : ℝ) in Set.Iic a, Real.exp (-(x - xi) ^ 2) * (1 + KR21Monoculture.theorem8Erf (x - xj))) *
        (2 / √Real.pi) *
        ((1 + KR21Monoculture.theorem8Erf (a - xi)) * Real.exp (-(a - xj) ^ 2) +
          (1 + KR21Monoculture.theorem8Erf (a - xj)) * Real.exp (-(a - xi) ^ 2))
```


Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC5_gaussian_source_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 39

Source locator: cited publication, lines 1874-1914

Verbatim source input

(C.5) becomes

$$\frac{\sqrt{2}}{\pi} \int_0^{\infty} (1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \exp(-x^2) \operatorname{erf}(x + \delta) dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^{\infty} (1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) \exp(-(t + \delta)^2) dt$$

$$= \frac{\sqrt{2}}{\pi} \int_0^{\infty} (1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2) \exp(-(t + \delta)^2) dt + \frac{\sqrt{2}}{\pi} \int_0^{\infty} (1 + \operatorname{erf}(t + \delta)) \exp(-t^2) \exp(-(t + \delta)^2) dt$$

To show that this is true, we will use the fact that $f(t) > 0$ whenever the following conditions are met:

- $f(t)$ is continuous and differentiable everywhere

```
[form-feed in source extraction]
```

$$2. \lim_{t \rightarrow -\infty} f(t) = 0$$

d

$$3. \, dt$$
$$f(t) > 0$$

We'll show that these conditions hold for the LHS of (C.6).

Expanded PaperInterface specification

$$\forall \{\text{delta } t : \mathbb{R}\},$$
$$\theta < \text{delta} \rightarrow$$
$$\theta <$$
$$(1 + \text{KR21Monoculture.theorem8Erf } t) * (1 + \text{KR21Monoculture.theorem8Erf } (t + \text{delta})) ^ 2 * \text{Real.exp } (-t ^ 2) /$$
$$((1 + \text{KR21Monoculture.theorem8Erf } t) * \text{Real.exp } (-(t + \text{delta}) ^ 2) +$$
$$(1 + \text{KR21Monoculture.theorem8Erf } (t + \text{delta})) * \text{Real.exp } (-t ^ 2)) -$$
$$(1 + \text{KR21Monoculture.theorem8Erf } t) -$$
$$2 / \sqrt{\text{Real.pi}} * \int (x : \mathbb{R}) \text{ in Set.Iic } t, \text{Real.exp } (-x ^ 2) * \text{KR21Monoculture.theorem8Erf } (x + \text{delta})$$

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC6 gaussian reduced expression pos

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 40

Source locator: cited publication, lines 1914-1951

Verbatim source input

We'll show that these conditions hold for the LHS of (C.6).
$$\frac{(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)}{2} - \frac{(1 + \operatorname{erf}(t)) - \sqrt{t \rightarrow -\infty (1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2)} + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2)}}{\pi}$$
$$\lim_{t \rightarrow -\infty} \frac{(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)}{(1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2)}$$
$$= \lim_{t \rightarrow -\infty} \int_{-\infty}^x \exp(-x^2) \operatorname{erf}(x + \delta) dx$$

(C.7)

Observe that both the numerator and denominator of (C.7) are positive, so this limit must be at least 0.

We can upper bound it by
$$\frac{(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)}{(1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))^2 \exp(-t^2)} \leq \lim_{t \rightarrow -\infty} \frac{(1 + \operatorname{erf}(t)) \exp(-(t + \delta)^2) + (1 + \operatorname{erf}(t + \delta)) \exp(-t^2)}{(1 + \operatorname{erf}(t + \delta)) \exp(-t^2)}$$
$$= \lim_{t \rightarrow -\infty} (1 + \operatorname{erf}(t))(1 + \operatorname{erf}(t + \delta))$$
$$\lim_{t \rightarrow -\infty} = 0$$

Thus, the limit is 0. Now, we must show that the derivative is positive. The derivative is

Expanded PaperInterface specification

```
∀ (delta : ℝ),
  Filter.Tendsto
    (fun t =>
      (1 + KR21Monoculture.theorem8Erf t) * (1 + KR21Monoculture.theorem8Erf (t + delta)) ^ 2 * Real.exp (-t ^ 2) /
        ((1 + KR21Monoculture.theorem8Erf t) * Real.exp (-(t + delta) ^ 2) +
          (1 + KR21Monoculture.theorem8Erf (t + delta)) * Real.exp (-t ^ 2)) -
        (1 + KR21Monoculture.theorem8Erf t) -
        2 / √Real.pi * ∫ (x : ℝ) in Set.Iic t, Real.exp (-x ^ 2) * KR21Monoculture.theorem8Erf (x + delta))
    Filter.atBot (nhds 0))
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC7_gaussian_reduced_expression_tendsto_atBot_zero

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 41

Source locator: cited publication, lines 1951-1985

Verbatim source input

Thus, the limit is 0. Now, we must show that the derivative is positive. The derivative is

```
[U+0015 control character]
[U+0014 control character]
Z t
2
(1 + erf(t))(1 + erf(t + δ))2 exp(-t2)
d
2
- (1 + erf(t)) - √
exp(-x) erf(x + δ) dx
dt (1 + erf(t)) exp(-(t + δ)2) + (1 + erf(t + δ)) exp(-t2)
π -∞
[U+0014 control character]
[U+0015 control character]
d
(1 + erf(t))(1 + erf(t + δ))2 exp(-t2)
2
2
=
- √ exp(-t2) - √ exp(-t2) erf(t + δ)
dt (1 + erf(t)) exp(-(t + δ)2) + (1 + erf(t + δ)) exp(-t2)
π
π
(C.8)
Taking this derivative and factoring out
√
```

```
2(1 + erf(t))(1 + erf(t + δ)) exp(4t2)
2
π (erf(t) + 1) et2 + (erf(δ + t) + 1) e(δ+t)2
[U+0001 control character]2 ,
we get that (C.8) is positive if and only if
```

Expanded PaperInterface specification

```
∀ (delta t : ℝ),
  HasDerivAt
    (fun u =>
      (1 + KR21Monoculture.theorem8Erf u) * (1 + KR21Monoculture.theorem8Erf (u + delta)) ^ 2 * Real.exp (-u ^ 2) /
      ((1 + KR21Monoculture.theorem8Erf u) * Real.exp (-(u + delta) ^ 2) +
      (1 + KR21Monoculture.theorem8Erf (u + delta)) * Real.exp (-u ^ 2)) -
      (1 + KR21Monoculture.theorem8Erf u) -
      2 / √Real.pi * ∫ (x : ℝ) in Set.Iic u, Real.exp (-x ^ 2) * KR21Monoculture.theorem8Erf (x + delta))
    (deriv
      (fun u =>
        (1 + KR21Monoculture.theorem8Erf u) * (1 + KR21Monoculture.theorem8Erf (u + delta)) ^ 2 *
        Real.exp (-u ^ 2) /
        ((1 + KR21Monoculture.theorem8Erf u) * Real.exp (-(u + delta) ^ 2) +
        (1 + KR21Monoculture.theorem8Erf (u + delta)) * Real.exp (-u ^ 2)))
        t -
        2 / √Real.pi * Real.exp (-t ^ 2) -
        2 / √Real.pi * Real.exp (-t ^ 2) * KR21Monoculture.theorem8Erf (t + delta))
      t
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationC8_gaussian_reduced_expression_hasDerivAt

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 42

Source locator: cited publication, lines 1801-1914

Verbatim source input

Theorem 8. For any $a \in \mathbb{R}$ and $X_i = x_i + \sigma \varepsilon_i$ where $\varepsilon_i \sim N(0, 1)$,

$$\frac{d}{da} \Pr[X_i > X_j \mid X_i < a, X_j < a] > 0.$$

Proof. Assume $\sigma = 1/2$. This is without loss of generality because for any instance with arbitrary σ , there is an instance with $\sigma = 1/2$ that yields the same distribution over rankings (simply by scaling all item values by σ/σ_0). First, we have

$$\frac{d}{da} \Pr[X_i > X_j \mid X_i < a, X_j < a] = \frac{\int_{-\infty}^a \exp(-(x - x_i)^2) / \pi \cdot (1 + \operatorname{erf}(x - x_j))/2 \, dx}{\int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) \, dx} \cdot \frac{d}{da} \int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) \, dx$$

The derivative with respect to a is positive if and only if

$$\frac{d}{da} \int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) \, dx > \int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) \, dx \cdot \frac{d}{da} \int_{-\infty}^a \exp(-(x - x_i)^2) (1 + \operatorname{erf}(x - x_j)) \, dx$$

(C.5)

Let $t = a - x_i$ and $\delta = x_i - x_j$. Then, using the fact that

$$\frac{d}{dt} \int_{-\infty}^t \exp(-x^2) (1 + \operatorname{erf}(x + \delta)) \, dx = \exp(-t^2) (1 + \operatorname{erf}(t + \delta))$$

(C.5) becomes

$$\frac{d}{dt} \int_{-\infty}^t \exp(-x^2) (1 + \operatorname{erf}(x + \delta)) \, dx > \int_{-\infty}^t \exp(-x^2) (1 + \operatorname{erf}(x + \delta)) \, dx \cdot \frac{d}{dt} \int_{-\infty}^t \exp(-x^2) (1 + \operatorname{erf}(x + \delta)) \, dx$$

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 43

Source locator: cited publication, lines 2050-2160

Verbatim source input

for some $t \leq t^* \leq t + \delta$. Thus, it suffices to show that

$$\frac{d}{dt} \frac{1}{g(t)} > -\pi$$

(C.10)

for all t . To do this, consider Mills Ratio [17]

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$\exp(-x^2/2) dx.$$

$$t$$

Note that this is quite similar in functional form to $g(t)$, and with some manipulation, we can relate the two:

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$\exp(-x^2/2) dx$$

$$t$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(2t) = \frac{\int_{2t}^\infty \exp(-x^2/2) dx}{\int_{2t}^\infty \exp(-x^2/2) dx}$$

$$2t$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$R(t) = \frac{\int_t^\infty \exp(-x^2/2) dx}{\int_t^\infty \exp(-x^2/2) dx}$$

$$\exp(-x^2/2) dx$$

$$t$$

$$R(-t) = \frac{\int_{-t}^\infty \exp(-x^2/2) dx}{\int_{-t}^\infty \exp(-x^2/2) dx}$$

$$R(-t) = \frac{\int_{-t}^\infty \exp(-x^2/2) dx}{\int_{-t}^\infty \exp(-x^2/2) dx}$$

$$R(-t) = \frac{\int_{-t}^\infty \exp(-x^2/2) dx}{\int_{-t}^\infty \exp(-x^2/2) dx}$$

$$R(-t) = \frac{\int_{-t}^\infty \exp(-x^2/2) dx}{\int_{-t}^\infty \exp(-x^2/2) dx}$$

$$\exp(-x^2/2) dx$$

$$(\exp(-x^2/2) \text{ is symmetric})$$

$$-\infty$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

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$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$\exp(-x^2/2) dx$$

$$-\infty$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$(1 + \operatorname{erf}(t))$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

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$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$g(t)$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

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$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

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$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

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$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

$$R(-2t) = \frac{\int_{-2t}^\infty \exp(-x^2/2) dx}{\int_{-2t}^\infty \exp(-x^2/2) dx}$$

which proves (C.10) and completes the proof. $\square$

$$\forall (t : \mathbb{R}), \exists d, \text{HasDerivAt } (\text{fun } u \Rightarrow ((1 + \text{KR21Monoculture.theorem8Erf } u) / \text{Real.exp } (-u^2))^{-1}) \, d \, t \wedge -\sqrt{\text{Real.pi}} < d$$

KR21Monoculture.PaperInterface.equationC10_gaussian_g_inv_derivative_lower_bound

Reason: not recorded

If left unchecked, explain the discrepancy or uncertainty below.

Review row 44

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

Theorem 9. The family of distributions F_θ produced by the Mallows Model with Kendall tau distance with $\theta = \phi - 1$ satisfies the conditions of Definition 1.

Proof. We must show that F_θ satisfies the differentiability, asymptotic optimality, and monotonicity conditions of Definition 1. Differentiability: Let Π be the set of all permutations on n candidates. The probability of a realizing a particular permutation π under the Mallows model is

$$\frac{\phi^{-d(\pi, \pi^*)}}{\sum_{\pi \in \Pi} \phi^{-d(\pi, \pi^*)}}$$

$$\Pr[\pi] = \frac{1}{Z}$$

27

[form-feed in source extraction]

Both the numerator and denominator are differentiable with respect to $\theta = \phi - 1$, so $\Pr_\theta[\pi]$ is differentiable with respect to θ .

Asymptotic optimality: For the correct ranking π^* ,

$$\frac{1}{Z} = \frac{\phi^{-d(\pi^*, \pi^*)}}{\sum_{\pi \in \Pi} \phi^{-d(\pi, \pi^*)}}$$

where the normalizing constant Z is

X

*

$$\sum_{\pi \in \Pi} \phi^{-d(\pi, \pi^*)}$$

In the limit,

$$\lim_{\phi \rightarrow \infty} Z = \lim_{\phi \rightarrow \infty} \frac{1}{Z}$$

X

*

$$= \lim_{\phi \rightarrow \infty} \phi^{-d(\pi, \pi^*)}$$

$$\theta \rightarrow \infty$$

$$\phi \rightarrow \infty$$

$$\pi \in \Pi$$

$$\phi \rightarrow \infty$$

$$\phi^{-d(\pi, \pi^*)}$$

$$\pi^* \in \Pi$$

*

X

$$= 1 +$$

*

X

$$= \lim_{\phi \rightarrow \infty} 1 +$$

$$\lim_{\phi \rightarrow \infty} \phi^{-d(\pi, \pi^*)}$$

$$\pi^* \in \Pi$$

$$\phi \rightarrow \infty$$

$$= 1$$

because for any $\pi^* \in \Pi$, $d(\pi, \pi^*) \geq 1$. Therefore,

$$\lim_{\phi \rightarrow \infty} Z$$

$$= 1$$

$$\theta \rightarrow \infty$$


```

28
Prφ0 [π1
= xi ]
(-S)
Prφ0 [π1
= xj ]

[form-feed in source extraction]
Rearranging,
(-S)

Prφ [π1
= xi ]

(-S)
Prφ0 [π1
= xi ]

(-S)

Prφ [π1
= xj ]
(-S)
Prφ0 [π1
= xj ]

<

(D.1)

For θ0 = φ0 - 1 and θ = φ - 1,
h
i X h
i
(-S)
-(S)
EFθ π1
=
Pr π1
= xi xi
φ

i∈S
/

h

(-S)

EFθ0 π1

i

=

X
i∈S
/

h
i
-(S)
Prθ π1
= xi xi
φ

By Lemma 4,
h
i
h
i
(-S)
(-S)
EFθ0 π1
> EFθ π1
,
which completes the proof.

```

Expanded PaperInterface specification

```

∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
  EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value →
    (∀ (phi theta : ℝ),
      1 < phi →
        theta = phi - 1 →
          have M := KR21Monoculture.concreteMallowsSpec center theta;
          M.q = phi-1 ∧
            ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
              (M.law pi).toReal =
                phi-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /

```

```

       $\sum \tau, \phi^{-1} \wedge \text{EconCSLib.SocialChoice.Ranking.kendallTau center tau} \wedge$ 
(∀ (theta : ℝ),
  0 < theta →
    ∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
      ContinuousAt (fun theta' => ((KR21Monoculture.concreteMallowsSpec center theta').law pi).toReal) theta ∧
      DifferentiableAt ℝ (fun theta' => ((KR21Monoculture.concreteMallowsSpec center theta').law pi).toReal)
      theta) ∧
Filter.Tendsto (fun theta => ((KR21Monoculture.concreteMallowsSpec center theta).law center).toReal)
  Filter.atTop (nhds 1) ∧
  ∀ (thetaA thetaH : ℝ),
    0 < thetaH →
      thetaH < thetaA →
        (∀ (remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate n)),
          remaining.Nonempty →
            KR21Monoculture.expectedBestInSet (KR21Monoculture.concreteMallowsSpec center thetaH).law value
              remaining ≤
            KR21Monoculture.expectedBestInSet (KR21Monoculture.concreteMallowsSpec center thetaA).law value
              remaining) ∧
            KR21Monoculture.expectedBestInSet (KR21Monoculture.concreteMallowsSpec center thetaH).law value
              Finset.univ <
            KR21Monoculture.expectedBestInSet (KR21Monoculture.concreteMallowsSpec center thetaA).law value
              Finset.univ

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.theorem9_source_mallows_definition1_literal_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 45

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

$$\Pr[\pi] = \frac{e^{-\phi d(\pi, \pi^*)}}{Z},$$

where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking π^* . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

Theorem 9. The family of distributions F_θ produced by the Mallows Model with Kendall tau distance with $\theta = \phi - 1$ satisfies the conditions of Definition 1.

Proof. We must show that F_θ satisfies the differentiability, asymptotic optimality, and monotonicity conditions of Definition 1. **Differentiability:** Let Π be the set of all permutations on n candidates. The probability of a realizing a particular permutation π under the Mallows model is

$$\Pr[\pi] = \frac{e^{-\phi d(\pi, \pi^*)}}{Z}$$

$$\Pr[\pi] = \frac{e^{-\phi d(\pi, \pi^*)}}{Z}$$

27

[form-feed in source extraction]

Both the numerator and denominator are differentiable with respect to $\theta = \phi - 1$, so $\Pr_\theta[\pi]$ is differentiable

[Next verbatim source excerpt]

Monotonicity: We must show that for any $S \subset [n]$, if π_1 denotes the value of the top-ranked candidate according to π excluding candidates in S ,

$$\pi_1 \geq \pi_1'$$

$$\pi_j \geq \pi_j'$$

$$\pi_j \geq \pi_j'$$

$$\pi_j \geq \pi_j'$$

For any $i \in$

S , let j be the smallest index such that $j > i$ and $j \notin S$. Consider any π such that π_1

Then, swapping i and j yields a permutation π' such that π_1

$$\Pr[\pi] = \Pr[\pi'] \cdot \phi^{\text{inv}(\pi) - \text{inv}(\pi')}$$

Since $i < j$, $\text{inv}(\pi) - \text{inv}(\pi') \geq 1$. Finally, note that swapping i and j is a bijection between $\{\pi : \pi_1$

$$\pi_1 \geq \pi_1'\}$$

$$\pi_j \geq \pi_j'\}$$

$$\pi_j \geq \pi_j'\}$$

$$\Pr[\pi] = \Pr[\pi'] \cdot \phi^{\text{inv}(\pi) - \text{inv}(\pi')}$$

$$\pi_j \geq \pi_j'\}$$

$$\pi_j \geq \pi_j'\}$$

$$\pi_j \geq \pi_j'\}$$

$\Pr[\pi_1]$
 1
 Note that the terms
 $\Pr[\pi]$
 X
 j
 $(-S)$
 $\Pr[\pi_1]$
 $= x_j]$
 $\cdot \phi^{\text{inv}(\pi) - \text{inv}(\pi^*)}$
 $\Pr[\pi]$
 sum to 1, so this is sum is some polynomial in ϕ with nonnegative weights
 $(-S)$
 $\Pr[\pi_1]$
 $= x_j]$
 and integer powers of ϕ . As a result, it must have a positive derivative with respect to ϕ , i.e., for $i < j$,
 $(-S)$
 $= x_i]$
 $\frac{d}{d\phi} \Pr[\pi_1]$
 > 0
 $\frac{d}{d\phi} \Pr[\pi] (-S) = x_j]$
 1
 0
 Let $\phi > \phi$. Then,
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_i]$
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_j]$
 $(-S)$
 $<$
 28
 $\Pr_{\phi} [\pi_1]$
 $= x_i]$
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_j]$
 [form-feed in source extraction]
 Rearranging,
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_i]$
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_i]$
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_j]$
 $(-S)$
 $\Pr_{\phi} [\pi_1]$
 $= x_j]$
 $<$
 $(D.1)$

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For a common center on exactly three candidates, $\phi_{\text{accurate}} > \phi_{\text{noisy}} > 1$, $\theta_r = \phi_r - 1$, and any nonempty remaining set
 $\hookrightarrow$ containing center-ordered candidates better before worse, the two source Mallows laws have $q_r = \phi_r^{-1}$ and normalized
 $\hookrightarrow$ mass $\phi_r^{-(\text{KendallTau}(\text{center}, \pi))} / \sum_{\tau} \phi_r^{-(\text{KendallTau}(\text{center}, \tau))}$. Writing $P_r(c)$ for the probability that c is
 $\hookrightarrow$ best in that remaining set under law r , $P_{\text{accurate}}(\text{better})P_{\text{noisy}}(\text{worse}) > P_{\text{accurate}}(\text{worse})P_{\text{noisy}}(\text{better})$. No
 $\hookrightarrow$ arbitrary-universe claim or printed reverse D.1 inequality is asserted.

Archival source anchor: cited publication, lines 2317-2376

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```

∀ (center : EconCSLib.SocialChoice.Ranking.Ranking 1) (phiNoisy thetaNoisy phiAccurate thetaAccurate : ℝ),
  1 < phiNoisy →
    phiNoisy < phiAccurate →
      thetaNoisy = phiNoisy - 1 →
        thetaAccurate = phiAccurate - 1 →
          ∀ {remaining : Finset (EconCSLib.SocialChoice.Ranking.Candidate 1)},
            remaining.Nonempty →
              ∀ {better worse : EconCSLib.SocialChoice.Ranking.Candidate 1},
                better ∈ remaining →
                  worse ∈ remaining →
                    EconCSLib.SocialChoice.Ranking.rankOf center better <
                      EconCSLib.SocialChoice.Ranking.rankOf center worse →
                        have MAccurate := KR21Monoculture.concreteMallowsSpec center thetaAccurate;
                        have MNoisy := KR21Monoculture.concreteMallowsSpec center thetaNoisy;
                        MAccurate.q = phiAccurate-1 ∧
                          MNoisy.q = phiNoisy-1 ∧
                            (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking 1),
                              (MAccurate.law pi).toReal =
                                phiAccurate-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                                  ∑ tau, phiAccurate-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                              (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking 1),
                                (MNoisy.law pi).toReal =
                                  phiNoisy-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                                    ∑ tau, phiNoisy-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                                0 <
                                  ((EconCSLib.pmfProb MAccurate.law fun pi =>
                                      better = KR21Monoculture.bestInSet pi remaining) *
                                    EconCSLib.pmfProb MNoisy.law fun pi =>
                                      worse = KR21Monoculture.bestInSet pi remaining) -
                                  (EconCSLib.pmfProb MAccurate.law fun pi =>
                                      worse = KR21Monoculture.bestInSet pi remaining) *
                                    EconCSLib.pmfProb MNoisy.law fun pi =>
                                      better = KR21Monoculture.bestInSet pi remaining)

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationD1_source_phi_likelihood_ratio_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 46

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

$$\begin{aligned} & * \\ & 1 \\ & (8) \\ & \Pr[\pi] = \frac{1}{Z} \phi^{-d(\pi, \pi^*)}, \end{aligned}$$

where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking r . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

E.1

Verifying Definition 2

We must show that when $\pi, \tau \sim F_\theta$,
 $E[\pi_1 - \pi_2 \mid \pi_1 \neq \tau_1] > 0$.

(E.1)

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For Candidate $n = \text{Fin}(n+2)$ with $n > 0$, a center ranking, $\phi > 1$, $\theta = \phi - 1$, and a value profile strictly ordered by that
↪ center, the source Mallows law has $q = \phi^{-1}$ and normalized mass $\phi^{-1} \cdot \text{KendallTau}(\text{center}, \pi) / \sum \tau$
↪ $\phi^{-1} \cdot \text{KendallTau}(\text{center}, \tau)$. For two iid draws π, τ , the top-disagreement event has positive probability and
↪ $E[(\text{value}(\text{first}(\pi)) - \text{value}(\text{second}(\pi))) \cdot 1_{\{\text{first}(\pi) \neq \text{first}(\tau)\}}]$ divided by that probability is strictly positive.

Archival source anchor: cited publication, lines 2444-2447

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
  theta = phi - 1 →
  0 < n →
  ∀ {value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ},
  EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value →
  have M := KR21Monoculture.concreteMallowsSpec center theta;
  M.q = phi⁻¹ ∧
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    (M.law pi).toReal =
      phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
      ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
  0 < EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent ∧
  0 <
    (EconCSLib.pmfPairIndicatorExp M.law M.law KR21Monoculture.disagreementEvent fun pair =>
      value (EconCSLib.SocialChoice.Ranking.firstChoice pair.1) -
      value (EconCSLib.SocialChoice.Ranking.secondChoice pair.1)) /
    EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationE1_source_phi_literal_conditional_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 47

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

```
*
1
(8)
Pr[ $\pi$ ] =  $\phi^{-d(\pi, \pi^*)}$  ,
Z
```

where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking r . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

We begin by expanding:

$E[\pi_1 - \pi_2 \mid \pi_1 \neq \pi_2] =$

$\sum_{i=1}^n \sum_{j=1}^n$

$(x_i - x_j) \Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \pi_2]$

$=$

$\sum_{i=1}^{n-1} \sum_{j=i+1}^n (x_i - x_j) (\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \pi_2] - \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \pi_2])$

Since $x_i > x_j$ for $i < j$, it suffices to show that for all $i < j$, $\Pr[\pi_1 = x_i \wedge \pi_2 = x_j \mid \pi_1 \neq \pi_2] \geq \Pr[\pi_1 = x_j \wedge \pi_2 = x_i \mid \pi_1 \neq \pi_2]$,

(E.2)

and that this holds strictly for some $i < j$. We simplify (E.2) as follows:

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For Candidate $n = \text{Fin}(n+2)$ with $n > 0$, a center ranking, $\phi > 1$, and $\theta = \phi - 1$, the source Mallows law has $q = \phi^{-1}$,
→ normalized mass $\phi^{-(\text{KendallTau}(\text{center}, \pi))} / \sum_{\tau} \phi^{-(\text{KendallTau}(\text{center}, \tau))}$, and positive iid top-disagreement
→ probability. Conditional on top disagreement, for every center-ordered pair c before d , the probability of first-two order
→ d, c is at most that of c, d ; for at least one center-ordered pair the inequality is strict.

Archival source anchor: cited publication, lines 2469-2474

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
  theta = phi - 1 →
  0 < n →
  have M := KR21Monoculture.concreteMallowsSpec center theta;
  M.q = phi⁻¹ ∧
  (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
    (M.law pi).toReal =
      phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
      ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
  0 < EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent ∧
  (∀ (c d : EconCSLib.SocialChoice.Ranking.Candidate n),
    EconCSLib.SocialChoice.Ranking.rankOf M.center c < EconCSLib.SocialChoice.Ranking.rankOf M.center d →
    (EconCSLib.pmfPairIndicatorExp M.law M.law KR21Monoculture.disagreementEvent fun pair =>
      if
        d = EconCSLib.SocialChoice.Ranking.firstChoice pair.1 ∧
        c = EconCSLib.SocialChoice.Ranking.secondChoice pair.1 then
      1
    )
  )
```

```

      else 0) /
      EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent ≤
      (EconCSLib.pmfPairIndicatorExp M.law M.law KR21Monoculture.disagreementEvent fun pair =>
      if
        c = EconCSLib.SocialChoice.Ranking.firstChoice pair.1 ∧
        d = EconCSLib.SocialChoice.Ranking.secondChoice pair.1 then
          1
        else 0) /
      EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent) ∧
    ∃ c d,
    EconCSLib.SocialChoice.Ranking.rankOf M.center c < EconCSLib.SocialChoice.Ranking.rankOf M.center d ∧
    (EconCSLib.pmfPairIndicatorExp M.law M.law KR21Monoculture.disagreementEvent fun pair =>
    if
      d = EconCSLib.SocialChoice.Ranking.firstChoice pair.1 ∧
      c = EconCSLib.SocialChoice.Ranking.secondChoice pair.1 then
        1
      else 0) /
    EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent <
    (EconCSLib.pmfPairIndicatorExp M.law M.law KR21Monoculture.disagreementEvent fun pair =>
    if
      c = EconCSLib.SocialChoice.Ranking.firstChoice pair.1 ∧
      d = EconCSLib.SocialChoice.Ranking.secondChoice pair.1 then
        1
      else 0) /
    EconCSLib.pmfPairProb M.law M.law KR21Monoculture.disagreementEvent

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationE2_source_phi_literal_conditional_semantic_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 48

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

```
*
1
(8)
Pr[ $\pi$ ] =  $\phi^{-d(\pi, \pi^*)}$ ,
Z
where  $Z$  is a normalizing constant. In this model,  $\phi > 1$  is the accuracy parameter: the larger  $\phi$  is, the more likely the ranking procedure is to output a ranking  $\pi$  that is close to the true ranking  $r$ . To instantiate this model, we need a notion of distance  $d(\cdot, \cdot)$  over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in  $\pi$  that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions  $F_\theta$  given by the Mallows Model satisfies Definition 1, defining  $\theta = \phi - 1$  (for consistency, so  $\theta$  is well-defined on  $(0, \infty)$ ).
```

[Next verbatim source excerpt]

and that this holds strictly for some $i < j$. We simplify (E.2) as follows:

```
Pr[ $\pi_1 = x_i \mid \pi_2 = x_j \mid \pi_1 \neq \tau_1$ ] > Pr[ $\pi_1 = x_j \mid \pi_2 = x_i \mid \pi_1 \neq \tau_1$ ]
Pr[ $\pi_1 = x_i \mid \pi_2 = x_j \mid \pi_1 \neq \tau_1$ ]
Pr[ $\pi_1 = x_j \mid \pi_2 = x_i \mid \pi_1 \neq \tau_1$ ]
>
Pr[ $\pi_1 \neq \tau_1$ ]
Pr[ $\pi_1 \neq \tau_1$ ]
 $\Leftrightarrow$  Pr[ $\pi_1 = x_i \mid \pi_2 = x_j \mid \pi_1 \neq \tau_1$ ] > Pr[ $\pi_1 = x_j \mid \pi_2 = x_i \mid \pi_1 \neq \tau_1$ ]
 $\Leftrightarrow$ 
 $\Leftrightarrow$  Pr[ $\pi_1 = x_i \mid \pi_2 = x_j \mid \tau_1 \neq x_i$ ] > Pr[ $\pi_1 = x_j \mid \pi_2 = x_i \mid \tau_1 \neq x_j$ ]
 $\Leftrightarrow$  Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] Pr[ $\tau_1 \neq x_i$ ] > Pr[ $\pi_1 = x_j \mid \pi_2 = x_i$ ] Pr[ $\tau_1 \neq x_j$ ]
```

(E.3)

We can simplify (E.3) using Lemmas 5 and 6. Let $|i - j|$ denote the difference in rank between x_i and x_j .

```
Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] Pr[ $\tau_1 \neq x_i$ ] - Pr[ $\pi_1 = x_j \mid \pi_2 = x_i$ ] Pr[ $\tau_1 \neq x_j$ ]
= Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] (1 - Pr[ $\tau_1 = x_i$ ]) -  $\phi^{-1}$  Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] (1 - Pr[ $\tau_1 = x_j$ ])
= Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] (1 - Pr[ $\tau_1 = x_i$ ]) -  $\phi^{-1}$  Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] (1 -  $\phi^{-|i-j|}$  Pr[ $\tau_1 = x_i$ ])
= Pr[ $\pi_1 = x_i \mid \pi_2 = x_j$ ] (1 - Pr[ $\tau_1 = x_i$ ]) -  $\phi^{-1}$  -  $\phi^{-|i-j|-1}$  Pr[ $\tau_1 = x_i$ ])
29
```

[form-feed in source extraction]

This is positive if and only if

```
1 - Pr[ $\tau_1 = x_i$ ] -  $\phi^{-1}$  -  $\phi^{-|i-j|-1}$  Pr[ $\tau_1 = x_i$ ] > 0
```

[U+0011 control character]

[U+0010 control character]

```
 $\Leftrightarrow$  Pr[ $\tau_1 = x_i$ ] 1 -  $\phi^{-|i-j|-1}$  < 1 -  $\phi^{-1}$ 
```

```
1 -  $\phi^{-1}$ 
```

```
1 -  $\phi^{-|i-j|-1}$ 
```

```
-1
```

```
1 -  $\phi$ 
```

```
1 -  $\phi^{-1}$ 
```

```
 $\Leftrightarrow$  i-1
```

```
<
```

```
-n
```

```
 $\phi$  (1 -  $\phi$ )
```

```
1 -  $\phi^{-|i-j|-1}$ 
```

```
 $\Leftrightarrow$  Pr[ $\tau_1 = x_i$ ] <
```

```
 $\Leftrightarrow$   $\phi i - 1$  (1 -  $\phi^{-n}$ ) > 1 -  $\phi^{-|i-j|-1}$ 
```

This is weakly true for any $i < j$ because $\phi i - 1 \geq 1$ and $|i - j| + 1 \leq n$, and it is strictly true for any i, j other than 1 and n . Thus, $E[\pi_1 - \pi_2 \mid \pi_1 \neq \tau_1] > 0$.

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For Candidate $n = \text{Fin}(n+2)$ with $n > 0$, a center ranking, $\phi > 1$, and $\theta = \phi - 1$, the source Mallows law has $q = \phi^{-1}$ and

- normalized mass $\phi^{-(\text{KendallTau}(\text{center}, \pi))} / \sum_{\tau} \phi^{-(\text{KendallTau}(\text{center}, \tau))}$. For every center-ordered pair c before
- d , $P(\text{first}=c, \text{second}=d)P(\text{first} \neq c) \geq P(\text{first}=d, \text{second}=c)P(\text{first} \neq d)$, and this inequality is strict for at least one
- center-ordered pair. The target uses the corrected positive final product term; it does not certify the printed negative-sign
- intermediate expansion.

Archival source anchor: cited publication, lines 2474-2515

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```

∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
    theta = phi - 1 →
      0 < n →
        have M := KR21Monoculture.concreteMallowsSpec center theta;
        M.q = phi⁻¹ ∧
          (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
            (M.law pi).toReal =
              phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
              ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
          (∀ (c d : EconCSLib.SocialChoice.Ranking.Candidate n),
            EconCSLib.SocialChoice.Ranking.rankOf M.center c < EconCSLib.SocialChoice.Ranking.rankOf M.center d →
              0 ≤
                M.firstSecondChoiceProb c d * KR21Monoculture.firstChoiceMissProb M.law c -
                M.firstSecondChoiceProb d c * KR21Monoculture.firstChoiceMissProb M.law d) ∧
          ∃ c d,
            EconCSLib.SocialChoice.Ranking.rankOf M.center c < EconCSLib.SocialChoice.Ranking.rankOf M.center d ∧
              0 <
                M.firstSecondChoiceProb c d * KR21Monoculture.firstChoiceMissProb M.law c -
                M.firstSecondChoiceProb d c * KR21Monoculture.firstChoiceMissProb M.law d

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationE3_source_phi_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 49

Source locator: cited publication, lines 2703-2850

Verbatim source input

Lemma 4. Let $\{y_i\}_{i=1}^n$, $\{p_i\}_{i=1}^n$, and $\{q_i\}_{i=1}^n$ be sequences such that

- y_i is strictly increasing.

$\prod_{i=1}^n$

$\prod_{i=1}^n$

•

$\sum_{i=1}^n p_i =$

$\sum_{i=1}^n q_i = 1$.

- p_{q_i} is decreasing.

$\prod_{i=1}^n$

$\prod_{i=1}^n$

Then, $\sum_{i=1}^n p_i y_i < \sum_{i=1}^n q_i y_i$.

Proof. First, note that there exists j such that $p_i > q_i$ for $i < j$ and $p_i \leq q_i$ for $i \geq j$. To see this, let j be the smallest index such that $p_j \leq q_j$. Such a j must exist because p_i and q_i both sum to 1, so it cannot be the case that $p_i > q_i$ for all i . This implies $p_i / q_i \leq 1$, and since p_i / q_i is decreasing, $p_i \leq q_i$ for $i \geq j$.

Next, note that

$\sum_{i=1}^n$

$\sum_{i=1}^n$

$\sum_{i=1}^n$

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$\sum_{i=1}^n$

$\sum_{i=1}^n$

$$\begin{aligned}
& \sum_{i=1}^n (p_i - q_i) y_{j-1} - \\
& \sum_{i=1}^{j-1} (p_i - q_i) y_j \\
& \sum_{i=j}^n (q_i - p_i) y_j \\
& \sum_{i=1}^n (p_i - q_i) y_{j-1} - \\
& \sum_{i=1}^j (p_i - q_i) y_j \\
& \sum_{i=1}^j (q_i - p_i) y_i \\
& \sum_{i=1}^j (p_i - q_i) y_j \\
& \sum_{i=1}^j (p_i - q_i) (y_{j-1} - y_j) \\
& < 0
\end{aligned}$$

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For Candidate $n = \text{Fin}(n+2)$, real sequences p, q , and a strictly increasing real sequence y , assume $\sum_i p_i = \sum_i q_i = 1$,
 $\hookrightarrow p_i * q_j - p_j * q_i \geq 0$ for every $i < j$, and there exist $i < j$ with $p_i * q_j - p_j * q_i > 0$. Then $\sum_i p_i * y_i < \sum_i q_i * y_i$. The
 $\hookrightarrow$ explicit strict cross-ratio witness is required; strictness is not inferred from the non-strict pairwise condition alone.

Archival source anchor: cited publication, lines 2703-2713

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

$$\begin{aligned}
& \forall (n : \mathbb{N}) \{p \ q \ y : \text{EconCSLib.SocialChoice.Ranking.Candidate } n \rightarrow \mathbb{R}\}, \\
& \quad \sum_i p_i = 1 \rightarrow \\
& \quad \sum_i q_i = 1 \rightarrow \\
& \quad (\forall (i \ j : \text{EconCSLib.SocialChoice.Ranking.Candidate } n), i < j \rightarrow 0 \leq p_i * q_j - p_j * q_i) \rightarrow \\
& \quad (\exists i \ j, i < j \wedge 0 < p_i * q_j - p_j * q_i) \rightarrow \text{StrictMono } y \rightarrow \sum_i p_i * y_i < \sum_i q_i * y_i
\end{aligned}$$

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma4_weighted_average_lt_of_cross_ratio

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation



Review row 50

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

$$\Pr[\pi] = \frac{1}{Z} \phi^{-d(\pi, \pi^*)},$$

where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking r . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

Lemma 5. For $x_i > x_j$,
 $\Pr[\pi_1 = x_i \wedge \pi_2 = x_j] = \phi \Pr[\pi_1 = x_j \wedge \pi_2 = x_i]$.

(F.1)

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
    theta = phi - 1 →
      ∀ {c d : EconCSLib.SocialChoice.Ranking.Candidate n},
        EconCSLib.SocialChoice.Ranking.rankOf center c < EconCSLib.SocialChoice.Ranking.rankOf center d →
          have M := KR21Monoculture.concreteMallowsSpec center theta;
            M.q = phi⁻¹ ∧
              (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                (M.law pi).toReal =
                  phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                    ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                M.firstSecondChoiceProb c d = phi * M.firstSecondChoiceProb d c
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationF1_source_phi_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 51

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

$$\Pr[\pi] = \frac{1}{Z} \phi^{-d(\pi, \pi^*)},$$
where Z is a normalizing constant. In this model, $\phi > 1$ is the accuracy parameter: the larger ϕ is, the more likely the ranking procedure is to output a ranking π that is close to the true ranking π^* . To instantiate this model, we need a notion of distance $d(\cdot, \cdot)$ over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in π that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions F_θ given by the Mallows Model satisfies Definition 1, defining $\theta = \phi - 1$ (for consistency, so θ is well-defined on $(0, \infty)$).

[Next verbatim source excerpt]

Lemma 6. For $1 \leq i \leq n$,
 $\Pr[\pi_1 = x_i] =$

$$\frac{1 - \phi^{-1}}{\phi^{i-1} (1 - \phi^{-n})} \quad (F.2)$$

Proof. Let π_{-i} be a permutation over all items except i . Then,
 $\Pr[\pi_1 = x_i] =$

$$\begin{aligned} & \sum \Pr[\pi_1 = x_i \mid \pi_{-i}] \Pr[\pi_{-i}] \\ &= \sum \phi^{-(i-1)} \Pr[\pi_{-i}] \\ &= \phi^{-(i-1)} \sum \Pr[\pi_{-i}] \\ &= \phi^{-(i-1)} \end{aligned}$$

Note that $\Pr[\pi_{-i}]$ doesn't depend on which $n-1$ items are being ranked, so this term appears for any i . Moreover, $\sum_{i=1}^n \Pr[\pi_1 = x_i] = 1$. Therefore, we have

[form-feed in source extraction]

$$\Pr[\pi_1 = x_i] \propto \phi^{-(i-1)}.$$

Normalizing, we get

$$\Pr[\pi_1 = x_i] = \frac{\phi^{-(i-1)}}{\sum_{j=1}^n \phi^{-(j-1)}} =$$

$$\frac{\phi^{-(i-1)}}{1 - \phi^{-n}} \frac{1 - \phi^{-1}}{\phi^{i-1} (1 - \phi^{-n})}$$

Intuitively, any permutation over $n-1$ items is equally likely regardless of what those items are, and inserting any item at the front of this permutation yields a likelihood proportional to the number of additional inversions this causes, which is equal to the item's position on the list.

Expanded PaperInterface specification

```

∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
1 < phi →
  theta = phi - 1 →
    ∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n),
      have M := KR21Monoculture.concreteMallowsSpec center theta;
      M.q = phi⁻¹ ∧
        (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
          (M.law pi).toReal =
            phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
            ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
          KR21Monoculture.firstChoiceProb M.law c =
            phi⁻¹ ^ ↑(EconCSLib.SocialChoice.Ranking.rankOf M.center c) /
            KR21Monoculture.candidateRankPowerSum n phi⁻¹

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma6_source_phi_rank_power_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 52

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

```
*
1
(8)
Pr[ $\pi$ ] =  $\phi^{-d(\pi, \pi^*)}$  ,
Z
where  $Z$  is a normalizing constant. In this model,  $\phi > 1$  is the accuracy parameter: the larger  $\phi$  is, the more likely the ranking procedure is to output a ranking  $\pi$  that is close to the true ranking  $r$ . To instantiate this model, we need a notion of distance  $d(\cdot, \cdot)$  over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in  $\pi$  that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions  $F_\theta$  given by the Mallows Model satisfies Definition 1, defining  $\theta = \phi - 1$  (for consistency, so  $\theta$  is well-defined on  $(0, \infty)$ ).
```

[Next verbatim source excerpt]

Lemma 6. For $1 \leq i \leq n$,
 $\Pr[\pi_1 = x_i] =$

$$\frac{1 - \phi^{-1}}{\phi^{i-1} (1 - \phi^{-n})}$$

(F.2)

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
    theta = phi - 1 →
      ∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n),
        have M := KR21Monoculture.concreteMallowsSpec center theta;
        M.q = phi⁻¹ ∧
          (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
            (M.law pi).toReal =
              phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
              ∑ tau, phi⁻¹ ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
            KR21Monoculture.firstChoiceProb M.law c =
              (1 - phi⁻¹) / (phi ^ ↑(EconCSLib.SocialChoice.Ranking.rankOf M.center c) * (1 - phi⁻¹ ^ (n + 2)))
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.equationF2_source_phi_closed_form_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 53

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

The Mallows Model also appears frequently in the ranking literature [8, 11], and is much more analytically tractable than RUMs. Under the Mallows Model, the likelihood of a permutation is related to its distance from the true ranking π^* :

```
*
1
(8)
Pr[ $\pi$ ] =  $\phi^{-d(\pi, \pi^*)}$ ,
Z
where  $Z$  is a normalizing constant. In this model,  $\phi > 1$  is the accuracy parameter: the larger  $\phi$  is, the more likely the ranking procedure is to output a ranking  $\pi$  that is close to the true ranking  $r$ . To instantiate this model, we need a notion of distance  $d(\cdot, \cdot)$  over permutations. For this, we'll use Kendall tau distance, another standard notion in the literature, which is simply the number of pairs of elements in  $\pi$  that are incorrectly ordered [10]. In Appendix D, we verify that the family of distributions  $F_\theta$  given by the Mallows Model satisfies Definition 1, defining  $\theta = \phi - 1$  (for consistency, so  $\theta$  is well-defined on  $(0, \infty)$ ).
```

[Next verbatim source excerpt]

Lemma 7. For the Mallows Model, $UH(\theta_A, \theta_H) > UHH(\theta_A, \theta_H)$.
Proof. Intuitively, this is because selecting first is better than selecting second. To prove this, let π and τ be ranking generated by independent human evaluators under the Mallows Model, i.e., $\pi, \tau \sim F_{\theta H}$.
 $UH(\theta_A, \theta_H) - UHH(\theta_A, \theta_H) = E[\pi_1] - E[\tau_1 \cdot 1_{\pi_1 \neq \tau_1} + \tau_2 \cdot 1_{\pi_1 = \tau_1}]$
 $= E[(\pi_1 - \tau_2) \cdot 1_{\pi_1 = \tau_1}]$

$= E[(\pi_1 - \pi_2) \cdot 1_{\pi_1 = \tau_1}]$
For any $i < j$, conditioned on $\pi_1 = \tau_1$, they are more likely to be correctly ordered than not:

```
X
E[( $\pi_1 - \pi_2$ ) · 1 $_{\pi_1 = \tau_1}$ ] =
(Pr[ $\pi_1 = x_i$  n  $\tau_1 = x_i$  n  $\pi_2 = x_j$ ] - Pr[ $\pi_1 = x_j$  n  $\tau_1 = x_j$  n  $\pi_2 = x_i$ ]) (x $_i$  - x $_j$ )
i<j
=
X
(Pr[ $\pi_1 = x_i$  n  $\pi_2 = x_j$ ] Pr[ $\tau_1 = x_i$ ] - Pr[ $\pi_1 = x_j$  n  $\pi_2 = x_i$ ] Pr[ $\tau_1 = x_j$ ]) (x $_i$  - x $_j$ )
i<j
>
X
(Pr[ $\pi_1 = x_i$  n  $\pi_2 = x_j$ ] Pr[ $\tau_1 = x_j$ ] - Pr[ $\pi_1 = x_j$  n  $\pi_2 = x_i$ ] Pr[ $\tau_1 = x_j$ ]) (x $_i$  - x $_j$ )
i<j
=
X
(Pr[ $\pi_1 = x_i$  n  $\pi_2 = x_j$ ] - Pr[ $\pi_1 = x_j$  n  $\pi_2 = x_i$ ]) (x $_i$  - x $_j$ )
i<j
≥
X
( $\phi^H$  Pr[ $\pi_1 = x_j$  n  $\pi_2 = x_i$ ] - Pr[ $\pi_1 = x_j$  n  $\pi_2 = x_i$ ]) (x $_i$  - x $_j$ )
i<j
>0
```

Expanded PaperInterface specification

```
∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) (phi theta : ℝ),
  1 < phi →
    theta = phi - 1 →
      ∀ {value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ},
        EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value →
          have M := KR21Monoculture.concreteMallowsSpec center theta;
            M.q = phi-1 ∧
              (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                (M.law pi).toReal =
                  phi-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                    ∑ tau, phi-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent M.law M.law value <
                  EconCSLib.SocialChoice.Ranking.expectedFirstMoverUtility M.law value
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma7_source_phi_complete

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 54

Source locator: cited publication, lines 133-138

Verbatim source input

More formally, we model the n candidates as having intrinsic values $x_1, \dots, x_n$, where any employer would derive utility x_i from hiring candidate i . Throughout the paper, we assume without loss of generality that $x_1 > x_2 > \dots > x_n$. These values, however, are unknown to the employer; instead, they must use some noisy procedure to rank the candidates. We model such a procedure as a randomized mechanism R that takes in the true candidate values and draws a permutation π over those candidates from some distribution. Our main results hold for families of distributions over permutations as defined below:

[Next verbatim source excerpt]

Lemma 8. Under the Mallows model, the probability that any two items $i < j$ are correctly ranked increases monotonically with the accuracy parameter ϕ .

1 Alternatively, we could prove this by showing that for any permutation with i in front, the permutation in which i and $i - 1$ are swapped is ϕ times more likely, and thus, $i - 1$ is ϕ times more likely to be in front than i .

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[form-feed in source extraction]

Proof. Let $\text{inv}(\pi)$ be the number of inversions in a permutation π . Under the Mallows model, the probability of observing π is proportional to $\phi^{-\text{inv}(\pi)}$. Let $S_i[U+001F \text{ control character}]j$ (resp. $S_j[U+001F \text{ control character}]i$) be the set of permutations where i is ranked before j (resp. j is ranked before i). Then, the probability i is ranked before j is

$$\begin{aligned} & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \Pr[i [U+001F \text{ control character}] j] = \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)} + \sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \text{We will show that } \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} > \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}}. \text{ Note that this is equivalent to showing } \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} > \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}}. \text{ Note that} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \end{aligned}$$

Let $\pi_{i:j}$ be the subsequence of π containing elements i through j . Then, we have

$$\begin{aligned} & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_i[U+001F \text{ control character}]j} \phi^{-\text{inv}(\pi)}} \\ & \frac{P}{\sum_{\pi \in S_j[U+001F \text{ control character}]i} \phi^{-\text{inv}(\pi)}} \end{aligned}$$

Intuitively, the term

$\phi^{\text{inv}(\pi_{i:j}) - \text{inv}(\pi)}$ does not depend on $\pi_{i:j}$ because for any $\pi_{i:j}$, if we fix

$\pi_{\setminus i:j}$, the order and positions of the remaining elements, the number of inversions involving at least one element outside of $i : j$ (i.e., $\text{inv}(\pi_{\setminus i:j}) - \text{inv}(\pi_{i:j})$) is a constant. For fixed $\pi_{i:j}$, there is a bijection between permutations


```

0
 $\pi \emptyset : \pi i : j$ 
=  $\pi i : j$  and a fixed order and position of the remaining elements (excluding  $i : j$ ), meaning this sum
does not depend on  $\pi i : j$ . Thus, for the remainder of this proof, we can assume without loss of generality
that  $i = 1$  and  $j = n$ . The quantity of interest becomes
P
-  $\text{inv}(\pi 1 : n)$ 
 $\text{Pr}[1 \text{ [U+001F control character] } n]$ 
 $\pi : \pi \in S1[\text{U+001F control character}] n \varphi$ 
=  $P \ 1 : n$ 
-  $\text{inv}(\pi 1 : n)$ 
 $\text{Pr}[n \text{ [U+001F control character] } 1]$ 
 $\pi i : j : \pi \in S n[\text{U+001F control character}] 1 \varphi$ 
P
-  $\text{inv}(\pi)$ 
 $\pi \in S1[\text{U+001F control character}] n \varphi$ 
= P
-  $\text{inv}(\pi)$ 
 $\pi \in S n[\text{U+001F control character}] 1 \varphi$ 
Next, we observe that we can similarly ignore inversions between two elements that are neither 1 nor n.
To see this, let  $\text{inv}_{1,n}(\pi)$  be the number of inversions involving at least one of 1 and n. Then, if we fix the
order and positions of 1 and n, all possible permutations of the remaining elements 2 through  $n - 1$  produce
the same number of inversions  $\text{inv}_{1,n}(\pi)$ . More formally, let  $\pi(1)$  and  $\pi(n)$  be the respective positions of
elements 1 and n. Then, this we have
X
X
X
 $\varphi - \text{inv}(\pi) =$ 
 $\varphi - \text{inv}(\pi)$ 
 $\pi \in S1[\text{U+001F control character}] n$ 

 $k < \pi : \pi(1) = k, \pi(n) = \backslash$ 

=

X

=

X

X

X

 $\varphi - \text{inv}_{1,n}(\varphi) \cdot \varphi \text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 

 $k < \pi : \pi(1) = k, \pi(n) = \backslash$ 

 $\varphi - (k-1) - (n-\backslash)$ 

 $k < \backslash$ 

X

 $\varphi \text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 

 $\pi : \pi(1) = k, \pi(n) = \backslash$ 

 $\text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 
As noted above,
does not depend on k or  $\backslash$ , since every permutation of
 $\pi : \pi(1) = k, \pi(n) = \backslash \varphi$ 
the remaining elements yields the same number of inversions among them regardless of k and  $\backslash$ . A similar
argument yields
X
X
X
 $\varphi - \text{inv}(\pi) =$ 
 $\varphi - (k-1) - (n-\backslash) + 1$ 
 $\varphi \text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 
P

 $\pi \in S n[\text{U+001F control character}] 1$ 

 $k > \backslash$ 

 $\pi : \pi(1) = k, \pi(n) = \backslash$ 

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[form-feed in source extraction]
Putting these together, we have
P
P
 $-(k-1) - (n-\backslash)$ 
 $\text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 
 $\text{Pr}[1 \text{ [U+001F control character] } n]$ 
 $k < \backslash \varphi$ 
 $\pi : \pi(1) = k, \pi(n) = \backslash \varphi$ 
P
= P
 $\text{inv}_{1,n}(\varphi) - \text{inv}(\pi)$ 
 $-(k-1) - (n-\backslash) + 1$ 
 $\text{Pr}[n \text{ [U+001F control character] } 1]$ 
 $\pi : \pi(1) = k, \pi(n) = \backslash \varphi$ 
 $k > \backslash \varphi$ 

```

$$\begin{aligned} & \varphi - (k-1) - (n-1) \\ &= \varphi - k < \varphi - (k-1) - (n-1) + 1 \\ & \varphi > \varphi \\ & \varphi - (k-1) - (n-1) \\ & \varphi - 1 \\ &= \varphi - k < \varphi - (k-1) - (n-1) + 1 \cdot n-1 \\ & \varphi \\ & \varphi > \varphi \\ & \varphi - k \\ &= \varphi - k < \varphi - k + 1 \\ & \varphi > \varphi \end{aligned}$$

Note that each term in the numerator is strictly increasing in φ , while each term in the denominator is weakly decreasing in φ . As a result, $d\varphi \Pr[n[U+001F \text{ control character}]] > 0$, meaning for any $i < j$, $d\varphi \Pr[i[U+001F \text{ control character}]] > 0$.

Expanded PaperInterface specification

```

∀ {n : ℕ} (center : EconCSLib.SocialChoice.Ranking.Ranking n) {phiMore phiLess : ℝ},
  1 < phiLess →
    phiLess < phiMore →
      ∀ {c d : EconCSLib.SocialChoice.Ranking.Candidate n},
        EconCSLib.SocialChoice.Ranking.rankOf center c < EconCSLib.SocialChoice.Ranking.rankOf center d →
          have Mless := KR21Monoculture.concreteMallowsSpec center (phiLess - 1);
          have Mmore := KR21Monoculture.concreteMallowsSpec center (phiMore - 1);
          Mless.q = phiLess-1 ∧
            Mmore.q = phiMore-1 ∧
              (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                (Mless.law pi).toReal =
                  phiLess-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                    ∑ tau, phiLess-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                (∀ (pi : EconCSLib.SocialChoice.Ranking.Ranking n),
                  (Mmore.law pi).toReal =
                    phiMore-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center pi /
                      ∑ tau, phiMore-1 ^ EconCSLib.SocialChoice.Ranking.kendallTau center tau) ∧
                  (EconCSLib.pmfProb Mless.law fun pi =>
                    EconCSLib.SocialChoice.Ranking.rankOf pi c < EconCSLib.SocialChoice.Ranking.rankOf pi d) <
                    EconCSLib.pmfProb Mmore.law fun pi =>
                    EconCSLib.SocialChoice.Ranking.rankOf pi c < EconCSLib.SocialChoice.Ranking.rankOf pi d)

```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.lemma8_source_mallows_phi_pairwise_correct_probability_lt

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 55

Source locator: cited publication, lines 497-500

Verbatim source input

First, we must define a family of RUMs that satisfies the conditions of Definition 1. Assume without loss of generality that the noise distribution E has unit variance. Then, consider the family of RUMs parameterized by θ in which candidates are ranked according to $x_i + \varepsilon \theta_i$. By this definition, the standard deviation of the noise for a particular value of θ is simply $1/\theta$. Intuitively, larger values of θ reduce the effect of the noise,

[Next verbatim source excerpt]

Finally, there exist noise distributions that violate Definition 2 for any candidate distribution D . In particular, the RUM family defined by the Gumbel distribution is well-known to be equivalent to the Plackett-Luce model of ranking, which is generated by sequentially selecting candidate i with probability

$\frac{\exp(\theta x_i)}{\sum_{j \in S} \exp(\theta x_j)}$

P

(7)

where S is the set of remaining candidates [12, 4]. Under the Plackett-Luce model, for any θ , $UAH(\theta, \theta) =$

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For any finite candidate set, common location, innovation scale $s > 0$, and $\theta > 0$, let $arrival_i$ be iid unit-rate exponential and set $\epsilon_i = location - s \cdot \log(arrival_i)$. The arrival law is the iid exponential product law, arrivals are almost surely positive, the innovation law is its coordinatewise pushforward, and ranking value $i + \epsilon_i / \theta$ has exactly the Plackett-Luce ranking law at inverse temperature θ/s . If the visible analytic premise $Var(-\log T) = \pi^2/6$ holds for unit-rate exponential T , then the specialization $s = \sqrt{6}/\pi$ has coordinate variance one and Plackett-Luce inverse temperature $\theta/(\sqrt{6}/\pi)$. The archival unit-variance/same- θ formula is not asserted.

Archival source anchor: cited publication, lines 497-548

Recorded basis User instruction 2026-07-25: assume whatever convention makes the result true, provided the convention is explicit and the archival claim is not represented as proved.

Expanded PaperInterface specification

```
∀ {n : ℕ} (location : ℝ) {s theta : ℝ},
  0 < s →
  0 < theta →
  ∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ),
    (KR21Monoculture.gumbelArrivalLaw n = MeasureTheory.Measure.pi fun x => ProbabilityTheory.expMeasure 1) ∧
    (∀" (arrival : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂KR21Monoculture.gumbelArrivalLaw n,
      ∀ (i : EconCSLib.SocialChoice.Ranking.Candidate n), 0 < arrival i) ∧
    (∀ (arrival : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
      (i : EconCSLib.SocialChoice.Ranking.Candidate n),
      KR21Monoculture.commonLocationPositiveScaleGumbelNoise location s arrival i =
        location + -s * Real.log (arrival i)) ∧
    (∀ (arrival : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
      (i : EconCSLib.SocialChoice.Ranking.Candidate n),
      KR21Monoculture.commonLocationPositiveScaleGumbelScores location s theta value arrival i =
        value i + (location + -s * Real.log (arrival i)) / theta) ∧
    (MeasureTheory.Measure.map (KR21Monoculture.commonLocationPositiveScaleGumbelNoise location s)
      (KR21Monoculture.gumbelArrivalLaw n) =
      MeasureTheory.Measure.pi fun x =>
        MeasureTheory.Measure.map (fun arrival => location + -s * Real.log arrival)
          (ProbabilityTheory.expMeasure 1)) ∧
    KR21Monoculture.commonLocationPositiveScaleGumbelRUMRankingPMF location s theta value =
      KR21Monoculture.plackettLuceRankingPMF (theta / s) value ∧
    (ProbabilityTheory.variance id KR21Monoculture.scaleOneGumbelMeasure = Real.pi ^ 2 / 6 →
      (KR21Monoculture.sourceUnitVarianceGumbelNoiseLaw location =
        MeasureTheory.Measure.pi fun x =>
          MeasureTheory.Measure.map (fun arrival => location + -(√6 / Real.pi) * Real.log arrival)
            (ProbabilityTheory.expMeasure 1)) ∧
      (∀ (epsilon : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ)
        (i : EconCSLib.SocialChoice.Ranking.Candidate n),
        KR21Monoculture.sourceUnitVarianceGumbelScores theta value epsilon i =
          value i + epsilon i / theta) ∧
      (∀ (i : EconCSLib.SocialChoice.Ranking.Candidate n),
        ProbabilityTheory.variance (fun epsilon => epsilon i)
          (KR21Monoculture.sourceUnitVarianceGumbelNoiseLaw location) =
          1) ∧
      KR21Monoculture.sourceUnitVarianceGumbelRUMRankingPMF location theta value =
        KR21Monoculture.plackettLuceRankingPMF (theta / (√6 / Real.pi)) value)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.section31_corrected_gumbel_plackett_luce_target

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 56

Source locator: cited publication, lines 176-182

Verbatim source input

Each firm in our model has access to the same underlying pool of n candidates, which they rank using a randomized mechanism R to get a permutation π as described above. Then, in a random order, each firm hires the highest-ranked remaining candidate according to their ranking. Thus, if two firms both rank candidate i first, only one of them can hire i ; the other must hire the next available candidate according to their ranking. In our model, candidates automatically accept the offer they get from a firm. For the sake of simplicity, throughout this paper, we restrict ourselves to the case where there are two firms hiring one candidate each, although our model readily generalizes to more complex cases.

[Next verbatim source excerpt]

where S is the set of remaining candidates [12, 4]. Under the Plackett-Luce model, for any θ , $U_A(\theta, \theta) = U_A(\theta, \theta)$. To see this, suppose the firm that hires first selects candidate i^* . Then, the firm that hires second

[form-feed in source extraction]
gets each candidate i with probability given by (7) with $S = \{1, \dots, n\} \setminus i^*$. As a result, by (3), if $\pi, \sigma \sim F_\theta$, $E[\pi_1 - \pi_2 \mid \pi_1 \neq \sigma_1] = 0$ for any candidate distribution D , meaning the Plackett-Luce model never meets Definition 2. Thus, under the Plackett-Luce model, monoculture has no effect—the optimal strategy is always to use the best available ranking, regardless of competitors' strategies.

Expanded PaperInterface specification

```
∀ {n : ℕ} (D : MeasureTheory.Measure (KR21Monoculture.ValueProfile n)) [MeasureTheory.IsProbabilityMeasure D]
(center : EconCSLib.SocialChoice.Ranking.Ranking n) {thetaA thetaH : ℝ},
  0 < thetaA →
  0 < thetaH →
  (∀ (c : EconCSLib.SocialChoice.Ranking.Candidate n), MeasureTheory.Integrable (fun value => value c) D) →
  (∀ (value : EconCSLib.SocialChoice.Ranking.Candidate n → ℝ) ∂D,
    EconCSLib.SocialChoice.Ranking.StrictlyOrderedBy center value) →
  have M := fun value =>
    { algorithmRanking := KR21Monoculture.plackettLuceRankingPMF thetaA value,
      humanRanking := KR21Monoculture.plackettLuceRankingPMF thetaH value, value := value };
  (∀ (self other : KR21Monoculture.Strategy),
    MeasureTheory.Integrable (fun value => (M value).payoffAgainst self other) D) ∧
  (thetaA ≤ thetaH →
    ∫ (value : KR21Monoculture.ValueProfile n),
      (M value).payoffAgainst KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm ∂D ≥
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.human KR21Monoculture.Strategy.algorithm ∂D ∧
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human ∂D ≥
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human ∂D) ∧
    (thetaA < thetaH →
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.human KR21Monoculture.Strategy.algorithm ∂D ≥
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.algorithm ∂D ∧
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.human KR21Monoculture.Strategy.human ∂D ≥
      ∫ (value : KR21Monoculture.ValueProfile n),
        (M value).payoffAgainst KR21Monoculture.Strategy.algorithm KR21Monoculture.Strategy.human ∂D)
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.section31_plackettLuce_outer_best_available_weakly_dominates

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 57

Source locator: cited publication, lines 139-165

Verbatim source input

Definition 1 (Noisy permutation family). A noisy permutation family F_θ is a family of distributions over permutations that satisfies the following conditions for any $\theta > 0$ and set of candidates x :

1. (Differentiability) For any permutation π , $\Pr F_\theta[\pi]$ is continuous and differentiable in θ .
2. (Asymptotic optimality) For the true ranking π^* , $\lim_{\theta \rightarrow \infty} \Pr F_\theta[\pi^*] = 1$.
3. (Monotonicity) For any (possibly empty) $S \subset x$, let $\pi(-S)$ be the partial ranking produced by removing

($-S$) the items in S from π . Let π_1 denote the value of the top-ranked candidate according to $\pi(-S)$. For

any $\theta > \theta$,

$\Pr F_\theta[\pi_1] \geq \Pr F_\theta[\pi_1]$

$\Pr F_\theta[\pi_1] \geq \Pr F_\theta[\pi_1]$

$\Pr F_\theta[\pi_1] \geq \Pr F_\theta[\pi_1]$

$\Pr F_\theta[\pi_1] \geq \Pr F_\theta[\pi_1]$

$\Pr F_\theta[\pi_1] \geq \Pr F_\theta[\pi_1]$

Moreover, for $S = \emptyset$, (1) holds with strict inequality.

θ serves as an “accuracy parameter”: for large θ , the noisy ranking converges to the true ranking over candidates. The monotonicity condition states that a higher value of θ leads to a better first choice, even if some of the candidates are removed after ranking. Removal after ranking (as opposed to before) is important because some of the ranking models we will consider later do not satisfy Independence of Irrelevant Alternatives. Examples of noisy permutation families include Random Utility Models [23] and the Mallows Model [14], both of which we will discuss in detail later.

[Next verbatim source excerpt]

B

3-candidate RUM Counterexamples

B.1

Violating Definition 2

Here, we provide a noise mode E , accuracy parameter θ , and candidate distribution D such that $U_{AH} < U_{AA}$. Choose the noise distribution E and accuracy parameter θ such that

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

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$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

$\Pr E[\pi_1] \geq \Pr E[\pi_1]$

```

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1
[U+F8F4 delimiter glyph]
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[U+F8F4 delimiter glyph]
[U+F8F2 delimiter glyph]
-1
ε
=
θ [U+F8F4 delimiter glyph]
10
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[U+F8F3 delimiter glyph]
-10

```

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w.p. 1-6
2
w.p. 1-6
2
w.p. 26
w.p. 26

```

Again, while this noise model doesn't satisfy Definition 1, we can approximate it arbitrarily closely with the sum of tightly concentrated Gaussians. Let the $\theta_A = 1.1\theta$ and $\theta_H = 0.9\theta$.

We will show that for these parameters, $UAH(\theta_A, \theta_H) > UHH(\theta_A, \theta_H)$, i.e., it is somehow better to choose after a better opponent than after a worse opponent. At a high level, the reasoning for this is as follows:

1. When choosing first, the only difference between the algorithm and the human evaluator is that the algorithm is more likely to choose x_2 than x_3 . Both strategies have identical probabilities of selecting x_1 .

2. When choosing second, the human evaluator's utility is higher when x_2 has already been chosen than when x_3 has already been chosen. This is because when x_2 is unavailable, the human evaluator is almost guaranteed to get x_1 ; when x_3 is unavailable, the human evaluator will choose x_2 with probability $\approx 1/4$.

Let τ and π be rankings generated by the algorithm and human evaluator respectively. First, we will show that

$$\Pr[\tau_1 = x_1] = \Pr[\pi_1 = x_1]$$

(B.1)

$$\Pr[\tau_1 = x_2] > \Pr[\pi_1 = x_2]$$

(B.2)

To do so, consider the realizations of $\epsilon_1, \epsilon_2, \epsilon_3$ that result in different rankings under θ_A and θ_H . In fact, the only set of realizations that result in different rankings are when $\epsilon_2/\theta = -1$ and $\epsilon_3/\theta = 1$. Thus, the algorithm and human evaluator always rank x_1 in the same position, conditioned on a realization, which proves (B.1); the only difference is that the algorithm sometimes ranks x_2 above x_3 when the human

[form-feed in source extraction]

evaluator does not. Moreover, whenever $\epsilon_1/\theta = -10$, x_2 is more strictly more likely to be ranked first under the algorithm than the human evaluator, which proves (B.2).

Next, we must show that when choosing second, the human evaluator is better off when x_2 is unavailable than when x_3 is unavailable. This is clearly true because for the human evaluator,

```

[U+0014 control character]
[U+0015 control character]
ε1 θH
ε3 θH
Pr x1 +
> x3 +
≈ 1 - 0(6)
θ
θ

```

```

[U+0014 control character]
[U+0015 control character]

```

```

ε1 θH
ε3 θH
3
Pr x1 +
> x2 +
≈
θ
θ
4

```

Thus, conditioned on x_2 being unavailable, the human evaluator gets utility ≈ 3 , whereas when x_3 is unavailable, the human evaluator gets utility ≈ 2.75 . Let u_i be the expected utility for the human evaluator when x_i is unavailable. Putting this together, we get

$$UAH(\theta_A, \theta_H) - UHH(\theta_A, \theta_H) = \sum_{i=1}^3 (\Pr[\tau_1 = x_i] - \Pr[\pi_1 = x_i])u_i$$

$$= (\Pr[\tau_1 = x_1] - \Pr[\pi_1 = x_1])u_1 + (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])u_2 + (\Pr[\tau_1 = x_3] - \Pr[\pi_1 = x_3])u_3$$

$$= (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])u_2 + (\Pr[\tau_1 = x_3] - \Pr[\pi_1 = x_3])u_3$$

$$= (\Pr[\tau_1 = x_2] - \Pr[\pi_1 = x_2])(u_2 - u_3)$$

$$= \sum_{i=1}^3 (\Pr[\tau_1 = x_i] - \Pr[\pi_1 = x_i])u_i$$

>0

The last step follows from (B.2) and because $u-2 > u-3$.

Expanded PaperInterface specification

```
(KR21Monoculture.appendixB1Value 0 = 7 / 4 ∧
  KR21Monoculture.appendixB1Value 1 = 1 / 2 ∧ KR21Monoculture.appendixB1Value 2 = 0) ∧
(KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.plusOne = 1 ∧
  KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.zero = 0 ∧
  KR21Monoculture.appendixB1NoiseValue KR21Monoculture.AppendixB1NoiseAtom.minusOne = -1) ∧
((KR21Monoculture.appendixB1NoisePMF KR21Monoculture.AppendixB1NoiseAtom.plusOne).toReal = 1 / 20 ∧
  (KR21Monoculture.appendixB1NoisePMF KR21Monoculture.AppendixB1NoiseAtom.zero).toReal = 9 / 10 ∧
  (KR21Monoculture.appendixB1NoisePMF KR21Monoculture.AppendixB1NoiseAtom.minusOne).toReal = 1 / 20) ∧
KR21Monoculture.appendixB1GaussianLatentMeasure =
  ((KR21Monoculture.appendixB1NoisePMF.toMeasure.prod KR21Monoculture.appendixB1NoisePMF.toMeasure).prod
    KR21Monoculture.appendixB1NoisePMF.toMeasure).prod
  ((ProbabilityTheory.gaussianReal 0 1).prod
    ((ProbabilityTheory.gaussianReal 0 1).prod (ProbabilityTheory.gaussianReal 0 1))) ∧
(∀ (s theta : ℝ),
  0 < s →
    0 < theta →
      (KR21Monoculture.appendixB1SourceGaussianMixtureFamily s).dist theta =
        KR21Monoculture.rumRankingPMFOfMeasure KR21Monoculture.appendixB1GaussianLatentMeasure
          (fun omega =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
              KR21Monoculture.appendixB1Value c +
                (KR21Monoculture.appendixB1NoiseValue
                  (KR21Monoculture.appendixB1NoiseTripleFunction omega.1 c) +
                    s * KR21Monoculture.appendixBGaussianTripleFunction omega.2 c) /
                  theta)
            )
          )
  )
(KR21Monoculture.appendixB2Value 0 = 3 ∧
  KR21Monoculture.appendixB2Value 1 = 2 ∧ KR21Monoculture.appendixB2Value 2 = 0) ∧
(KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusOne = 1 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusOne = -1 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.plusTen = 10 ∧
  KR21Monoculture.appendixB2NoiseValue KR21Monoculture.AppendixB2NoiseAtom.minusTen = -10) ∧
((KR21Monoculture.appendixB2NoisePMF KR21Monoculture.AppendixB2NoiseAtom.plusOne).toReal = 9 / 20 ∧
  (KR21Monoculture.appendixB2NoisePMF KR21Monoculture.AppendixB2NoiseAtom.minusOne).toReal = 9 / 20 ∧
  (KR21Monoculture.appendixB2NoisePMF KR21Monoculture.AppendixB2NoiseAtom.plusTen).toReal = 1 / 20 ∧
  (KR21Monoculture.appendixB2NoisePMF KR21Monoculture.AppendixB2NoiseAtom.minusTen).toReal =
    1 / 20) ∧
KR21Monoculture.appendixB2GaussianLatentMeasure =
  ((KR21Monoculture.appendixB2NoisePMF.toMeasure.prod
    KR21Monoculture.appendixB2NoisePMF.toMeasure).prod
    KR21Monoculture.appendixB2NoisePMF.toMeasure).prod
  ((ProbabilityTheory.gaussianReal 0 1).prod
    ((ProbabilityTheory.gaussianReal 0 1).prod (ProbabilityTheory.gaussianReal 0 1))) ∧
(∀ (s theta : ℝ),
  0 < s →
    0 < theta →
      (KR21Monoculture.appendixB2SourceGaussianMixtureFamily s).dist theta =
        KR21Monoculture.rumRankingPMFOfMeasure KR21Monoculture.appendixB2GaussianLatentMeasure
          (fun omega =>
            EconCSLib.SocialChoice.Ranking.rankByScore fun c =>
              KR21Monoculture.appendixB2Value c +
                (KR21Monoculture.appendixB2NoiseValue
                  (KR21Monoculture.appendixB2NoiseTripleFunction omega.1 c) +
                    s * KR21Monoculture.appendixBGaussianTripleFunction omega.2 c) /
                  theta)
            )
          )
  )
  )
(∃ delta,
  0 < delta ∧
  ∀ (s : ℝ),
    0 < s →
      s < delta →
        EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
          ((KR21Monoculture.appendixB1SourceGaussianMixtureFamily s).dist 1)
          ((KR21Monoculture.appendixB1SourceGaussianMixtureFamily s).dist 1)
          KR21Monoculture.appendixB1Value -
            EconCSLib.SocialChoice.Ranking.expectedSecondMoverShared
              ((KR21Monoculture.appendixB1SourceGaussianMixtureFamily s).dist 1)
              KR21Monoculture.appendixB1Value <
                0)
  )
  )
  )
(∃ delta,
  0 < delta ∧
  ∀ (s : ℝ),
    0 < s →
      s < delta →
        EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
          ((KR21Monoculture.appendixB2SourceGaussianMixtureFamily s).dist (9 / 10))
          ((KR21Monoculture.appendixB2SourceGaussianMixtureFamily s).dist (11 / 10))
          KR21Monoculture.appendixB2Value -
            EconCSLib.SocialChoice.Ranking.expectedSecondMoverIndependent
              ((KR21Monoculture.appendixB2SourceGaussianMixtureFamily s).dist (9 / 10))
              ((KR21Monoculture.appendixB2SourceGaussianMixtureFamily s).dist (9 / 10))
              KR21Monoculture.appendixB2Value
  )
  )
  )
```

Lean proof endpoint (built separately)

KR21Monoculture.PaperInterface.appendixB_smoothing_source_core

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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